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Zero Closure Was Four-Dimensional — Claim Draft v4

The basic representation is the four-dimensional (r, t, R, Q) , and zero closure imposes on it the light cone $r^2 - t^2 - R^2 - Q^2 = 0$

Author: Noriaki Kihara **ORCID:** 0009-0004-6753-4020 **Date:** 12 August 2026 **Version DOI:** 10.5281/zenodo.21902806 **Concept DOI:** 10.5281/zenodo.21902805 **Position:** “Generative Structure of Dimension” series — Geometry and Algebra of Zero Closure, v4 **License:** CC BY 4.0

Position of this note: **an interpretive paper**. It contains no new theorems. It rearranges known theorems under the single closure condition $\sum x_n^2 = 0$ and fixes what is determined by it and what is not. In v3 we measured what actually happens in the numerical model of the series and compared the geometric statements with the numerical ones.

Changes v1 → v2: Claim 4 was completely revised. In v1 we said “zero closure selects centrally symmetric convex configurations” and left the assignment of signs for $N \geq 5$ unresolved. In v2 we showed (a) that the sign is uniquely determined from length data by “the dimension of the smallest face containing both endpoints”, and (b) that for **parallelotopes** ($N = 2^d$) the alternating sum vanishes, and that centrally symmetric plus convex is not enough (the converse necessity is proved only for $d = 2$; see the v4 corrections). Claim 6B was newly introduced to show that this strong constraint does not contradict the ellipsoid structure of Claim 6.

Changes v2 → v3: Claims 10–14, based on measurements of the numerical model, were added.

- **Claim 10:** $\sum x_n^2 = 0$ is an identity that holds only in the steady state; it is broken in transient and metastable states. We measured this as deviation from the ellipsoidal surface.
- **Claim 11:** A seed is required for space and matter to be generated, and the seed strength directly affects the size of the deviation. The seed $\delta = 0.1$ of this run is large, and the deviation persisted even after entry into the metastable state. **Whether the deviation decreases when the seed is weakened or when τ is extended has not been confirmed.**
- **Claim 12:** The double-centred readout of the system has at most $N - 1$ non-trivial principal axes, and the spectral concentration into the top three directions A, B, C rapidly expands the observed scale. The middle directions barely change in magnitude, and the lower directions move to the imaginary side. **Whether these can be identified with physical spacetime, or whether another map is required, is an open question.** (In v3 we wrote 11 dimensions for $N = 12$; in v4 it is 15 for $N = 16$.)

- **Claim 13:** What is conserved is the **signed** trace; the total real content and the total imaginary content both increase and cancel inside the signed sum. This claim explicitly forbids the misreading “a conservation law holds, therefore no expansion occurs”.
- **Claim 14:** Imaginary directions do not appear in the vacuum control. They appear together with the generation of matter.
- **Claim 15:** A quasi-oscillation on the scale of 10^2 steps is observed, but its period varies as τ evolves. **Whether this is due to the second axiom $U^n = I$ or to some other phenomenon is an open question.**

The naming of the axes was also changed in v3 (§0B).

Changes v3 → v4: We raised the constraint on the resolution N itself to the status of a claim, and replaced the numerical runs with $N = 16$. We also raised the content of the title’s “four dimensions” and the stability of the principal-axis orientation to the status of claims.

- **Claim 16:** If zero closure is realised as a parallelotope family, the resolution is $N = 2^d$ (Claim 4-b). This note takes $N = 16 = 2^4$, $d = 4$. **This is not a sufficient condition** (rank = d alone is not enough either, and satisfying both does not put the vertices on an ellipsoidal surface). **Nor has it been derived that it is a necessary condition for all zero-closure solutions** (Claim 4-e).
- **Claim 17:** If the sign rule is removed, zero closure carries no information. With all-real quantities and no signs there is only the trivial solution, and if **both signs and** configuration may be chosen freely, non-trivial solutions exist from $N \geq 3$ on. **What selects parallelotopes is the sign rule (face dimension), not the form of the equation $\sum x_n^2 = 0$.**
- **Supplement to Claim 4:** We executed the procedure of 4-a (lengths → configuration → convex hull → minimal face dimension) using lengths alone as input, and confirmed exact agreement with the true classification for $d = 2, 3, 4$ (4-f). We also made explicit the exact all-real identity $\sum_{\text{edges}} d^2 = \sum_{\text{main diagonals}} d^2$ and the breakdown of which classes become imaginary.
- **Claim 0:** We made explicit, as a claim, the starting point that **central projection reduces the many-body zero-closure problem to a single quadratic equality constraint**. In the real case $\sum x_n^2 = R^2$ represents an arbitrary point P on the surface of a spherical shell, and the constraint reduces to the single equation $X^T F(X) = 0$. That the complex extension $\sum x_n^2 = 0$ can also be written as $x^2 + y^2 + z^2 - t^2 = R^2 + Q^2$ was already known, but **whether this preserves the same reduction as an equation representing points on a projection surface had not been derived**. Under assumptions (S), (C fixed), (U) and (R), we showed that the level set of the three visible components at fixed $C = t^2 + R^2 + Q^2$ is a closed surface (an ellipsoidal surface) defined by a positive-definite quadratic form G . However, what is mathematically required for the right-hand side to be conserved is only $\dot{C} = 0$, and in the component layer of this numerical model that can be expected only for steady solutions (in transient and metastable states it deviates and oscillates; 0-d). **A conserved quantity does not determine the motion; it only determines the space of permitted motions.**

Corrections within v4 (following third-party review): the following were errors or incomplete derivations and have been corrected. In every case the modification is **not to weaken a claim but to close a hole in the logic**.

How to read the table (important): review was carried out several times, and **some corrections were themselves corrected again by later review**. The “after” column has been unified to the content of the final version. Intermediate proposals have not been left in place. For example, “a section at fixed t ” was corrected once in that form, but was later re-corrected to “a section at fixed $C = t^2 + R^2 + Q^2$ ”. Similarly “a sphere under (S)” was re-corrected to “a sphere under (S) + (C fixed)”.

Location	Before	After
Claim 6	The vertices lie on an ellipsoid only if $n \leq d(d+1)/2$	Wrong. Counterexamples in both directions (one of them is Claim 6B itself). Demoted to a rule of thumb for over-determination

Location	Before	After
Claim 15, §3B	If $U^n = I$ then there are no fixed points	Wrong. $U = I$ is a counterexample. The period divides n (including period 1)
Claim 0-c	It appears as an ellipsoidal surface because the readout is not isotropic	Not a derivation. The readout map $X = \Lambda u$ is made explicit and the result derived from $G = (\Lambda^{-1})^\top \Lambda^{-1} > 0$. It is also made explicit that this is the level set of the three visible components at fixed $C = t^2 + R^2 + Q^2$
Claim 0 title	A conserved quantity determines the motion	A conserved quantity determines the space of permitted motions
Claim 18-e	Non-decay of the deviation follows from the conserved quantity	This contradicted Claim 0. Non-decay is a measured fact, not a derived consequence
Claim 4 title, 16-a	Zero-closing configurations are parallelotopes	Necessity for $d \geq 3$ is unproved (4-e). Restricted to the condition for realisation as a parallelotope family
Claim 17-b	If signs are free it is always solvable	False for a fixed distance set (the equilateral triangle is a counterexample). Restricted to the case where both signs and configuration are free
Claim 5	Zero closure fixes $\text{tr}(T)$	By homogeneity the absolute value is not determined. What is imposed is the single signed condition $S(D) = 0$; the unsigned $U(D) = N\text{tr}(B)$ is not determined (re-corrected in a later round)
Claim 8	$\sum q_n p_n$ is a generator, equivalent to the meaninglessness of the overall scale	No symplectic structure has been introduced. Replaced by the cone argument from homogeneity (which is more rigorous)
Claim 9	Compactness hence a discrete spectrum	An operator must be specified.
Claim 2	The unit is two-dimensional because the relation is two-body	Restricted to compactness alone
Notation	The T of (r, T, R, Q) collided with the inertia tensor T	Made explicit that the map “edge $\rightarrow$ two-dimensional state unit” is not given
Claim 0-b	There is one constraint	The base-layer degree of freedom was changed to lower-case t
		For general complex numbers there are two (equality plus orthogonality). Only under assumption (S: separation of the supports of the real and imaginary parts) does it reduce to one. (S) is the content of Claim 7 but does not hold in the numerical model

Location	Before	After
Claim 0-b	$\sum_n x_n^2$ was written as both 0 and $t^2 + R^2 + Q^2$	The index ranges differ. Organised with the $\mathcal{I}_\pm$ split as $\sum_{n \in \mathcal{I}_+} x_n^2 = t^2 + R^2 + Q^2$
Claim 0-c	The reduction $\mathbb{R}^M \rightarrow \mathbb{R}^3$ was performed silently	A general linear surjection maps a sphere to a solid ellipsoid . Made explicit that restriction to a three-dimensional subspace U is required first, and decomposed the underived part into three stages
Claim 0-c	$\frac{d}{d\tau}(X^\top GX) = 0$	This contradicted 0-d. The correct form is $\dot{C}$. $\dot{G}$ (shape and orientation), $\dot{C}$ (scale) and $\dot{X}$ (motion on the surface) separate
Claim 0-c	Diffusion of the orientation is direct evidence that G is not fixed	It presupposes identifying the eigendirections of G with the top three principal axes. Since that identification is itself unresolved, the statement is conditional
Notational remark in Claim 5	T and B “coincide when B is positive semi-definite”	They cannot coincide, since the matrices have different sizes. $B = VV^\top$ ($N \times N$), $T = V^\top V$ ($d \times d$); the non-zero eigenvalues agree with multiplicity and $\text{tr}(B) = \text{tr}(T)$
Body of Claim 8	Scale symmetry follows from the complex convention	It does not depend on complexity. It follows from the equation being homogeneous (the conclusion had been corrected but the body had not)
§3	“Dynamics, time evolution and interaction have not been measured”	Wrong. Claims 10–18 measure time evolution extensively. The correct statement is “ the dynamical law F has not been derived ”
Claim 0-c	“Unconditionally: the section at fixed t is a sphere”	Not unconditional. No assumption $\rightarrow$ two constraints / (S) $\rightarrow$ one equality constraint (a family of spheres) / (S) + (C fixed) $\rightarrow S^{ \mathcal{I}_+ -1}$ / + (U) $\rightarrow S^2$ / + (R) $\rightarrow$ ellipsoidal surface: organised into five stages . What must be fixed is not t but $C = t^2 + R^2 + Q^2$
Claim 0-b	$\sum_n x_n^2 = t^2 + R^2 + Q^2$ reappeared immediately after the $\mathcal{I}_\pm$ split	Unified to $\sum_{n \in \mathcal{I}_+} x_n^2$. Also made explicit that the form $x^2 + y^2 + z^2 - t^2$ presupposes the choice of “three visible components” and therefore belongs after assumption (U)
Claim 5	“The constraint is one condition” in general	This clashed with Claim 0-b. For general complex numbers there are two. Restricted to the layer of real distance geometry , and to statements under assumption (S)

Location	Before	After
Claim 0-c	$\dot{G} = \text{shape}, \dot{C} = \text{scale}$	Not unique because of the redundancy $G \mapsto \alpha G, C \mapsto \alpha C$. Made unique by $G = g\hat{G}, \det \hat{G} = 1, \rho^2 = C/g$
Heading of 4-d, 6B, 16-b	Parallelotopes are “necessary”, “required”, “the only ones that lie on it”	Necessity is unproved for $d \geq 3$. Changed to “a stronger constructive family exists”, “which Claim 4-b guarantees”, “in this test only parallelotopes gave 1, but uniqueness is unproved”
Claim 8 title	Scale symmetry is not an independent axiom (unconditional)	Distinguished reading A (invariance of the solution set) and reading B (identification of states) of axiom 0.5. Under A it can be dropped; under B a projectivisation is separately required. Deciding which is intended is an open task
Claim 1	“The motion of the system decomposes into a direct sum of two-dimensional rotation planes”	The decomposition is per generator , and since the algebra is non-commutative different generators cannot be block-diagonalised simultaneously. The rank is the dimension of a maximal abelian subalgebra, not a permanent number of units
Claim 2B	The explanation of four dimensions had become excessively conditional	Returned to the skeleton. The basic representation is the four-dimensional (r, t, R, Q) , and zero closure imposes on it the light cone $r^2 - t^2 - R^2 - Q^2 = 0$. That is enough. x, y, z belong to the later three-dimensional readout, and the Stiefel manifold, the product of spheres and the number of hidden components are not needed for the four-dimension claim
Claim 0-c	$\mathbb{R}^M, S^{M-1}, U \subset \mathbb{R}^M$ in the reduction argument	After (S) the object is the visible side . Unified to $\mathbb{R}^{ \mathcal{I}_+ }, S^{ \mathcal{I}_+ -1}, U \subset \mathbb{R}^{ \mathcal{I}_+ }$
Claim 0 title	“This reduction is not lost under complex extension” (unconditional)	For general complex numbers there are two constraints. “ Under assumption (S) ” was added
End of 0-c, Claim 2B	Jumped directly to $S(D) = 0$	Made explicit that assumption (Γ) is used
Claim 0-c	A varying C was called a conserved quantity	Do not call something conserved when it is not. Unified to “the quadratic scale C ”, reserving “conserved quantity” for the case $\dot{C} = 0$. “ $\dot{C} = 0$ only in the steady state” was also changed to “ what can be expected to persist over an interval is ”, since it can occur at isolated times

Location	Before	After
Claim 0-b	The Stiefel 2-plane as “a partial answer to the two-dimensionality of Claims 1 and 2”	The three “2”s live in different spaces. Stiefel is in the relation-coefficient space $\mathbb{R}^M$, Claim 1 is in the vertex space $\mathbb{R}^N$, Claim 2 is an abstract state unit. The identifying map is underived
Claim 0-c	The full state set under (S) written as $\bigcup_C S^{ \mathcal{I}_+ -1}$	It is a product of spheres. At fixed C the full state set is $S^{ \mathcal{I}_+ -1} \times S^{ \mathcal{I}_- -1}$; the projection to the visible side is the family of spheres. If all C are allowed, the union on the visible side is all of $\mathbb{R}^{ \mathcal{I}_+ }$ and no information remains
Claim 0-b	“In the model there are still two constraints” connected to the three quantities measured in 18-b	Different questions. The former is the number of target conditions in the component layer; the latter is which quadratic quantities are conserved in the readout layer. $\sum_e x_e^2$ is non-zero in the readout layer
§6 tasks	Assumption (Γ : geometric sign correspondence) was missing	Restored to five items. The agreement of the $\mathcal{I}_\pm$ split with the face-dimension sign is exactly the gap of §3B ; without it complex zero closure and $S(D) = 0$ are not connected
Claim 18-f	The δ sweep can discriminate option 1 from 2 and 3	Logically wrong. Even if the diffusion stops, 1 cannot be denied. What can be measured is only whether the diffusion is transient/seed-dependent or persists in the steady limit. An experiment discriminating the three readings has not been designed
Notation	L (index set and readout map), R (index set and physical quantity), G (assumption and positive-definite matrix) collided	Index sets renamed to $\mathcal{I}_\pm$, the readout map to Λ , and the geometric-sign assumption to (Γ)
Claim 2B	The λ of the projectivisation $\sim \lambda(r, t, R, Q)$ was unrestricted	r is a length with $r \geq 0$, so $\lambda > 0$. To include signs, introduce $\tilde{r} \in \mathbb{R}$ and set $r = \tilde{r} $
Claim 14	Vacuum rank 15 described as “ that is, no degeneracy ”	The opposite. Rank 15 only says that the 15 non-trivial eigenvalues are non-zero. In fact they are all 0.0645, i.e. maximally degenerate . Changed to “the 15 eigenvalues other than the trivial zero are non-zero; in the late regime the spectrum is completely degenerate and isotropic”

Location	Before	After
Claim 14	The negative eigenvalues of B identified with the imaginary symbol of Claim 7	This re-connected the three layers separated in Claim 5 (real/imaginary of the complex form, face-dimension sign, sign of the eigenvalues of B). Changed to “the identifying map is underived; it remains a qualitative candidate correspondence”
End of Claim 10	“The scope of Claims 1–9 is limited to the steady state ”	Claims 4 and 6B are static geometric theorems and do not require stationarity. Changed to “the propositions themselves do not require the steady state; to apply them to this model the premises must hold, and in transient and metastable states they do not”
Claim 13	“The structure is the same when N is changed”	Only two points were checked. By the same standard as Claim 12, changed to “ N -independence is underived”
Claim 14	Asserted the cause of the larger vacuum deviation at $N = 16$	Not a comparison under identical conditions. Changed to “qualitatively consistent with the trend of Claim 16-b; the cause has not been established”
Claim 9	The real spinor square map is “2:1”	2:1 away from the origin (the preimage of the origin is a single point). In the projectivisation the origin is removed, so this causes no problem
§5.4 heading	“What are the 15 dimensions ”	The measured rank is 8–11. Changed to “what are the 15 principal axes ” (because “15 dimensions” reads as configuration dimension 15)
Claim 12	“If the rank falls to $d = 4$ then a parallelotope has been reached ”	Wrong. It contradicts the counterexample of Claim 16-b (a random configuration with $N = 16$, rank = 4 is neither a parallelotope nor on an ellipsoidal surface). Changed to “a four-dimensional parallelotope must have rank 4; the measured rank is 8–11, so it is not a parallelotope. The converse, however, does not hold”
Claim 18-e	The non-zero constant of the readout layer described as “ zero closure is not broken”	Zero closure is defined as $\sum x_n^2 = 0$. The readout-layer $\sum_e x_e^2$ is non-zero , hence not zero closure. Changed to “it exists as a non-zero conserved quantity”

Location	Before	After
Claim 0-d	“State in which layer 0-c is being used”	Neither layer satisfies all the assumptions of 0-c (the component layer has varying C ; the readout layer fails (S)). Changed to “it is necessary to establish in which layer and under which map (S), (U) and (R) hold”
Claim 0-d	“The ‘solvability’ of 0-c is a statement about the steady state ”	0-c does not require stationarity, only $\dot{C} = 0$. The steady state is needed because in the component layer of this model C can be expected to be fixed only for steady solutions
Claim 18 title	“What is conserved is an inner product ”	$\sum x_e^2$ is a quadratic form built from a complex symmetric bilinear form, not an inner product. Changed to “orientation-independent quadratic forms are conserved”
Heading of 18-b	“There are two ” conserved quantities	It has not been shown that there are no others. Changed to “ at least two kinds are confirmed”
§6 Question 1	The complex representation “saves one axiom”	What reduces the axiom is the homogeneity of the zero-closure equation , not the complex representation. What the complex representation adds is the orthogonality condition $\sum q_n p_n = 0$
Misreading 3, Claim 12	“This is not compactification”	What the numbers can deny is the shrinking picture. Restricted to “it is not compactification of the type in which extra directions shrink and become invisible”
Claims 6, 2B and Figure 4	“Zero closure constrains $l = 0$ (the magnitude)”	A chain that became wrong through the correction to Claim 5. Zero closure $S(D) = 0$ does not fix the scale (for the parallelotope family it is an identity, so it holds under any choice of generators). What determines $l = 0$ is fixing C or a normalisation
Claim 5 title	“What is constrained is one trace-type condition”	$S(D) = 0$ is not a trace-type condition. Changed to “it imposes the single signed condition $S(D) = 0$; the unsigned $U(D) = N\text{tr}(B)$ is not determined by zero closure alone”
Claim 5	Written as if the face-dimension sign rule followed from assumption (S)	Assumption (Γ: geometric sign correspondence) is separately required. (S) gives only the separation of supports; it does not give $\mathcal{I}_+ = \{s_e = +1\}$, $\mathcal{I}_- = \{s_e = -1\}$. This is exactly the gap of §3B

Location	Before	After
Claim 7	The group generated by i is $\mathbb{Z}_2$	At the amplitude level it is $\mathbb{Z}_4$ ($1 \rightarrow i \rightarrow -1 \rightarrow -i \rightarrow 1$). What becomes $\mathbb{Z}_2$ is the induced action on the sign of the squared quantity
Claim 12 title	“The system has $N - 1$ degrees of freedom ”	Not degrees of freedom, but at most $N - 1$ non-trivial principal axes (signed spectral directions) . The same reason as the distinction between ambient dimension and independent degrees of freedom in Claim 2B
Claim 5	From the unsigned identity $\sum d^2 = N \text{tr}(B)$ to “zero closure constrains one trace-type condition”	Confusion of the signed $S(D) = \sum s_e d_e^2$ with the unsigned $U(D) = \sum d_e^2$. $S(D) = 0$ does not determine the value of $U(D)$. Only with C fixed do we obtain $U(D) = 2C$ and $\text{tr}(B) = 2C/N$. The “sign of the eigenvalues of B ” in Claim 13 is a third, different thing
Claims 4-b and 6B	The generators were written u_i in both sections	A factor-2 discrepancy (factor 2 in edge length, factor 4 in Σd^2 ; verified numerically). 4-b uses edge vectors e_i ; 6B uses $A \equiv \frac{1}{2}[e_1 \cdots e_d]$
Claim 0-c	“Fix (t, R, Q) , equivalently fix C ”	Not equivalent . Even with C constant, (t, R, Q) can be redistributed. What is required is fixing C ; fixing (t, R, Q) individually is a stronger sufficient condition. Also “three-dimensional section” is inaccurate; the correct phrase is “ the level set of the three visible components at fixed C ”
Claim 2	“The number of common components is 3” asserted immediately before stating that the aggregation rule is underived	Separated into three layers: 3 for a single block (proved) / 3 for the whole system (numerical fact) / the aggregation rule connecting them (underived)
Claim 0-c	Assumption (R) had swallowed (U)	Separately defined (U: the state lies in a three-dimensional subspace) and (R: $\Lambda : U \rightarrow \mathbb{R}^3$ is linear and regular)
Claim 6	“Saturation is caused by the closure condition being quadratic”	Not accurate . The cause is that the readout is the second-order tensor T . Fourth moments do distinguish the cases (the numerical checks in the text show this)
Claim 6	$l = 2$ is “shape 2 + orientation 3”	The 5 components are always correct, but that decomposition holds in the non-degenerate generic case . At degeneracies the eigendirections lose meaning

Location	Before	After
Claim 4-a	“Splits into exactly d classes”	For a general polytope some classes may be empty (a simplex has only $k = 1$). Changed to “each pair is assigned a unique k ; for parallelotopes all classes appear”
§3B	The numerical model “ imposes ” $\sum x_e^2 = 0$	A hard constraint could not be broken. In fact it is non-zero (Claim 10). Changed to “ it is placed as a steady-state closure condition ; it is not enforced by projection at each step”
Claim 0-c	“Under assumption (S): the visible side is a sphere $S^{ \mathbb{Z}_+ -1}$ ”	(S) alone does not give a sphere. C can vary, so the permitted states form a family (cone) of spheres $\bigcup_C S_{\sqrt{C}}^{ \mathbb{Z}_+ -1}$. It becomes a sphere only when (C fixed) is added. Four stages $\rightarrow$ five stages
Claim 6	The multipole table contained $l = 1$ (centre offset, zero by central symmetry)	Wrong twice. $l = 1$ is the first moment and is not a component of T (the decomposition of Sym^2 is $l = 0 \oplus l = 2$ only). Moreover it vanishes not by central symmetry but by the definition of barycentric coordinates
Opening of 6B	“The circumscribed ellipsoid and the inertia ellipsoid are identical”	Unified to identical with the normalised inertia ellipsoid ($c = d/N$) (the conclusion had been corrected but the opening had not)
Claim 6B	“A parallelotope is not a generic configuration ”	The word can be read either way. Changed to “ it is not a configuration in general position ; the vertices are built from d generators, so there are dependencies among the conditions”
Claim 17-b	With free signs and configuration it is “ almost always ” satisfiable	A measure-theoretic statement, unproved. Deleted and replaced by “a non-trivial family of configurations exists; hence the zero-closure equation alone cannot select uniquely”
Claim 3 title	“The only thing undetermined is the sign of the signed volume ”	For degenerate configurations the volume is 0 and the binary choice does not exist. Changed to “ for a non-degenerate maximal-dimensional configuration , the only thing undetermined is the $\mathbb{Z}_2$ of orientation (reflection)”

Location	Before	After
Claim 0 title and text	“Reduces to motion on a single conserved quadratic form”	What zero closure gives is an equality constraint . Unified to “reduces to a single quadratic equality constraint; if C is conserved, the motion closes on a fixed quadric”
Claim 0-c	“A section at fixed t ”	Fixing only t leaves R, Q free, so C changes and the surface does not close. What must be fixed is $C = t^2 + R^2 + Q^2$. Fixing (t, R, Q) individually is merely a stronger sufficient condition
Claim 2 title	The three components are “determined by the relation being two-body ”	Only as far as selecting $d = 2$ within $\text{Sym}^2(\mathbb{R}^d)$. The bridge to two-body-ness is underived. Title and text unified
Claim 2B title, §6	“The independent degrees of freedom are 4”	Old text remained in the heading, the misreading section, Claim 16 and the conclusion. All unified to “a four-dimensional ambient space with coordinates (r, t, R, Q) ; by the null-cone constraint the independent degrees of freedom are 3”
Claim 2B	Wording that allowed four dimensions to be read unconditionally	Made explicit that it is a statement within the central-projection formulation that adopts assumption (S) . The general complex solution space is $\sqrt{C} V_2(\mathbb{R}^M)$, which is high-dimensional
Figure 4, §6	“The circumscribed ellipsoid coincides with the inertia ellipsoid ”	Unified to coincidence with the normalised inertia ellipsoid ($c = d/N$)
Claim 2B	(r, t, R, Q) are four independent degrees of freedom	There is one constraint $r^2 = t^2 + R^2 + Q^2$, so the independent number is 3 . The correct statement is “ the null cone inside a four-dimensional ambient space ”. The four dimensions of the title refer to the ambient dimension
Claim 2B	x, y, z are determined by r	For one r there are infinitely many points. $r \mapsto S_r^2$ (it determines an orbit). What selects the point is the readout or the dynamics
§0 definitions, Claim 6B	Inertia ellipsoid $x^T T^{-1} x = 1$	With this definition the coincidence of Claim 6B fails. Since $Q^{-1} = (d/N)T$, normalise with $c \equiv k/N$. This unifies the definition, 6B, the deviation s_i and 16-c

Location	Before	After
Claim 0-b	It reduces to one conservation condition	What zero closure gives is a single equality constraint . $\dot{C} = 0$ does not follow. The algebraic reduction and the requirement on the dynamics were separated
Claim 1 title	The relation layer is completely reducible	A term from representation theory that does not match what is proved. Changed to “each antisymmetric generator decomposes orthogonally into two-dimensional rotation blocks”
Claim 2	The measured 3 is direct evidence that the relation is two-body	Only as far as selecting $d = 2$ under the assumption of $\text{Sym}^2(\mathbb{R}^d)$. The connection to two-body-ness is underived
Table in Claim 9	The state space for $(3, 3)$ given as $\mathfrak{so}(3, 3)$	That is the Lie algebra of the symmetry. The state space is $\text{Gr}(2, 4; \mathbb{R})$. The columns were separated
Claim 9	With real signatures all the main structures hold	Not deducible from that table. Restricted to the structures examined (null cone, projectivisation, 2:1 cover, Klein type)
Claim 2	The three components follow immediately from Claim 1	An aggregation rule from “the 2D block of each generator” to “three common components” is separately required. This is a different gap from the “edge $\rightarrow$ two-dimensional state unit” one already noted

- **Claim 2B: The content of the title’s “four dimensions” was made explicit.** The basic representation is the **four-dimensional** (r, t, R, Q) , and zero closure imposes on it the **light cone** $r^2 - t^2 - R^2 - Q^2 = 0$. x, y, z belong to the subsequent three-dimensional readout and are not needed for the four-dimension claim. The degrees of freedom on the null cone are 3, and 2 after projectivisation. **This is a different thing from the parallelotope dimension $d = 4$ of Claim 16; the numerical coincidence is accidental.**
- **Claim 18: The orientation of the principal axes is not conserved, but orientation-independent quadratic forms are.** The subspace spanned by the top three principal axes loses correlation down to the random baseline in about 2000 steps. On the other hand $\sum_e |x_e|^2$ (Hermitian) and $\sum_e x_e^2$ (bilinear, complex) are both **conserved within numerical precision**, and the latter is **invariant as a complex number right through the transition** (unproved as an analytic conservation law). Both are of trace type and are therefore unaffected by the diffusion of orientation. **However, conservation holds only in the readout layer, not in the component layer.**
- **Restriction of Claim 10:** It is made explicit that Claim 10 is a statement about the component layer (Claim 18-e). **The term “zero closure” is not applied to the readout layer.** There $\sum_e x_e^2$ takes a **non-zero** value and is conserved within numerical precision.
- **Replacement of the numerical sections:** up to v3 the numerical sections were based on runs at $N = 12$. Since 12 is not of the form 2^d , by Claim 16 it is not an object to be compared with the geometric statements. They were replaced by runs under the same conditions at $N = 16$, $T = 40000$. Accordingly **the number of principal axes changes from $N - 1 = 11$ to 15** (Claim 12).

0. Definitions

The following words are used in this note only in the senses given here. Words that are not defined are not used.

Term	Definition
Resolution N	The number of vertices placed in the system. A positive integer.
Vertex	An endpoint of a relation. It carries no state quantity of its own.
Relation (edge, segment)	An unordered pair of distinct vertices. Total number $M = N(N - 1)/2$.
Relation quantity x_n	A scalar assigned to each relation. Whether it is real or complex is stated in each claim.
Length d_{ij}	The relation quantity read as a non-negative real number and interpreted as the distance between vertices i, j .
Closure	A configuration of N vertices in Euclidean space realising all the given lengths d_{ij} exactly. Its existence is determinate.
Zero closure	The condition $\sum_n x_n^2 = 0$.
Centre	The barycentre of all vertices. It is not a vertex.
Centrally symmetric	The configuration is invariant under point reflection about the centre.
Main diagonal	In a centrally symmetric configuration, the segment joining two vertices exchanged by the point reflection. It passes through the centre.
Inertia tensor	$T = \sum_i v_i v_i^\top$ (v_i is the vertex position with the centre as origin).
Inertia ellipsoid	The level surface of T : $\{x : x^\top T^{-1} x = c\}$, $c \equiv k/N$ (k is the rank of the readout, N the number of vertices). Normalised in v4 (see below).
Semi-axis	The principal half-length of the normalised inertia ellipsoid. $a_i = \sqrt{c \lambda_i(T)}$.
Deviation s_i	$s_i = \sqrt{v_i^\top T^{-1} v_i / c}$. If all vertices lie on the inertia ellipsoid then $s_i = 1$ for every i .

Normalisation of the inertia ellipsoid (corrected in v4): up to v3 the inertia ellipsoid was defined by $x^\top T^{-1} x = 1$, but **with that definition the statement of Claim 6B, “the circumscribed ellipsoid and the inertia ellipsoid coincide”, does not hold.** For a parallelotope $T = 2^d A A^\top = N A A^\top$ and $Q^{-1} = d A A^\top$, so

$$Q^{-1} = \frac{d}{N} T \quad \Longleftrightarrow \quad x^\top Q x = 1 \quad \Longleftrightarrow \quad x^\top T^{-1} x = \frac{d}{N}$$

and the right-hand side is not 1. With $c = 1$ the two remain **similar, coaxial and of the same shape but of different size**.

We therefore normalise with $c \equiv k/N$ (k the rank of the readout). If the readout rank is d then $c = d/N$ and

$$E_Q = E_T^{(d)}$$

becomes an **exact identity**. When measuring in the three-dimensional projection $c = 3/N$; when measuring in full dimension (rank = $N - 1$) $c = (N - 1)/N$.

This normalisation is already the one used on the computational side of this note. The definition of the deviation s_i uses $c = 3/N$ (the figures of §5, Claim 10), and the degeneracy argument

of Claim 16-c uses $c = (N - 1)/N$. **The definitions, Claim 6B, the deviation s_i and Claim 16-c are unified by $c = k/N$.**

Term	Definition
Rotation plane	The two-dimensional invariant subspace appearing in the orthogonal canonical form of an element of $\mathfrak{so}(N)$.
Readout	The operation of extracting a quadratic form from the state. The number of independent components extractable is called the number of readout components.
Subjective space	A local coordinate system that does not include the central direction among its coordinates; the coordinate system available to an inhabitant of the surface of the spherical shell.
Deviation s_i (restated)	How far vertex i departs from the normalised inertia ellipsoid. $s_i = \sqrt{v_i^T T^{-1} v_i / c}$, $c = k/N$. The figures of §5 and Claim 10 measure in the three-dimensional projection, so $c = 3/N$.
Seed δ	The strength of the perturbation added to the initial state in order to make the system generate matter.
Imaginary direction	A principal axis corresponding to a negative eigenvalue of the double-centred Gram matrix B .

0B. Naming convention for the principal axes

The principal axes are called by **ordinal abstract names assigned in decreasing order of eigenvalue**. The names carry no physical meaning.

Name	Position	Content
A, B, C	1st–3rd principal axes	Appear in the three-dimensional projection; used for plotting
D, E, F	4th–6th principal axes	Real, but do not appear in the three-dimensional projection
$h, i, j, k, l, m, n, o, p$	7th principal axis onwards	The sign may pass back and forth between real and imaginary as τ evolves

Double centring produces one trivial zero ($B\mathbf{1} = 0$), so there are $N - 1$ principal axes. For the $N = 16$ adopted in this note there are **15** (A through p). For $N = 12$ there would be 11 (up to l). **This number is the $N - 1$ that comes out of the resolution, not a number chosen from outside** (Claims 12 and 16-d).

This naming replaces the earlier names t, R, Q . We abandoned $t/R/Q$ because the following three measurements established that the distinction had no basis.

1. The labels were assigned merely in decreasing order of eigenvalue; there was not a single independent criterion distinguishing t , R and Q .
2. There are times at which the eigenvalues are nearly degenerate. The measured minimal gaps in the $N = 16$ run are 5.1×10^{-5} between the 1st and 2nd axes ($\tau = 3112$), 3.4×10^{-6} between the 2nd and 3rd ($\tau = 2786$), 2.6×10^{-5} between the 3rd and 4th ($\tau = 3045$) and 1.8×10^{-5} between the 4th and 5th ($\tau = 3803$). At the degeneracies the ordering swaps.
3. The principal-axis vectors are not continuous in time. The absolute value of the inner product of the first principal-axis vector is 0.4689 between $\tau = 0$ and $\tau = 39991$, 0.0450 between $\tau = 4000$ (before the transition)

and $\tau = 39991$, and 0.0233 between $\tau = 9487$ (just after the transition) and $\tau = 39991$. One cannot attach labels to fixed physical axes.

1. Background (how this study came about)

The series starts from **central projection**, $\sum x_n^2 = R^2$. On the surface of a spherical shell of curvature radius R , the subjective space cannot observe the following three things.

- The direction of the central projection
- The number of projection axes
- The breakdown of the contributions of the individual projection axes (the separation of R and Q in $R'^2 = R^2 + Q^2$)

What can be observed is only the **magnitude and sign** of the composite curvature. In this real system we move R^2 to the left-hand side,

$$\sum_n x_n^2 - R^2 = 0$$

and, in order to bring it into the form $\sum x_n^2 = 0$ required by the zeroth axiom of namelessness, complex numbers were introduced under the policy of **attaching the imaginary symbol to unobservable quantities** [S1]. The imaginary unit here was introduced as an operational symbol for moving the unobservable right-hand side to the left, and is given no further meaning.

In the subsequent papers of the series x_n is treated as complex, but, unlike standard quantum theory, **complex conjugation is not used**. For $x_n = a + ib$ we take $x_n^2 = (a + ib)^2$, not $|x_n|^2$ [S1].

Further, scale symmetry was placed as axiom 0.5 and “the only parameter is the resolution” as an additional axiom; with resolution N the number of relations is $M = N(N - 1)/2$, and the research proceeded on the assumption that there are as many complex-represented waves as relations [S1][S2]. The main results obtained are as follows.

- With a single-frequency wave as initial condition, even when the fluctuation is at the limit of computational precision, a geometric-series development occurs after a run-up period, after which the system moves autonomously into a metastable state [S8][S14]. At the stage of transition to the metastable state three directions appear in the system [S6][S7][S11]. The cessation of splitting, the emergence of new orthogonal rotation planes, and the causal separation of that temporal structure were treated separately [S10][S12].
- The number of waves is not fixed inside the system; it is determined by the resolution supplied from outside [S9].
- To reproduce a localised particle-like wave, harmonics of equal amplitude are required [S18].
- For linear waves only boson-like transmission occurs. When an interaction with a computed reflection coefficient (a rotation operation) is introduced, fermion-like elastic reflection and collapse-like wave-packet reactions are reproduced [S3]. The readout of the mixed white-cat/black-cat/grey-cat state is also reproduced [S4].
- There is a special solution near reflectivity 0.7, where the exact roots of finite-order recurrence lie near $\alpha^{-1} \approx 137$ and $\alpha^{-1} \approx 128.946$ [S5]. A predictor near the same 0.70 also appeared in the many-body embedding [S16]. **We do not claim to have derived the fine-structure constant** [S5].
- In this process it was found that the second axiom $U^n = I$ is required [S5].
- The generation mechanism of fermion-like structure, and the genesis of the three spatial axes and proper time, were treated [S13][S15][S17].
- Organising the above, a periodic table of 62 particle-like waves was produced [S18]. The extension to field readout was treated separately [S19]. The conditions under which a seed produces a particle, and the lower bound on the resolution, were treated in the most recent paper [S20].

At the stage of proceeding to the dynamics of the waves, it became clear that the understanding of the foundations was insufficient. Specifically, **whether representing waves by complex numbers is mandatory**, and what form central projection takes under a complex representation, were not settled. This note was written to settle these two points.

On the latter, let us be more precise. The real central projection $\sum_n x_n^2 = R^2$ is an equation for an arbitrary point on the surface of the spherical shell, and it has the advantage that **instead of following the many-body interactions individually, the problem can be reduced to the single constraint of conserving R** . This is the starting point of the series. That after complexification one can still write $x^2 + y^2 + z^2 - t^2 = R^2 + Q^2$ was known, but **whether this equation preserves the same reduction as an equation for points on a projection surface had not been derived**. Claim 0 treats this.

2. Claims

Claim 0 (Central projection reduces the many-body zero-closure problem to a single quadratic equality constraint. If C is conserved, the motion closes on a fixed quadric. Under assumption (S) this reduction is not lost under complex extension)

This is why the series started from central projection. All the claims below rest on it.

0-a. The real case. The many-body problem reduces to a single constraint.

Central projection can be written entirely in real quantities as

$$\sum_n x_n^2 = R^2$$

This is **an equation representing an arbitrary coordinate P** lying on the surface of a spherical shell of radius R in n -dimensional space.

From this the following follows at once. Writing the dynamics as $\dot{X} = F(X)$, the condition imposed on the system is

$$\frac{d}{d\tau} (X^\top X) = 0 \iff X^\top F(X) = 0$$

a single equation.

There is no need to solve the many-body problem of $\sum_n x_n^2$ individually. As long as the single constraint of conserving R is satisfied, the system may move freely on the surface of the shell.

Let us be precise about the wording. A conserved quantity **does not determine the motion**. What it determines is **the space of permitted motions**. There are infinitely many motions on $X^\top X = R^2$, and a single conserved quantity cannot select among them. The advantage of central projection is not that “the motion is uniquely determined” but that **instead of following the many-body interactions individually, the problem is reduced to a single constraint that conserves one right-hand side**. This is why central projection was chosen as the starting point.

0-b. The complex case. The equality of the real squared quantities survives unchanged.

The series takes x_n to be complex and adopts $\sum_n x_n^2 = 0$ as the basic equation (§1, Claim 7).

First the shortest proof. Writing $x_n = q_n + ip_n$ (q_n, p_n real),

$$\sum_n x_n^2 = \sum_n (q_n^2 - p_n^2) + 2i \sum_n q_n p_n$$

so setting the real and imaginary parts separately to zero,

$$\boxed{\sum_n q_n^2 = \sum_n p_n^2}, \quad \boxed{\sum_n q_n p_n = 0}$$

This is algebraically exact and requires no assumption.

This is the most direct proof that “the map to the surface survives complexification”. $\sum q_n^2 = \sum p_n^2$ says that **in each state q and p have the same radius** (fixing C , both lie on the sphere of the same radius $\sqrt{C}$). It is a condition of the same type as the real central projection $\sum x_n^2 = R^2$.

Here we must count how many constraints there are. For a general complex $z_n = q_n + ip_n$, zero closure imposes the **two** real conditions above: the equality condition $\sum q_n^2 = \sum p_n^2$ and the **orthogonality condition** $\sum q_n p_n = 0$. **Hence, for general complex numbers, one cannot say that “there is one constraint”.**

Using the separation of supports (the content of Claim 7) it drops to one. Claim 7 states that “the imaginary symbol is attached only to the unobservable central direction, and the lengths of the segments are all real”. Written in terms of indices, splitting into the observed side $\mathcal{I}_+$ and the unobserved side $\mathcal{I}_-$,

$$z_n = a_n \quad (n \in \mathcal{I}_+, a_n \in \mathbb{R}), \quad z_m = i b_m \quad (m \in \mathcal{I}_-, b_m \in \mathbb{R})$$

That is, **no single component has both a real and an imaginary part.** Then $q = (a_{\mathcal{I}_+}, 0)$ and $p = (0, b_{\mathcal{I}_-})$, so

$$\sum_n q_n p_n = 0$$

becomes **an identity of the structure, not a constraint.** What remains is the single equality condition.

Assumption (S: separation of supports): the supports of the real and imaginary parts are disjoint, $\text{supp}(q) \cap \text{supp}(p) = \emptyset$.

Under assumption (S), the independent real constraints of complex zero closure reduce to the single equality of the sums of squares.

$$\boxed{\sum_{n \in \mathcal{I}_+} a_n^2 = \sum_{m \in \mathcal{I}_-} b_m^2 \equiv C}$$

This is the core of this note. The many-body zero-closure problem drops to a **single equality constraint** on one real scalar.

Do not confuse “equality constraint” with “conservation” (separated in v4). What zero closure gives is

$$|a|^2 - |b|^2 = 0$$

a **single algebraic equality constraint.** $\dot{C} = 0$ **does not follow from it.** The equality can hold while C varies with τ (indeed, as in 0-d, C itself varies in transient and metastable states).

We therefore separate two stages.

Stage one (algebraic reduction; follows from zero closure alone): $\sum z_n^2 = 0$ and assumption (S) $\Rightarrow |a|^2 = |b|^2 = C$. **Stage two (a requirement on the dynamics; does not follow from zero closure):** if in addition $\dot{C} = 0$ is imposed, the motion closes on a fixed quadric.

Zero closure reduces the many-body system to a single scalar equality constraint. If one chooses dynamics that conserve C , the motion closes on a fixed quadric. The first half is a theorem; the second is a design policy for the dynamics.

Even in the general complex case (without assumption (S)) something exact can be said. The two constraints can be written

$$|q|^2 = |p|^2 = C, \quad q \cdot p = 0$$

For $C > 0$, setting

$$E \equiv \frac{1}{\sqrt{C}} \begin{pmatrix} | & | \\ q & p \\ | & | \end{pmatrix} \in \mathbb{R}^{M \times 2}$$

these two are exactly equivalent to

$$\boxed{E^\top E = I_2}$$

That is, **the solution set of general complex zero closure at fixed C is precisely the space of orthonormal 2-frames (the Stiefel manifold)**

$$\sqrt{C} \cdot V_2(\mathbb{R}^M)$$

Moreover the phase transformation $z \mapsto e^{i\theta} z$ merely rotates (q, p) within their plane, so identifying phases gives

$$V_2(\mathbb{R}^M)/SO(2) = \text{Gr}_2^+(\mathbb{R}^M)$$

the **space of oriented two-dimensional planes**.

This must not be identified with the two-dimensionality of Claims 1 and 2. The three “2”s live in different spaces.

Origin	Two-dimensional object	Ambient space
This item (complex zero closure)	Oriented 2-plane (Stiefel/Grassmann)	Relation-coefficient space $\mathbb{R}^M$
Claim 1	Rotation block of an antisymmetric generator	Vertex space $\mathbb{R}^N$
Claim 2	State unit $d = 2$ of the quadratic readout	Abstract state unit

This is as far as one can say rigorously: complex zero closure selects, exactly, an oriented 2-plane inside the relation-coefficient space $\mathbb{R}^M$. **The map identifying this 2-plane with the rotation plane of Claim 1 in $\mathbb{R}^N$, or with the two-dimensional state unit of Claim 2, is underived.**

If a map connecting them is found, the three appearances of two-dimensionality become one. This is one of the tasks this note leaves open.

Assumption (S) corresponds to the special case of selecting, among these 2-frames, those for which the supports of q and p are disjoint.

- Assumption (S) is exactly the content of Claim 7, and it holds in the central-projection formulation (because the imaginary symbol is attached only to the central direction).
- **It does not, however, hold in the numerical model.** The relation amplitudes of the model, $x_e = \sum_{k,\eta} C_2[e, k, \eta]$, are general complex numbers whose real and imaginary supports are not separated. Hence **as long as the steady-state closure condition of the component layer is read as a general complex number, the target condition consists of two real equations** (though during the run that condition itself does not hold exactly; Claim 10). **This is a different question from the conserved quantities of the readout layer in Claim 18-b.** What 18-b measures is not “how many constraints zero closure imposes” but “which quadratic quantities are conserved”, and there $\sum_e x_e^2$ is **non-zero** (Claim 18-c). Do not confuse the number of constraints with the number of conserved quantities.

- **Whether a structure corresponding to assumption (S) appears in the model is an open question.** If it does not, another map is required between the model and the central-projection formulation.

Tidying the notation (corrected in v4): from the above, when $\sum_n x_n^2$ is written as “equal to 0” and when it is written as “equal to $t^2 + R^2 + Q^2$ ”, **the index ranges differ**. With the same range the two are incompatible. Correctly, for the whole set $(\mathcal{I}_+ \cup \mathcal{I}_-)$,

$$0 = \sum_{n \in \mathcal{I}_+} x_n^2 - (t^2 + R^2 + Q^2)$$

and therefore

$$\boxed{\sum_{n \in \mathcal{I}_+} x_n^2 = t^2 + R^2 + Q^2}$$

The left-hand side may have any number of bodies and any number of components. The right-hand side is the quadratic quantity C assembled by the unobserved side. **Whether C is a conserved quantity depends on the dynamics and on the layer** (0-d, Claim 18-d). What zero closure gives is equality, not conservation.

The sign appearing when t^2 moves from the left to the right comes from the policy of §1 of attaching the imaginary symbol to unobservable quantities ($((it)^2 = -t^2)$). The right-hand side $R^2 + Q^2 = R'^2$ is the composite curvature that the subjective space cannot observe, and the separation of R and Q within it is likewise unobservable (§1).

The three-component expansion of the hidden side does not follow from zero closure.

What 0-b derived rigorously is, under assumption (S),

$$\sum_{n \in \mathcal{I}_+} a_n^2 = \sum_{m \in \mathcal{I}_-} b_m^2 = C$$

Just as the visible side can be gathered into a single radius $r^2 = \sum_{\mathcal{I}_+} a_n^2$, **the hidden side likewise gathers first into a single composite radius $h^2 = \sum_{\mathcal{I}_-} b_m^2$** . **From zero closure alone, $r^2 = h^2$.**

Above, however, h^2 is expanded into the **three components $t^2 + R^2 + Q^2$** . **From zero closure one may decompose $C = \sum_{\alpha=1}^k h_\alpha^2$ into any number of components, and no reason to choose $k = 3$ follows.** This is exactly what §1 itself says: the subjective space can observe neither the number of projection axes nor the breakdown of their contributions.

$$(\text{H: three-component representation of the hidden side}) : \quad C = \sum_{m \in \mathcal{I}_-} b_m^2 = t^2 + R^2 + Q^2$$

This three-component representation comes from the central-projection convention of the series (the t, R, Q of §1). **The “four dimensions” of the title is a statement made on top of that convention, namely that (r, t, R, Q) are taken as coordinates, and that is sufficient** (Claim 2B). Whether a formulation with a different number of hidden components is possible may be examined where it becomes necessary.

Furthermore, when the visible side $\mathcal{I}_+$ has three components (this assumption is assumption (U) of 0-c; it is not yet justified here), writing the left-hand side as $x^2 + y^2 + z^2$ and moving t^2 back to the left gives the familiar form

$$x^2 + y^2 + z^2 - t^2 = R^2 + Q^2$$

This form already presupposes the choice “the visible side has three components” and is not justified at the stage of 0-b. In logical order it belongs after assumption (U) of 0-c.

That it can be written in this form was known before. What had not been derived is the single point:

Can this, like central projection in the real case, be interpreted as **an equation representing an arbitrary point P lying on a projection surface centred on the projection axis?**

0-c. Result of the derivation. The level set of the three visible components at fixed C is a closed surface defined by a positive-definite quadratic form.

First check the signature. Regarded as four variables (x, y, z, t) , the left-hand side of $x^2 + y^2 + z^2 - t^2 = R^2 + Q^2$ has signature $(3, 1)$; this is a **one-sheeted hyperboloid and not a closed surface**. It closes only when a section is taken. **If this point is dropped, everything below collapses, so we state it explicitly.**

What must be fixed is not t but C (corrected in v4). Fixing t alone leaves R, Q free, so

$$C \equiv t^2 + R^2 + Q^2$$

changes, the radius changes, and **one does not obtain a single closed surface**. As stated in Claim 2B, t, R, Q are all ambient components, and t is not special. **Correctly, it is the section at fixed C .** Fixing (t, R, Q) individually is merely a **stronger sufficient condition**, since with C constant (t, R, Q) can still be redistributed on the sphere $t^2 + R^2 + Q^2 = C$.

This distinction is theoretically meaningful. Even if the breakdown of t, R, Q changes, the same composite quadratic quantity C gives the same readout surface. This is consistent with the subjective space being unable to observe the separation of R and Q (§1).

Fixing $C > 0$ gives

$$x^2 + y^2 + z^2 = t^2 + R^2 + Q^2 \equiv C > 0$$

In these coordinates this is a sphere of radius $\sqrt{C}$, not an ellipsoid. To obtain an ellipsoid from here one must insert a readout map.

The step down to three dimensions is another bridge. Since we have already used assumption (S), what we are about to map to an ellipsoidal surface is the **visible side**. The visible side is an element of $\mathbb{R}^{|\mathcal{I}_+|}$, and u is three-dimensional. **This reduction $\mathbb{R}^{|\mathcal{I}_+|} \rightarrow \mathbb{R}^3$ is not trivial.**

Why it is not trivial. Under a general linear surjection $P : \mathbb{R}^{|\mathcal{I}_+|} \rightarrow \mathbb{R}^3$ (with $|\mathcal{I}_+| > 3$), the image of the sphere $S^{|\mathcal{I}_+|-1}$ is a **solid ellipsoid, not an ellipsoidal surface**. A surface does not necessarily map to a surface.

To preserve the surface, a **restriction to a three-dimensional subspace** is needed first: one must show that the state lies in a three-dimensional subspace $U \subset \mathbb{R}^{|\mathcal{I}_+|}$ ($\dim U = 3$) and then that $\Lambda : U \rightarrow \mathbb{R}^3$ is regular. The existence of U should connect with “the number of readout components is 3” of Claim 2, but **the argument connecting them is not given.**

The unresolved part therefore decomposes into three stages:

$$\text{complex zero closure} \xrightarrow{\text{underived}} \text{3-dim state subspace } U \xrightarrow{\text{underived}} \Lambda \longrightarrow \text{ellipsoidal surface}$$

Important (the standing of this note’s result): the reduction itself does not require three-dimensionalisation. Even in $|\mathcal{I}_+|$ dimensions, $\sum_{n \in \mathcal{I}_+} x_n^2 = C$ is a single equality constraint, and fixing C makes it motion on $S^{|\mathcal{I}_+|-1}$. That is, the conclusion of 0-a and 0-b, “the many-body problem reduces to a single quadratic equality constraint”, **already holds before the three-dimensional readout**. Three-dimensionalisation is needed to make contact with observation, not for the reduction.

What follows assumes a three-dimensional readout. This is a conditional theorem, not a derivation.

Assumption (U: three-dimensional subspace): the state lies in a three-dimensional subspace $U \subset \mathbb{R}^{|\mathcal{I}_+|}$, $\dim U = 3$.

Assumption (R: linear regular readout): the readout $\Lambda : U \rightarrow \mathbb{R}^3$ is linear and regular, i.e. $X = \Lambda u$, $u \in U \cong \mathbb{R}^3$, $\Lambda \in GL(3, \mathbb{R})$.

(U) and (R) are different assumptions (separated in v4). Up to v3 the definition of (R) had swallowed (U), so it did not correspond to the five-stage organisation. (U) concerns where the state lies; (R) concerns the map from there to the observed coordinates.

Under (R), substituting $u = \Lambda^{-1}X$ into $u^\top u = C$ gives

$$X^\top G X = C, \quad G \equiv (\Lambda^{-1})^\top \Lambda^{-1}$$

G is the Gram matrix of a regular matrix and hence **positive definite**. The surface traced by X is therefore a **closed ellipsoidal surface**, with principal axes the eigenvectors of G and semi-axes $\sqrt{C/\lambda_i(G)}$.

This is the content of 0-c, stated without dropping any assumption.

No assumption (general complex zero closure): there are **two** independent real constraints (0-b). The surface is not yet determined. At fixed C the solution set is $\sqrt{C} V_2(\mathbb{R}^M)$.

Under (S): the orthogonality condition becomes an identity, leaving the **single equality constraint** $\sum_{n \in \mathcal{I}_+} a_n^2 = \sum_{m \in \mathcal{I}_-} b_m^2$. **This is not yet a sphere.** At fixed C the **full state set** is

$$S_{\sqrt{C}}^{|\mathcal{I}_+|-1} \times S_{\sqrt{C}}^{|\mathcal{I}_-|-1}$$

a **product of two spheres**. Since C may vary with τ , the whole set of permitted states is

$$\mathcal{N}_S = \bigcup_{C \geq 0} \left(S_{\sqrt{C}}^{|\mathcal{I}_+|-1} \times S_{\sqrt{C}}^{|\mathcal{I}_-|-1} \right)$$

a **cone over a product of spheres**. Projecting to the visible side $\mathcal{I}_+$ alone gives a family of **spheres**, but if all C are allowed the union is all of $\mathbb{R}^{|\mathcal{I}_+|}$ and no constraint information remains.

Under (S) + (C fixed): the **full state set becomes the product of spheres** $S^{|\mathcal{I}_+|-1} \times S^{|\mathcal{I}_-|-1}$, and **looking only at the visible side**, $S^{|\mathcal{I}_+|-1}$. Fixing C does not follow from zero closure; it is a **requirement on the dynamics** (stage two of 0-b). **Here the meaning of central projection becomes clear: the visible projection obtained by discarding the freedom of the hidden side $\mathcal{I}_-$ is a sphere** — the structure corresponding to $\sum x_n^2 = R^2$ in 0-a.

Under (S) + (C fixed) + (U): if the state lies in a three-dimensional subspace, $u \in \mathbb{R}^3$, $u^\top u = C$, i.e. the **2-sphere** S^2 .

Under (S) + (C fixed) + (U) + (R): its linear image is $X^\top G X = C$, i.e. an **ellipsoidal surface**. The $GL(3, \mathbb{R})$ orbit exhausts all centred ellipsoids.

Unresolved: (S) holds in the central-projection formulation but not in the numerical model (0-b). What (C fixed) mathematically requires is only $\dot{C} = 0$, and in the component layer of this model that can be expected to persist over an interval only for steady solutions (0-d). (U) is underived. The Λ of (R) is not given either. Hence only the *type* “it is an ellipsoid” is determined; **which ellipsoid is not**.

This much can be said; beyond this it cannot. The G above is only the general form following from the assumption “the readout is linear and regular”, and there is no guarantee that G is **proportional to the inverse of the inertia tensor** T . Only by showing

$$G^{-1} \propto T$$

can one say that this ellipsoidal surface is **the inertia ellipsoid itself**. This is underived.

- For parallelotopes $Q^{-1} \propto T$ is shown exactly (Claim 6B, $T = 2^d A A^\top$, maximal numerical discrepancy 10^{-14}). **It is not shown for general configurations.**
- Moreover the numerical model does not reach a parallelotope (rank $15 \rightarrow 8-11$; it does not fall to $d = 4$; Claims 12 and 16-d). **Hence even if $G^{-1} \propto T$ were shown for parallelotopes, that alone would not reach the numerical model.** The bridge would be conditional on “only if a parallelotope is reached”.

- The readout map Λ itself is not given (Claim 2B). Giving Λ and giving the map that reads x, y, z out of r are the same problem.

Therefore what can be said at present is the following two-part statement.

Can be said (under S, C fixed, U, R): complex zero closure gives, on the section at fixed C , a real closed quadric (an ellipsoidal surface defined by a positive-definite G). **The reduction itself holds under (S) alone and requires neither (C fixed) nor (U) nor (R) — but its meaning is “it reduces to a single quadratic equality constraint”, not “it becomes motion on a fixed sphere”.** To become motion on a fixed surface, (C fixed) is required, and that is a requirement on the dynamics. (U) and (R) are needed to make contact with the observed coordinates.

Cannot be said: that this ellipsoidal surface is the inertia ellipsoid itself. For that one must give the readout map Λ and show $G^{-1} \propto T$. **This is the largest underived part of this note.**

Structurally this is consistent with Claims 5, 6 and 6B. **If, under assumption (Γ), we pass to real distance geometry, what zero closure imposes is the single signed scalar condition $S(D) = 0$; what determines the unsigned $U(D)$ (hence $l = 0$, hence the magnitude) is fixing C or a normalisation (Claims 5, 6).** The individual semi-axes A, B, C are not fixed (Claim 6). **The structure “one conserved quantity determines the surface, and the distribution of points on it is free” is the same in the real and the complex case.**

Hence, under assumption (S), the reduction of 0-a is not lost under complexification. As long as the single constraint of conserving the right-hand side is satisfied, the system may move freely on the surface. **Here too, however, a conserved quantity does not determine the motion; it only determines the space of permitted motions. Without (S) there are two constraints and this statement fails.**

0-c. When the surface depends on time. The reduction is still to a single equation.

The argument above was written with G fixed. In that case the constraint is

$$\frac{d}{d\tau} (X^\top G X) = 2 X^\top G \dot{X} = 0$$

(tangential motion on a fixed ellipsoidal surface).

But in the numerical model the orientation of the principal axes diffuses (Claim 18-a: the subspace of the top three principal axes loses correlation down to the random baseline in about 2000 steps). If the readout depends on time, $\Lambda = \Lambda(\tau)$ and hence $G = G(\tau)$, the constraint becomes

$$\boxed{\frac{d}{d\tau} (X^\top G X) = 2 X^\top G \dot{X} + X^\top \dot{G} X = \dot{C}}$$

The right-hand side must not be written as 0. As in 0-d, in transient and metastable states **the quadratic scale C itself varies** (C may be called a conserved quantity only when $\dot{C} = 0$). In this model, $\dot{C} = 0$ can be **expected to persist over an interval** only for steady solutions, or in a layer where conservation holds (that $\dot{C} = 0$ occurs at isolated times is a separate matter).

This general form separates into three contributions.

Term	Meaning
$X^\top \dot{G} X$	change of the shape and orientation of the ellipsoidal surface
$\dot{C}$	change of the quadratic scale C itself (overall scale)
$2X^\top G \dot{X}$	motion of the point on that surface

As it stands, however, the separation of shape and scale is not unique, because $X^\top G X = C$ gives the same surface under

$$G \mapsto \alpha G, \quad C \mapsto \alpha C \quad (\alpha > 0)$$

To read “ $X^\top \dot{G} X$ is shape and $\dot{C}$ is scale” a **normalisation is required**. Splitting

$$G = g \hat{G}, \quad g \equiv (\det G)^{1/3}, \quad \det \hat{G} = 1$$

$\hat{G}$ is a positive-definite symmetric matrix of determinant 1 (**purely shape and orientation**) and g carries the size. The surface can be written

$$X^\top \hat{G} X = \rho^2, \quad \rho^2 \equiv \frac{C}{g}$$

and $\hat{G}$ (**shape and orientation**) and ρ (**overall scale**) **separate uniquely**. Use this form when constructing the dynamics. Since $\det \hat{G} = 1$ implies $\text{tr}(\hat{G}^{-1} \dot{\hat{G}}) = 0$, $\dot{\hat{G}}$ automatically represents only volume-preserving deformations.

This does not break the conclusions of 0-a and 0-c. It is still a single scalar condition. What breaks is only the picture of “tangential motion on a fixed ellipsoidal surface”; correctly, it is **motion on a time-dependent family of ellipsoidal surfaces. Start from this form when constructing the dynamics.**

Relation to Claim 18-a (conditional): Claim 18-a shows that the orientation of the top three principal axes diffuses. **If one identifies the eigendirections of G with the top three principal axes of the numerical model, this is evidence that $\dot{G} \neq 0$.** That identification is itself unresolved (the readout map Λ is not given; $G^{-1} \propto T$ is unproved). **Without presupposing the identification, $\dot{G} \neq 0$ cannot be deduced from Claim 18-a.**

0-d. Scope of the conservation. What is mathematically required is only $\dot{C} = 0$. In this model that can be expected for steady solutions.

This is already known, but it must always be attached when using 0-c.

What 0-c mathematically requires is only $\dot{C} = 0$. In the component layer of this numerical model that can be expected when the system has reached a steady state. In transient and metastable states C deviates and may oscillate.

Measurement supports this. The deviation from the ellipsoidal surface does not decrease monotonically with τ ; it increases through the transition and stays around $\max/\min = 11$ for 30,000 steps after it (Claim 10). The closure residual has median 1.338×10^{-2} even in the metastable state, and the number of steps at which it was exactly zero is 0. A quasi-oscillation on the scale of 10^2 steps is superposed, and its period varies with τ (Claim 15).

Here we must separate the mathematical proposition from the facts of the numerical model.

The proposition 0-c itself does not require a steady state. It requires only $\dot{C} = 0$; as long as C is constant the system **may move non-stationarily on a fixed surface**. This is exactly the central idea of 0-a (a conserved quantity determines only the space of permitted motions, not the motion).

The steady state is needed because, in the component layer of this numerical model, fixing C can be expected only for steady solutions. That is, it is not that “0-c holds only in the steady state” but that “in the component layer of this model, $\dot{C} = 0$ can be expected only for steady solutions”.

However, **the layers must be distinguished** (Claim 18-d). The deviation above concerns the **component layer**. In the **readout layer** ($\sum_e x_e^2$ with the harmonics summed first, the aggregation that gives distances), this quantity is fixed at the complex value given by the initial condition and is invariant to a precision of 10^{-7} right through the transition (Claim 18-c). **“The conserved quantity holds only in the steady state” applies to the component layer; the readout layer has a conserved quantity that survives the transition.**

At present, however, there is no layer to which 0-c can be applied directly. The ellipsoid theorem of 0-c requires (S) + (C fixed) + (U) + (R). In the component layer both the closure and C vary; in the readout layer the quadratic quantities are conserved but the numbers are **general complex**, so assumption (S) does not hold (0-b). Hence “state in which layer 0-c is used” is not enough. **Correctly: in the present numerical model it is unconfirmed that either the component layer or the readout layer satisfies all the assumptions of 0-c; it is necessary to establish in which layer, and under which map, (S), (U) and (R) hold.**

- Status: 0-a is known (real central projection). The $\mathcal{I}_\pm$ split of 0-b, the reduction to one constraint by assumption (S), and the fact that the general complex solution set is $\sqrt{C} V_2(\mathbb{R}^M)$ (a Stiefel manifold) are **newly derived in this note. The ellipsoidal surface of 0-c is a conditional theorem under (S, C fixed, U, R); (C fixed) holds only in the steady state and (U), (R) are underived.** 0-d rests on the measurements of Claims 10, 15 and 18.
- **Unresolved 1 (the largest):** the readout map Λ is not given and $G^{-1} \propto T$ is unproved. Until these are shown one cannot say “the readout surface is the inertia ellipsoid”.
- **Unresolved 2:** which measured quantities t, R, Q correspond to is not given. As long as the identification of the principal axes $A, \dots, p$ with physics is unresolved (Claims 12, 18-f), the assignment of t, R, Q in 0-b is also undetermined. This is the same problem as the readout map from r not being given (Claim 2B).
- **Derivation program: not written.** 0-c is an analytic statement; confirmation on the numerical model (whether the point P actually lies on the ellipsoidal surface once the conserved quantity is fixed) has not been carried out. The deviation measurements of Claim 10 show that it does *not* lie on it, but that is because the steady state has not been reached (0-d), not a refutation of 0-c. **Confirmation is needed in a state where (S), (U), (R) and fixed C all hold; in this numerical model the candidate for that is the steady state.**

Claim 1 (Each antisymmetric generator decomposes orthogonally into two-dimensional rotation blocks)

[Title corrected in v4] The title up to v3 was “the relation layer is completely reducible”. “Completely reducible” is a term of representation theory and differs from what is actually proved here (the orthogonal canonical decomposition of each generator). Moreover the decomposition is **per generator** and is not a decomposition common to the whole layer (see below). The title was matched to what is proved.

The edge set of K_N corresponds one-to-one with a basis of $\mathfrak{so}(N)$, and $\dim \mathfrak{so}(N) = M = N(N-1)/2$. Any $A \in \mathfrak{so}(N)$ decomposes, under an orthogonal transformation, into a direct sum of $\lfloor N/2 \rfloor$ two-dimensional rotation planes. The number of rotation planes that can be moved independently at once does not exceed $\text{rank } \mathfrak{so}(N) = \lfloor N/2 \rfloor$.

Hence, even if the resolution N is increased, the internal structure (two-dimensionality) of the blocks appearing in the canonical decomposition of each generator does not change. Only the number of blocks $\lfloor N/2 \rfloor$ changes.

However, one cannot say that “the system is made of a collection of two-dimensional units”, because the decomposition is per generator, as follows.

[Gap in the derivation, made explicit in v4, no. 1] The canonical decomposition above is **per generator**. Precisely, for any $A \in \mathfrak{so}(N)$

$$\exists O_A \in O(N) : \quad O_A^T A O_A = J(\omega_1) \oplus J(\omega_2) \oplus \dots$$

and **the planes of the decomposition change with A .** Since $\mathfrak{so}(N)$ is non-commutative, different A, B cannot be block-diagonalised **simultaneously** by one fixed two-dimensional plane decomposition.

Also, what $\text{rank } \mathfrak{so}(N) = \lfloor N/2 \rfloor$ counts is **the dimension of a maximal abelian subalgebra**, i.e. the number of mutually commuting rotations, not that “the system is permanently made of $\lfloor N/2 \rfloor$ two-dimensional units”.

Hence what can be said rigorously is only that “each antisymmetric generator can be canonically decomposed, generator by generator, into two-dimensional rotation blocks”. One cannot say that “the motion of the system decomposes into a direct sum of fixed two-dimensional planes”. This distinction matters when passing to Claim 2 (below).

- Basis: the orthogonal canonical form of a real antisymmetric matrix (known).
- Numerical check: for $N = 3, 4, 5, 6, 8, 12$ the number of independent rotation planes = $\lfloor N/2 \rfloor$.

Claim 2 (The number of components of the quadratic readout is 3. This value selects the state unit $d = 2$. The connection to two-body relations is underived)

The number of independent components of a quadratic form on a two-dimensional real vector space is 3:

$$\dim \text{Sym}^2(\mathbb{R}^2) = \frac{2 \cdot 3}{2} = 3$$

By Claim 1 **each generator** decomposes into two-dimensional rotation planes. **For a single two-dimensional block** $\dim \text{Sym}^2(\mathbb{R}^2) = 3$, so the number of components readable as quadratic quantities from that block is 3.

Three things must be separated here.

	Content	Status
Mathematical consequence	The quadratic readout of a single 2D block has 3 components	Proved
Numerical fact	The common readout of the whole system also has 3 components (measured for $N = 3 \dots 200$)	Measured
Underived	The aggregation rule passing from the blocks of different generators to 3 components common to the whole system	Not given

The statement about the whole system, “it is 3 regardless of the number of planes”, is a numerical fact and not a derived consequence.

[Gap in the derivation, made explicit in v4, no. 2] What Claim 1 gives is a **per-generator** 2D block decomposition, not a fixed plane decomposition (gap 1 of Claim 1). Hence passing from “the 2D block of each generator at each time” to “a 3-component readout common to the whole system” requires **one aggregation rule**.

$$\{\text{2D block of each generator}\} \xrightarrow{\text{aggregation rule (underived)}} 3 \text{ common components}$$

This note does not give that aggregation rule. The numerical checks ($N = 3, \dots, 200$, always 3) suggest that the aggregation does hold, but do not explain why. This is a **different gap** from the “edge $\rightarrow$ two-dimensional state unit” gap discussed below.

In this series we have associated this two-dimensionality with the fact that a relation is a pair of two vertices (a segment has two endpoints). The map connecting the two is, however, underived (see the gap below). Note that Claim 0-b shows that complex zero closure itself builds an isometric orthogonal 2-frame from the real and imaginary parts. **At least one origin of two-dimensionality comes directly from the algebra of complex zero closure, without passing through two-body-ness.**

[Gap in the derivation, made explicit in v4] Claim 1 says that elements of $\mathfrak{so}(N)$ decompose canonically into two-dimensional rotation planes. That is correct. But “**the unit of the state is a two-dimensional real vector space because a relation has two vertices**” does not follow immediately. That the rotation plane is two-dimensional and that the state unit is a two-dimensional real vector space use the same “2” but are different statements.

To connect them, one map — or an axiom giving it —

edge (relation) $\longrightarrow$ two-dimensional state unit

is required. **This note does not give it.** At present Claim 2 is stated assuming this correspondence. **Supplying this one step would make Claim 2 considerably stronger.** Conversely, until it is supplied, the “3” of Claim 2 may also be read as coming from “the rotation plane is two-dimensional” rather than from “the relation is two-body”.

- Numerical check: for $N = 3, 4, 6, 8, 12, 20, 50, 200$ (1 to 100 independent planes) the number of components of the common readout is always 3.
- **Refutation condition:** if the unit were d -dimensional the number of readout components would be $d(d+1)/2$. A measured 6 would mean a three-dimensional unit, 10 a four-dimensional one. **Within the assumption that the quadratic readout has the form $\text{Sym}^2(\mathbb{R}^d)$, a component count of 3 selects $d = 2$ uniquely.**
- **However, “therefore the relation is two-body” is underived,** because the map connecting the unit dimension $d = 2$ with “a relation is a pair of two vertices” is not given (gap 2 above). The measured 3 is evidence for $d = 2$, **not direct evidence for two-body-ness.**

Claim 2B (The basic representation is the four-dimensional (r, t, R, Q) . Zero closure imposes on it the light cone $r^2 - t^2 - R^2 - Q^2 = 0$)

This is the “four dimensions” of the title. The skeleton is exhausted by the following two lines.

$(r, t, R, Q) \text{ — four dimensions}$

$r^2 = t^2 + R^2 + Q^2 \iff r^2 - t^2 - R^2 - Q^2 = 0$

The basic representation is the four-dimensional (r, t, R, Q) , and zero closure gives a light-cone-type null cone inside it. That is sufficient.

x, y, z belong to the subsequent three-dimensional readout and are not needed for the four-dimension claim.

What relationality gives is only the relation between two bodies. If real, only the distance, i.e. the length r . What the system possesses is therefore r , not x, y, z .

$$r^2 = x^2 + y^2 + z^2$$

This equation is **the relation satisfied when x, y, z are read out of r ;** it is not that x, y, z exist first and r is defined from them. **x, y, z are not entities. They are dependent quantities determined by the readout.**

Claim 2 states that this readout has 3 components ($\dim \text{Sym}^2(\mathbb{R}^2) = 3$; the map connecting $d = 2$ with two-body relations is underived). Claim 2 and the present claim are two sides of one coin: Claim 2 says “three components can be read out”, and the present claim says “hence the base layer is r , not x, y, z ”.

On the central-projection side, what the subjective space cannot observe is the direction of the projection and the number of axes; what it can observe is only the magnitude and sign of the composite curvature (§1). This is where t, R, Q appear.

Hence the basic components are

$(r, t, R, Q) \quad \mathbf{4 \text{ components}}$

and not the 6 components (x, y, z, t, R, Q) . Since x, y, z arise from the readout of r , they must not be counted independently.

A note on counting: the four dimensions are the dimension of the space with coordinates (r, t, R, Q) . There is one constraint, the null cone $r^2 - t^2 - R^2 - Q^2 = 0$, so the states on it have 3 degrees of freedom. If in

addition axiom 0.5 is taken in reading B $((r, t, R, Q) \sim \lambda(r, t, R, Q), \lambda > 0$; Claim 8), the projective state space is 2-dimensional. **“Four dimensions” refers to the dimension of the space, not to the number of degrees of freedom.**

r does not determine a point. It determines a surface. The phrase “ x, y, z are determined by the readout of r ” is also inaccurate, since for one r there are infinitely many points with $x^2 + y^2 + z^2 = r^2$. Precisely,

$$r \mapsto S_r^2 = \{(x, y, z) : x^2 + y^2 + z^2 = r^2\}$$

and what selects the point P on that surface is the readout or the dynamics.

$$r \mapsto S_r^2 \xrightarrow{\text{readout / dynamics}} P$$

This is exactly the structure of Claim 0 (a conserved quantity determines the surface; the point on the surface is free). **The base layer is the radius r , and the readout gives not a point but the orbit that r permits.**

Notational note (corrected in v4): the t here is the same quantity as the t of 0-b, and is a different object from the inertia tensor $T = \sum_i v_i v_i^T$ (§0 definitions). Up to v3 both were written T and collided. Henceforth the base-layer degree of freedom is lower-case t and the inertia tensor upper-case T .

This is consistent with Claim 5. **If, under assumption (Γ) , we pass to real distance geometry, what zero closure gives is the signed condition $S(D) = 0$.** The unsigned quantity corresponding to the magnitude of r^2 , $U(D) = \sum_e d_e^2 = N \text{tr}(B)$, is a different thing, and its value is fixed only by fixing C or by a normalisation (Claim 5). The breakdown of t, R, Q is not determined by zero closure.

A misreading this claim forbids

The “four dimensions” of the title must not be identified with the parallelotope dimension $d = 4$ (Claim 16, $N = 16 = 2^4$). **They are different things.** The former is the dimension of the ambient space with coordinates (r, t, R, Q) (with 3 independent degrees of freedom because of the null-cone constraint); the latter is the dimension of the Euclidean space into which a zero-closing configuration is embedded. The numerical coincidence is accidental, and neither follows from the other.

Unresolved (the same question as Claims 12 and 18)

How to read out of r — that map is not given.

This is very likely **the same map problem** as the task raised in Claim 12, “can $A, B, C, D, E, F, h, \dots, p$ be identified with physical spacetime, or is another map needed?”. Indeed the principal axes A, B, C are obtained by double-centring the distance matrix and taking the top three components, which as a procedure is exactly “reading three components out of r ”. That is, A, B, C are **one concrete candidate** for the readout $r \rightarrow x, y, z$.

The component counts agree (the 3 of Claim 2 and the 3 of x, y, z). What is unknown is the single point of **whether the concrete form of the map may be double centring plus top-three projection**. Claim 18 is a direct test of this candidate.

Claim 3 (Lengths determine the configuration up to congruence and reflection. For a non-degenerate maximal-dimensional configuration, the only thing undetermined is the $\mathbb{Z}_2$ of orientation (reflection))

[Title made precise in v4] Up to v3 this read “the only thing undetermined is the sign of the signed volume”. **This fails for degenerate configurations:** the volume is 0, so the binary choice $V = \pm\sqrt{\dots}$ does not exist. Restricted to non-degenerate maximal-dimensional configurations, “the single $\mathbb{Z}_2$ of reflection” is correct.

The totality of the M lengths determines the configuration uniquely up to Euclidean motions and reflection. What the Cayley–Menger determinant gives is the **square** of the volume, and the sign in

$$V = \pm \sqrt{\text{Cayley-Menger determinant}}$$

cannot be recovered from length data.

- Basis: Schoenberg's theorem (1935), the Cayley-Menger determinant (known).
- Numerical check: the distance matrix of a mirrored configuration agrees exactly with the original, and only the signed volume flips sign.

Claim 4 (Parallelotopes satisfy signed zero closure. Centrally symmetric and convex is not enough. The converse is proved only for $d = 2$)

[Title corrected in v4] The title up to v3 was “signed zero closure selects **only** parallelotopes”. That exceeds the current range of proof. What is proved is

- **parallelotope $\Rightarrow$ zero closure**: all d (4-b, binomial theorem)
- **zero closure $\Leftrightarrow$ parallelogram**: only $d = 2$ (4-c, Euler's quadrilateral theorem)

and **zero closure $\Rightarrow$ parallelotope** is unproved for $d \geq 3$ (4-e). The counterexample table of 4-d shows that “centrally symmetric and convex is not enough”, but is not a proof of necessity. The title was matched to this.

4-a. The assignment of signs is uniquely determined.

Classify segments by the **dimension k of the smallest face containing both endpoints** ($k = 1$ edges, $k = 2$ face diagonals, $k = 3$ space diagonals, and so on). **Each vertex pair is assigned a unique $k \in \{1, \dots, d\}$.** Give the segment the sign $(-1)^{k+1}$.

[Corrected in v4] Up to v3 we wrote “it splits into exactly d classes”, but **for a general polytope some classes may be empty**. For instance in a d -dimensional simplex every vertex pair is an edge, so only $k = 1$ occurs. **For parallelotopes all classes $k = 1, \dots, d$ occur** (the breakdown table of 4-g). The uniqueness of the classification is unchanged, so the uniqueness of the sign rule is not weakened.

This classification is **uniquely determined by length data**: lengths $\rightarrow$ configuration (unique up to congruence and reflection by Schoenberg's theorem, Claim 3) $\rightarrow$ convex hull $\rightarrow$ the dimension of the smallest face containing each segment. Reflection does not change the face structure, so the (1:2) ambiguity of Claim 3 does not affect the classification. **It is not an arbitrary convention.**

The classification must not be made by length. In a skew parallelepiped there are 13 distinct values of length², whereas the classification by face dimension gives exactly 3 classes. Only for highly symmetric figures such as the cube or the octahedron do the two happen to coincide.

4-b. For parallelotopes the alternating sum vanishes identically.

For a parallelotope built as the Minkowski sum of d **edge vectors** $e_1, \dots, e_d$ (number of vertices $N = 2^d$), the centred vertices are $v_s = \frac{1}{2} \sum_i s_i e_i$ ($s \in \{\pm 1\}^d$). The contribution of class k is

$$\sum_{k\text{-class}} d_{ij}^2 = 2^{d-1} \binom{d-1}{k-1} \sum_{i=1}^d |e_i|^2$$

Notational convention (unified in v4): e_i are **edge vectors** (the lengths of the $k = 1$ class are exactly $|e_i|$). In Claim 6B we set $A \equiv \frac{1}{2}[e_1 \cdots e_d]$ and write $v_s = As$. **Taking $A = [e_1 \cdots e_d]$ would make the edge length $2|e_i|$ and disagree with the coefficient above by a factor 4** (verified numerically). Up to v3 both sections used the same symbol u_i and this factor was inconsistent.

Hence the alternating sum is

$$\sum_{k=1}^d (-1)^{k+1} 2^{d-1} \binom{d-1}{k-1} E = 2^{d-1} E \sum_{j=0}^{d-1} (-1)^j \binom{d-1}{j} = 2^{d-1} E (1-1)^{d-1} = 0 \quad (d \geq 2), \quad E \equiv \sum_i |e_i|^2$$

The generators need be neither orthogonal nor of equal length. **It is an identity following from the binomial theorem.**

- Numerical check: for $d = 2, \dots, 7$ ($N = 4, \dots, 128$), with random skew generators, relative error of order 10^{-16} ; maximal discrepancy with the closed formula below 10^{-11} .
- Numerical check (face-dimension classification by convex hull): for a skew parallelepiped $N = 8$, $k = 1$: 12 segments / 30.670, $k = 2$: 12 / 61.341, $k = 3$: 4 / 30.670, alternating sum -3.6×10^{-15} . For a skew four-dimensional parallelotope $N = 16$, alternating sum -2.0×10^{-13} .

4-c. For $d = 2$ necessity also holds.

For each of the three ways of splitting the 6 segments of 4 vertices into “4 edges and 2 diagonals”, Euler’s quadrilateral theorem gives

$$\sum(\text{edges})^2 - \sum(\text{diagonals})^2 = 4LM^2$$

identically (L, M are the midpoints of the two diagonals). The right-hand side is non-negative, so the left-hand side vanishes if and only if $LM = 0$. $LM = 0$ is equivalent to the diagonals bisecting each other, i.e. to the figure being a parallelogram.

- Numerical check: 0 for a parallelogram, +4 for a convex non-parallel quadrilateral; for non-convex configurations none of the three splittings gives zero (minimum 16).

4-d. Centrally symmetric and convex is not enough. A stronger constructive family, the parallelotopes, exists.

All of the following are **centrally symmetric and convex**, yet the alternating sum under the face-dimension classification does not vanish.

Configuration	N	Counts and $\sum d^2$ by class	Alternating sum
Octahedron	6	$k=1$: 12 / 24.000, $k=3$: 3 / 12.000	+36.000
Cuboctahedron	12	$k=1$: 24 / 48.000, $k=2$: 12 / 48.000, $k=3$: 30 / 192.000	+192.000
Icosahedron	12	$k=1$: 30 / 120.000, $k=3$: 36 / 400.997	+521.000
Dodecahedron	20	$k=1$: 30 / 45.836, $k=2$: 60 / 240.000, $k=3$: 100 / 914.164	+720.000
(control) skew parallelepiped	8	$k=1$: 12 / 30.670, $k=2$: 12 / 61.341, $k=3$: 4 / 30.670	-3.6×10^{-15}

Hence the constraint imposed by zero closure is strictly stronger than “centrally symmetric and convex”. The octahedron and the icosahedron are centrally symmetric and convex, yet do not zero-close. **One must not, however, conclude from this that “zero-closing configurations are limited to parallelotopes”.** The table is a collection of counterexamples to “centrally symmetric and convex is enough”, not a proof of necessity (4-e). What can be said is only “parallelotopes zero-close” (4-b, all d) and “for $d = 2$ the converse also holds” (4-c). Only when zero closure is realised as a parallelotope family does the vertex number become $N = 2^d$.

4-e. Unresolved.

Necessity for $d \geq 3$, i.e. “alternating sum zero $\Rightarrow$ parallelotope”, is not proved. The table gives counterexamples among centrally symmetric convex polytopes but is not a proof of necessity. Necessity is proved only for $d = 2$ (4-c).

4-f. The classification can be recovered from lengths alone (execution check of the procedure of 4-a).

4-a states that the classification is uniquely determined by length data, but the derivation program up to v3 (face_dim_classes in make_figs.py) **took the configuration V as input**. We implemented the path taking lengths alone as input,

$$d_{ij} \longrightarrow B = -\frac{1}{2}JD^{\circ 2}J \longrightarrow V \longrightarrow \text{convex hull} \longrightarrow k$$

and checked it. For skew parallelotopes with $d = 2, 3, 4$ ($N = 4, 8, 16$), all 10 trials each gave **exact agreement with the true classes** (the rank also agreed with d). This confirms in execution the argument of 4-a that, since reflection does not change the face structure, the (1:2) ambiguity of Claim 3 does not affect the classification.

- Derivation program: figures_v1/check_real_solutions_v1.py (test F).

4-g. Which classes become imaginary. Edges and main diagonals are always equal.

In the closed formula of 4-b, $\sum_{k\text{-class}} d^2 = 2^{d-1} \binom{d-1}{k-1} E$, taking $k = 1$ and $k = d$ gives $\binom{d-1}{0} = \binom{d-1}{d-1} = 1$, so for any parallelotope

$$\sum_{\text{edges}} d_{ij}^2 = \sum_{\text{main diagonals}} d_{ij}^2$$

holds **exactly** (the generators may be skew). Numerical check: relative residual of order 10^{-16} (machine precision) for $d = 2, \dots, 6$.

This form means that **“if only the main diagonals are read as imaginary, then $\sum x_n^2 = 0$ holds exactly, with all lengths real, within the range of edges and main diagonals”**. For $d = 2$ ($N = 4$) the 4 edges and 2 diagonals exhaust all 6 pairs, so this alone completes zero closure (exactly Euler’s quadrilateral theorem). **Edges and main diagonals exhaust all pairs only when $d + 1 = 2^d - 1$, i.e. only for $d = 2$.**

For $d \geq 3$ face diagonals remain, and the sign is determined by the parity of the face dimension: odd k real, even k imaginary. The breakdown for $d = 4$ ($N = 16$, the case adopted in this note) is as follows.

k	Role	Count	$\sum d^2$ (example)	Sign
1	edge	32	73.3868	+ (real)
2	diagonal of a 2-face	48	220.1604	– (imaginary)
3	diagonal of a 3-face	32	220.1604	+ (real)
4	main diagonal	8	73.3868	– (imaginary)

Alternating sum = 0.000. 64 real / 56 imaginary (total $M = 120$). The main diagonals become imaginary only when d is even; for $d = 3$ the main diagonals are real and the imaginary ones are the 12 face diagonals. **Hence “what becomes imaginary is the diagonals” is exactly right for $d = 2$, and in general the accurate statement is “what becomes imaginary is the even face-dimension classes”.**

- Derivation program: figures_v1/check_real_solutions_v1.py (tests D and E).

Claim 5 (In real distance geometry, zero closure imposes the single condition $S(D) = 0$ on a signed quadratic scalar. The value of the unsigned total square $U(D) = N \text{tr}(B)$ is not determined by zero closure alone)

Scope (made explicit in v4): this claim concerns the **layer of real distance geometry**, i.e. the layer after the relation quantities have been read as lengths d_{ij} . **General complex zero closure imposes two real conditions** (Claim 0-b). It drops to one when assumption (S: separation of the supports of the real and imaginary parts) holds, and **what then remains is the single equality of the sums of squares. To make this correspond to $S(D) = 0$ by the face-dimension sign, a further assumption (Γ : geometric sign correspondence) is required** (below). Do not confuse the layers.

For any N and any dimension,

$$\sum_{i < j} d_{ij}^2 = N \operatorname{tr}(B), \quad B = -\frac{1}{2} J D^{\circ 2} J$$

holds. When B is positive semi-definite (i.e. when the configuration can be realised in Euclidean space), $\operatorname{tr}(B) = \sum_i R_i^2 = \operatorname{tr}(T)$, where R_i is the distance from the centre to vertex i and $T = \sum_i v_i v_i^\top$ is the inertia tensor. This is an identity and does not constrain the configuration.

[Corrected in v4] Signed and unsigned sums must be distinguished. The Lagrange identity above concerns the **unsigned sum** $U(D) \equiv \sum_e d_e^2 = N \operatorname{tr}(B)$, whereas the zero closure imposed by the geometry of Claim 4 is the **signed sum** $S(D) \equiv \sum_e s_e d_e^2 = 0$, $s_e = (-1)^{k_e+1}$. **These are not the same equation.**

Quantity	Definition	Role
$S(D)$	$\sum_e s_e d_e^2$ (signed)	the one condition imposed by zero closure (Claim 4)
$U(D)$	$\sum_e d_e^2 = N \operatorname{tr}(B)$ (unsigned)	total square; tied to $\operatorname{tr}(B)$ by the identity

$S(D) = 0$ **does not determine the value of $U(D)$** . Only when, under (S), $\sum_L a_n^2 = \sum_R b_m^2 = C$ holds do we get $U(D) = 2C$, $\operatorname{tr}(B) = 2C/N$. **Unless C is fixed, zero closure alone does not determine $\operatorname{tr}(B)$** (Claim 0-b, second stage).

A separate assumption (Γ : geometric sign correspondence) is required. Assumption (S) gives only the **separation of supports** — which components lie on the real side $\mathcal{I}_+$ and which on the imaginary side $\mathcal{I}_-$. The sign rule of Claim 4, $s_e = (-1)^{k_e+1}$ (odd face dimension $\rightarrow$ real, even $\rightarrow$ imaginary), is a **concrete geometric classification**. **Separation of supports does not imply $\mathcal{I}_+ = \{e : s_e = +1\}$, $\mathcal{I}_- = \{e : s_e = -1\}$.**

Assumption (Γ): the partition $\mathcal{I}_\pm$ coincides with the partition by the face-dimension sign s_e

This note does not derive (Γ). This is exactly the gap stated in §3B: nothing connects the closure of complex amplitudes with signed closure by face dimension. The full reduction is therefore

$$\text{complex zero closure} \xrightarrow{S} \mathcal{I}_\pm \text{ separation} \xrightarrow{\Gamma \text{ (underived)}} \text{face-dimension sign} \longrightarrow S(D) = 0$$

Do not confuse a third sign either. The “signed trace” of Claim 13 is a sign given by the **sign of the eigenvalues of B** , a different object from the **face-dimension sign s_e** of Claim 4. No bridge connects those two either (§3B). This note handles three layers, $\sum x_e^2$ of complex zero closure $\leftrightarrow S(D)$ of real distance geometry $\leftrightarrow \operatorname{tr}(B)$ of the numerical model, and **they are not identified with one another**.

Notational caution (corrected in v4): T and B must not be identified. **They are not even the same size.** With V the centred coordinate matrix ($N \times d$, rows = vertices),

$$B = V V^\top \ (N \times N), \quad T = V^\top V \ (d \times d)$$

These are **not the same matrix**, but by the $V V^\top / V^\top V$ relation the **non-zero eigenvalues agree with multiplicity** and $\operatorname{tr}(B) = \operatorname{tr}(T)$ (when the configuration is realisable in Euclidean space). When it is not, B has negative eigenvalues, V cannot be taken real, and T itself is undefined. **In contexts involving signed sums we always write $\operatorname{tr}(B)$** . Writing $\operatorname{tr}(T)$ up to v3 (e.g. in Claim 13) was an error.

Hence **zero closure constrains exactly one independent scalar** (the signed sum $S(D) = 0$). Individual semi-axes, the radius of each vertex, and the number of hidden axes are not determined. **This one condition does not determine the value of the unsigned trace $\operatorname{tr}(B)$** (caution above).

[**Made precise in v4**] Up to v3 this read “what zero closure fixes is the single quantity $\text{tr}(T)$ ”, which is **doubly wrong**. First, $\sum x_n^2 = 0$ is **homogeneous**, so it survives scaling the whole configuration by λ (Claim 8), and “fixes” overstates. Second, zero closure imposes the **signed** $S(D) = 0$, not a condition on the **unsigned** $U(D)$ that is tied to $\text{tr}(B)$.

Precisely: zero closure imposes the one condition $S(D) = 0$. The value of $U(D)$ (hence $\text{tr}(B)$, hence $l = 0$) is determined only by fixing C or by a normalisation. This is the structure of Claim 0 itself (an equality constraint gives a family of surfaces; fixing C selects one).

- Basis: Lagrange’s identity (known).
- Numerical check: for all combinations $N = 3, 4, 5, 6, 8, 12, 20, 50$ and dimensions 1–7, relative error below 10^{-15} .
- Numerical check: many configurations exist with $\sum d^2$ fixed but widely dispersed radii (2434 out of 20000 trials).

Claim 6 (A quadratic form can carry multipoles only up to $l \leq 2$, where it saturates)

The second-order information of a configuration is exhausted by the inertia tensor T . In d dimensions T has $d(d+1)/2$ independent components, and its decomposition under the rotation group has exactly two pieces, $l = 0$ and $l = 2$:

$$T = \underbrace{\frac{\text{tr } T}{d} I}_{l=0} + \underbrace{\left(T - \frac{\text{tr } T}{d} I\right)}_{l=2}$$

Order	Components	Content	Fate
$l = 0$	1	magnitude (trace part)	not fixed by $S(D) = 0$ alone; fixed by fixing C or by normalisation (Claim 5)
$l = 2$	$d(d+1)/2 - 1$	shape and orientation (traceless part)	remains
$l \geq 3$	—	—	not represented by a quadratic form

[**Corrected in v4**] $l = 1$ is **not a component of T** . Up to v3 the table had a row “ $l = 1$: d components, displacement of the centre, identically zero for centrally symmetric configurations”. **This is wrong on two counts.** First, $l = 1$ corresponds to the **first moment** $\sum_i v_i$, not to a component of the second-rank symmetric tensor T ; the irreducible decomposition of $\text{Sym}^2(\mathbb{R}^d)$ under rotations is $l = 0 \oplus l = 2$ only. Second, the reason $\sum_i v_i = 0$ is **not central symmetry**: by the §0 definition v_i are coordinates with the centroid at the origin, so $\sum_i v_i = 0$ **by definition for any configuration**.

Object	Order	Components	Status
first moment $\sum_i v_i$	$l = 1$	d	zero by definition in centroid coordinates
second central moment T	$l = 0$	1	determined by fixing C or by normalisation
”	$l = 2$	$d(d+1)/2 - 1$	remains

This correction does not change the conclusion of Claim 6. For $d = 3$, $6 = 1 + 5$; the one $l = 0$ component is set by fixing C or by normalisation, and the five $l = 2$ components remain.

For $d = 3$, $6 = 1 + 5$. **The single $l = 0$ component is not determined by zero closure.** Only on adding the fixing of C or a normalisation do we get $U(D) = 2C$, $\text{tr}(B) = 2C/N$, hence (if Euclidean-realisable) $\text{tr}(T)$. What remains is then **5 components**.

[Corrected in v4] **Zero closure does not fix the scale.** Up to v3 we wrote “zero closure constrains the one magnitude component”, but as in Claim 5 what zero closure imposes is the signed $S(D) = 0$, which does not determine the unsigned $U(D)$. **For the parallelotope family $S(D) = 0$ is an identity, so it holds however the generators e_i are replaced. That is, zero closure does not fix the scale of T at all.**

$$\boxed{\text{zero closure} \neq \text{scale fixing}}$$

That $l = 2$ has 5 components is always true. Splitting it into “2 of shape and 3 of orientation” is valid in general position with three non-degenerate eigenvalues; at degeneracy points part of the eigendirections lose meaning (§0B item 2; the measured minimal gap is of order 10^{-6} – 10^{-5}). There are 3 semi-axes, but since the single magnitude component is set by fixing C or by normalisation, the shape has 2 degrees of freedom.

The cause of the saturation is that the readout uses the second central moment T , a rank-two tensor.

[Corrected in v4] Up to v3 this read “the cause of the saturation is that the closure condition is quadratic”. **That is inaccurate.** Zero closure being quadratic does not mean that the $l \geq 3$ information of a configuration **does not exist**. Indeed, as the numerical checks below show, the cube, octahedron and cuboctahedron **are distinguished by fourth moments**. The information is not annihilated; T simply does not pick it up. Precisely: **“a quadratic-form readout saturates at $l \leq 2$. Zero closure being quadratic is consistent with this quadratic readout, but does not by itself annihilate higher moments.”**

- Numerical check: the inertia ellipsoids of the cube, octahedron and cuboctahedron are perfectly isotropic (normalised semi-axes² = (1, 1, 1)) and indistinguishable from a sphere.
- Numerical check: the distinction first appears at fourth moments. $\langle x^4 + y^4 + z^4 \rangle$ is 0.333 (cube), 1.000 (octahedron), 0.500 (cuboctahedron), 0.600 (sphere).
- $d(d+1)/2$ is the number of parameters of a quadratic form Q . When the number of pairs n exceeds it, the conditions $v_i^T Q v_i = 1$ are overdetermined for a point set in general position.

[Corrected in v4] Up to v3 this read “all vertices lie on one ellipsoid only if $n \leq d(d+1)/2$ ”. **This is wrong;** parameter counting gives neither a necessary nor a sufficient condition.

Counterexample to sufficiency: in $d = 2$ take $v_1 = (1, 0)$, $v_2 = (2, 0)$ (centrally symmetric, so $\pm v_1, \pm v_2$). Here $n = 2 \leq 3 = d(d+1)/2$, yet $v_1^T Q v_1 = 1$ gives $q_{11} = 1$ while $v_2^T Q v_2 = 1$ gives $4q_{11} = 1$ — contradiction. No ellipsoid exists.

Counterexample to necessity: the parallelotopes of Claim 6B. The number of pairs is 2^{d-1} , which for $d = 5$ is $16 > 15$ and for $d = 7$ is $64 > 28$, exceeding the parameter count, yet **the circumscribed ellipsoid still exists** (verified numerically for $d = 5, 6, 7$).

So the inequality **holds in neither direction**. It is only a rough guide to overdetermination for point sets in general position. Parallelotopes exceed the guide because they are not in general position (Claim 6B).

Claim 6B (The parallelotope constraint of Claim 4 is consistent with the ellipsoid of Claim 6)

The parallelotopes that Claim 4-b guarantees to satisfy zero closure are **always inscribed in an ellipsoid**. The vertices are As ($s \in \{\pm 1\}^d$, $A \equiv \frac{1}{2}[e_1 \cdots e_d]$, e_i the edge vectors), so setting $Q = (A^{-1})^T A^{-1}/d$ gives $s^T s/d = 1$ at every vertex. That is, a parallelotope lies on the image under the linear map A of the circumscribed sphere of a hypercube.

Moreover **this circumscribed ellipsoid and the normalised inertia ellipsoid ($c = d/N$) are identical** (see the normalisation in the §0 definitions):

$$T = \sum_s (As)(As)^T = A \left(\sum_s ss^T \right) A^T = 2^d AA^T, \quad Q^{-1} = d AA^T \propto T$$

- Numerical check: for $d = 2, \dots, 7$, the maximal deviation of $x^T Q x$ from 1 over all vertices is below 10^{-13} .

- Numerical check: for $d = 2, 3, 4, 5$, $T = 2^d AA^\top$ (max difference 10^{-14}), and $Q^{-1}T^{-1}$ is a constant multiple of the identity (deviation 10^{-15}).

Furthermore, parallelotopes lie on an ellipsoid even beyond the general counting bound. The number of pairs is 2^{d-1} ; for $d = 5$, $16 > 15 = d(d+1)/2$ and for $d = 7$, $64 > 28$, beyond the range of Claim 6. The circumscribed ellipsoid nonetheless exists (verified for $d = 5, 6, 7$), because parallelotopes are **not in general position**: the 2^d vertices are not independent points but are built from d generators via $v_s = As$, so there are many dependencies among the conditions $v_s^\top Q v_s = 1$.

Hence Claims 4 and 6 are compatible. The parallelotopes for which 4-b guarantees zero closure carry the ellipsoid structure of Claim 6, and in fact the stronger property that **the circumscribed ellipsoid and the normalised inertia ellipsoid** ($c = d/N$) **coincide exactly**, namely $Q^{-1} = (d/N)T$. **With the old $c = 1$ definition they do not coincide; the correct statement there is that they are similar, coaxial and of the same shape but of different size.**

Claim 7 (The imaginary unit is the operation symbol for transposition)

In the real closure equation

$$\sum_{n \in \mathcal{I}_+} x_n^2 = \sum_{m \in \mathcal{I}_-} y_m^2$$

moving a quantity from the right to the left **reverses the sign of its square**. Written as a symbol acting on amplitudes, that operation is i , and $i^2 = -1$ is the definition of the operation itself.

Two groups must be distinguished here. As multiplication on amplitudes, $1 \rightarrow i \rightarrow -1 \rightarrow -i \rightarrow 1$, so i **generates $\mathbb{Z}_4$** . What zero closure reads, however, is the squared quantity, and the induced action on its sign is the $\mathbb{Z}_2$ exchanging $(+) \leftrightarrow (-)$.

[Corrected in v4] Up to v3 we wrote “the $\mathbb{Z}_2$ generated by this symbol”, but at the amplitude level it is $\mathbb{Z}_4$. Precisely: **“at the amplitude level i generates $\mathbb{Z}_4$; the induced action on the sign of the squared quantities read by zero closure is $\mathbb{Z}_2$, recording only the parity of the number of crossings of the boundary between the observed and unobserved sides.”**

Writing L for quantities observable in the subjective space and R for the unobservable ones, the imaginary unit attaches only to the R side, and all segment lengths remain real.

- Status: a consequence of the convention. Numerical check: for an amplitude A , $(iA)^2 = -A^2$, $(i^2A)^2 = +A^2$, $i^2A = -A$.

Claim 8 (Read as “scale invariance of the solution set”, axiom 0.5 is not an independent axiom. Read as “identification of different scales”, a separate projectivisation is required)

The definition of axiom 0.5 must be settled (made explicit in v4). §1 merely says “scale symmetry as axiom 0.5” without distinguishing:

Reading A (invariance of the solution set): the solution set of zero closure is invariant under $x \mapsto \lambda x$. **Reading B (identification of physical states):** x and λx denote the same physical state (gauge equivalence).

Under reading A the axiom can be dropped, as this claim shows (it follows from homogeneity). **Reading B does not follow from homogeneity**; projectivisation $x \sim \lambda x$ must be posited separately. **Which reading is intended is undecided, and settling it is an open task.** What follows concerns reading A.

Under the convention $x_n = q_n + ip_n$ with $x_n^2 = (q_n + ip_n)^2$ (not $|x_n|^2$), $\sum_n x_n^2 = 0$ is equivalent to the two real equations

$$\sum_n q_n^2 = \sum_n p_n^2, \quad \sum_n q_n p_n = 0$$

This is as far as one can go rigorously. Since $\sum x_n^2 = 0$ is homogeneous, $\sum (\lambda x_n)^2 = \lambda^2 \sum x_n^2 = 0$, so **the solution set is a cone, invariant under scaling.** This is algebraically exact.

[Corrected in v4] Up to v3 this continued “the left side of the second equation is the generator of dilatation; its vanishing is equivalent to the overall scale of the system having no physical meaning”. **This derivation is incomplete.** For $\sum q_n p_n$ to act as the dilatation generator, a **canonically conjugate symplectic structure** on q, p must first be introduced, which this note does not provide. Moreover the constraint $D = 0$ and the gauge equivalence “states of different scale are the same physical state” are **different matters**; the latter does not follow automatically from the former, and requires a separate projectivisation.

The cone argument by homogeneity needs none of this and suffices for the conclusion that scale symmetry is not an independent axiom. **Claim 8 is therefore strengthened.** We drop the generator argument and replace it by the homogeneity argument.

Hence axiom 0.5 (scale symmetry) follows from the homogeneity of $\sum x_n^2 = 0$; it need not be posited as an independent axiom. The axioms are reduced by one. Note that it does not follow from the complex convention (any homogeneous equation has a cone as solution set, complex or not).

However, “the solution set is a cone” and “different scales denote the same physical state” are different statements. The latter requires the separate assumption of projectivisation. Settling which of these axiom 0.5 means is an open task.

- Status: algebraically exact.
- Numerical check: real signature- (n, n) zero closure requires only $\sum q^2 = \sum p^2$ and leaves $\sum q_n p_n$ free; complex zero closure drives it to zero.

Claim 9 (Complex numbers are not indispensable)

The argument that “ $\sum x_n^2 = 0$ needs imaginary numbers to have a non-trivial solution” **presupposes that all terms have the same sign.** With indefinite signs, non-trivial real solutions exist ($x^2 - y^2 = 0 \Rightarrow x = \pm y$).

For the structures examined in this note, everything goes through with real signs: the structure of the null cone, projectivisation, the 2:1 double cover of the squaring map, and the Klein-type quadric. **One may not conclude from the table that “all the principal structures of the series hold”;** structures not examined remain unchecked.

Components	Real signature	Factorisation	Projective state space	Lie algebra of symmetry
3	(2, 1)	$\sigma_{\mathbb{R}}(a, b) = (a^2 - b^2, a^2 + b^2, 2ab)$	$\mathbb{RP}^1$	$\mathfrak{so}(2, 1)$
4	(2, 2)	$\det X = 0 \iff X = \xi \eta^T$ (real)	$\mathbb{RP}^1 \times \mathbb{RP}^1$	$\mathfrak{so}(2, 2)$
6	(3, 3)	real Klein quadric	$\text{Gr}(2, 4; \mathbb{R})$	$\mathfrak{so}(3, 3) \cong \mathfrak{sl}(4, \mathbb{R})$

[Corrected in v4] Up to v3 the “state space” entry for (3, 3) read $\mathfrak{so}(3, 3) \cong \mathfrak{sl}(4, \mathbb{R})$. **That is not a state space but the Lie algebra of the symmetry.** The state space is the real Klein quadric, i.e. the projective null quadric, read canonically as the real Grassmannian $\text{Gr}(2, 4; \mathbb{R})$. The columns have been separated.

The real spinor squaring map $\sigma_{\mathbb{R}}$ is also **2:1 (away from the origin)**, so the double cover does not originate from imaginary numbers: (a, b) and $(-a, -b)$ have the same image. The preimage of the origin is the single point 0, but **projectivisation removes the origin anyway**, so this causes no problem. The projectivised real null cone $(S^{p-1} \times S^{q-1})/\mathbb{Z}_2$ is **compact**. This fact holds without imaginary numbers.

[Corrected in v4] Up to v3 this continued “hence the quantisation argument ‘a closed surface implies a discrete spectrum’ also holds without imaginary numbers”. **This derivation is incomplete.**

Compactness of the state space alone does not give discreteness of a spectrum, because the operator is unspecified. Even on a compact space, multiplication operators have continuous spectrum.

Discreteness requires self-adjointness, ellipticity, or compact resolvent. In geometric quantisation an integrality condition and a choice of polarisation are separately required.

So what this note can say is only: “**the projectivised state space is compact, and this requires no imaginary numbers.**” Deriving a discrete spectrum from that requires specifying the operator and applying the spectral theorem. **Identifying which operator is claimed to have discrete spectrum is an open task.** This correction does not affect the conclusion of Claim 9.

What complex numbers add, relative to the real indefinite form, is only the single equation $\sum q_n p_n = 0$ (Claim 8).

- Numerical check: residual of the image of $\sigma_{\mathbb{R}}$, 1.7×10^{-13} ; $\sigma_{\mathbb{R}}(a, b) = \sigma_{\mathbb{R}}(-a, -b)$ exactly; agreement of $\det X$ with the signature-(2, 2) form to residual 3.6×10^{-15} .

Claim 10 (At the component layer, zero closure is an identity holding only in the steady state; it is violated in transient and metastable states)

This series has long stated that $\sum x_n^2 = 0$ is **the answer in the stable state**, failing temporarily during transients and just after a change of state. In v3 we measured this from the geometric side.

Scope (restricted by Claim 18-e): this claim concerns the **component layer** (the model-internal closure diagnostic $\sum_{e,k,j} C_{ekj}^2$). At the **readout layer** (summing harmonics first, $\sum_e x_e^2$, the aggregation that gives distances), zero closure is not violated but is **a conserved quantity fixed at a complex value set by the initial conditions**, unmoved through the transition. See Claims 18-c, 18-d, 18-e.

Measurement conditions: mode = electron, resolution $N = 16 = 2^4$ (Claim 16), seed $\delta = 0.1$, $T = 40000$ steps. Two series, matter and vacuum control. Lengths are read from $M = 120$ relation quantities and restored to $N - 1 = 15$ principal axes by double centring. The deviation s_i from the ellipsoid is measured at all 16 vertices. **The deviation is measured in the projection onto the top three principal axes (rank 3).** Measured in the full-dimensional configuration with rank = $N - 1 = 15$, $s_i \equiv 1$ holds trivially by Claim 16-c and the test loses discriminating power; rank 3 here is the condition for the test to have content.

Choice of τ : the $N = 16$ transition occurs in $\tau \in [8991, 9487]$ (top-3 occupancy $0.287 \rightarrow 0.579$, imaginary directions $0 \rightarrow 5$, rank $15 \rightarrow 10$), roughly **four times later** than the $N = 12$ transition ($\tau \approx 2500$). The four time points are therefore start $\tau = 0$ / pre-transition $\tau = 4000$ / just after transition $\tau = 9487$ / end $\tau = 39991$. Using the $N = 12$ values $\tau = 2000/4000$ directly would place both before the transition at $N = 16$.

Results

τ	Series	dev. max/min	dev. CV	median s	imag. dirs	top-3 occupancy
0 (start)	matter	4.10	0.271	0.977	2	0.384
4000 (pre)	matter	4.25	0.349	0.926	0	0.227
9487 (just after)	matter	11.74	0.487	1.000	5	0.579
39991 (end)	matter	11.16	0.614	0.738	6	0.812
0	vacuum	6.92	0.393	1.082	0	0.343
4000	vacuum	1.98	0.225	0.922	0	0.240
9487	vacuum	4.31	0.370	1.009	0	0.204
39991	vacuum	4.61	0.378	0.945	0	0.200

If all vertices lay on the ellipsoid, max/min = 1.00, CV = 0, median s = 1.00.

What can be read off

- The matter-side deviation does not decrease monotonically in τ . From 4.25 at $\tau = 4000$ (pre-transition) it goes to 11.74 at $\tau = 9487$ and **stays high** at 11.16 at $\tau = 39991$.

- For 30,000 steps after the transition the deviation stays around 11 and does not return to the ellipsoid. The CV keeps growing, $0.349 \rightarrow 0.614$.
- The vacuum control stays at 4.3–4.6 after the transition, showing no jump. The increase occurs simultaneously with the creation of matter.
- The 2 imaginary directions at $\tau = 0$ are a transient right after seeding; they disappear by $\tau \approx 1000$ and 0 persists until the transition (the vacuum side has 0 throughout).

Direct measurement of the closure residual in the same run: at $N = 16$ the median closure residual is 1.338×10^{-2} , the conditional closure residual 1.056×10^{-1} , and the number of steps with exact zero is 0 (`closure_med / cond_closure_med / odd_exact_zero_steps` in the run script). Even in the metastable state $\sum x_n^2 = 0$ does not hold exactly. The corresponding values at $N = 12$ are $1.796 \times 10^{-2} / 1.436 \times 10^{-1} / 0$; raising N does not change the conclusion.

Here the mathematical propositions and their application to the numerical model must be separated.

[Corrected in v4] Up to v3 this read “the range of applicability of the geometric statements of Claims 1–9 is limited to the steady state”, which is **wrong**. Claim 4 (parallelotope $\Rightarrow$ alternating sum zero) and Claim 6B (parallelotopes inscribe in an ellipsoid) are **static geometric theorems** requiring no dynamical steady-state condition. As Claim 0-d says, as long as C is constant the state may move non-stationarily on the fixed surface.

Precisely: “**the mathematical propositions of Claims 1–9 do not themselves require a steady state. Applying them to the present numerical model, however, requires the premises of each claim (zero closure, assumption (S), fixed C , ...) to hold, and in the transient and metastable states of this run they do not.**”

- Status: measured. $T = 40000$ run, two-stage sampling ($\tau < 4000$ every step, thereafter every 31 steps; 5162 frames total).
- Determinism verified via `determinism_max_abs` in the run script.
- **Sampling limitation (newly arising at $N = 16$, resolved by a dedicated run):** the dense region is set at $\tau < 4000$ by a design matched to the $N = 12$ transition time ($\tau \approx 2500$). Since the $N = 16$ transition is at $\tau \approx 9000$, in this run (`...rep-dump40k16_N16`) **the transition itself falls in the coarse region sampled every 31 steps**. To measure the period in the transition region, a dedicated run with the dense region widened to $\tau < 10000$ (`...rep-dump40k16d_N16`, `DUMP_TAUC=10000`) was performed separately. Claim 15 uses that run. The ellipsoid figures and spectra of this section come from `...rep-dump40k16_N16`; the two runs share conditions and random seed and differ only in the dense-region setting.

Claim 11 (The deviation depends on the seed strength. The seed in this run is large)

For space and matter to appear, the system must be given a seed perturbation. Below a threshold, matter does not grow at all. This run used $\delta = 0.1$ because existing measurements showed it to be the only level reaching the final value within $T = 4000$.

δ	growth of f_{seed} , $\tau = 0 \rightarrow 4000$	attainment of the $T = 42000$ final value
10^{-15}	1.69	59.3 %
10^{-8}	1.00	100.0 % (nothing happens)
10^{-4}	1.00	100.0 % (nothing happens)
10^{-3}	1.00	100.0 % (nothing happens)
10^{-2}	1.001	94.1 % (matter does not move)
3.16×10^{-2}	1.054	0.2 % (onset too slow)
4.36×10^{-2}	1.218	0.5 %
1.0×10^{-1}	51.05	102.0 % $\leftarrow$ adopted

$\delta = 0.1$ is not small relative to typical relation quantities. **Hence the deviation 11.16 measured in Claim 10 contains a component originating in the largeness of the seed.**

This sweep table was measured at $N = 12$ and has not been remeasured at $N = 16$. Its criterion, “reaches the final value within $T = 4000$ ”, presupposes the $N = 12$ transition time ($\tau \approx 2500$). Since the $N = 16$ transition is at $\tau \approx 9000$, **at $N = 16$ nothing happens within $T = 4000$ even at $\delta = 0.1$.** Re-establishing the choice of δ at $N = 16$ requires a sweep with the decision window moved to $T \gtrsim 20000$. This note inherits $\delta = 0.1$ as an $N = 12$ -derived setting, and this point is unresolved.

Unverified

Whether the deviation from the ellipsoid decreases towards 1.00 when the seed strength δ is lowered, or when τ is extended further, **has not been checked**.

All that can be said from this run is that at $\delta = 0.1$, $\tau \leq 40000$ the deviation does not decay. A two-axis sweep in δ and τ is the next experiment needed to settle this claim.

A failed prediction, recorded. From data up to $T = 4000$ the envelope decay time constant was estimated at 1190 and the deviation was predicted to return at $\tau \approx 9300$. The $T = 40000$ measurement **falsified this**. The deviation is flat over 36,000 steps, and the “decay” visible up to $T = 4000$ was a one-off settling from the transition peak, not an envelope decay.

Claim 12 (The double-centring readout has at most $N - 1$ non-trivial principal axes, and spectral concentration into the top three directions magnifies the observed scale. At $N = 16$ there are 15)

[Title corrected in v4] Up to v3 this read “the system has $N - 1$ degrees of freedom”. **These are not degrees of freedom.** What is shown is that, by $B\mathbf{1} = 0$, the double-centred Gram matrix can have **at most $N - 1$ non-trivial principal axes (signed spectral directions)** — not the number of independent degrees of freedom of the configuration. The same reason as the strict separation of ambient dimension from independent degrees of freedom in Claim 2B.

The $N = 16$ system has $N - 1 = 15$ principal axes. The signed $\sqrt{\lambda}$ of each was measured before the transition (mean over $\tau < 2000$) and in the late period (mean over 968 points with $\tau \geq 10000$).

Name	pre-transition	late	late / pre	real / imaginary, late
<i>A</i>	+0.08605	+0.24084	2.799	real (100 %)
<i>B</i>	+0.08234	+0.14226	1.728	real (100 %)
<i>C</i>	+0.07913	+0.11234	1.420	real (100 %)
<i>D</i>	+0.07615	+0.09646	1.267	real (100 %)
<i>E</i>	+0.07333	+0.08339	1.137	real (100 %)
<i>F</i>	+0.07074	+0.07116	1.006	real (100 %)
<i>h</i>	+0.06812	+0.05944	0.872	real (100 %)
<i>i</i>	+0.06496	+0.04615	0.711	real (100 %)
<i>j</i>	+0.06198	+0.02808	0.477	alternating (94.7 % real)
<i>k</i>	+0.05889	−0.01082	0.380	alternating (31.1 % real)
<i>l</i>	+0.05531	−0.03993	0.725	alternating (0.6 % real)
<i>m</i>	+0.05147	−0.05779	1.123	imaginary (100 %)
<i>n</i>	+0.04494	−0.07524	1.674	imaginary (100 %)
<i>o</i>	+0.03771	−0.09894	2.624	imaginary (100 %)
<i>p</i>	+0.03360	−0.18711	5.568	imaginary (100 %)

Distribution of the number of imaginary directions ($\tau \geq 10000$, 968 points): 4 directions at 6 points, 5 at 295, **6 at 616**, 7 at 51. The rank of the Gram matrix is 8–11.

Structure

- A, B, C grow by factors 1.42–2.80. These three directions were used for the figures. The top-3 occupancy rises $0.384 \rightarrow 0.812$ (isotropic value $3/15 = 0.200$).
- D, E, F, h, i barely change or shrink gently (factors 0.71–1.27). They stay real and do not appear in the three-dimensional projection.
- m, n, o, p move to imaginary directions, but **they do not shrink and vanish**: p grows by 5.568, o by 2.624, n by 1.674 while crossing to the imaginary side.
- j, k, l oscillate across the real/imaginary boundary. These three account for the number of imaginary directions fluctuating between 4 and 7.

Hence this is not a compactification of the “extra directions shrink and become invisible” type. The extra dimensions did not curl up and disappear; **the three-direction side expanded.** (What the numbers refute is the shrinking picture. We have not refuted compactification in general, including topological compactness, from eigenvalue magnitudes alone.) The directions moving to the imaginary side keep or increase their magnitude while changing sign. This conclusion is the same at $N = 12$ and $N = 16$.

N dependence (comparison with $N = 12$)

	$N = 12$	$N = 16$
number of principal axes $N - 1$	11	15
always real	6 ($A-F$)	8 ($A-i$)
alternating	2 (h, i)	3 (j, k, l)
always imaginary	3 (j, k, l)	4 (m, n, o, p)
modal number of imaginary directions	4	6
transition time	$\tau \approx 2500$	$\tau \approx 9000-9500$
top-3 occupancy	$0.485 \rightarrow 0.809$	$0.384 \rightarrow 0.812$
growth ratio of the first axis	2.508	2.799

The counts follow $N - 1$, but **the qualitative structure of concentration into the top three directions appeared at both $N = 12$ and $N = 16$.** The agreement of the final top-3 occupancy at 0.81 is notable. **This is reproduction at two points only, and grounds neither N -independence nor N -dependence.**

Open tasks

1. Whether the 15 directions A through p **can be identified with physical spacetime (time, three spatial axes, $R, Q, \dots$), cannot be, or require an intervening map, is an open task. There is at present no ground for identification.** The names are merely ordinals by eigenvalue magnitude (§0B). In particular, the principal-axis vectors are not continuous in time (§0B item 3), a direct obstacle to matching them with fixed physical axes.
2. How this structure — 3 directions expanding rapidly, 5 staying nearly constant, the rest moving to imaginary — **relates to existing higher-dimensional theories including superstring theory is also an open task.** That the number 11 obtained at $N = 12$ coincided with the 11 of superstring theory is **an accident of the choice of resolution**; at $N = 16$ it is 15. The mechanism also differs (this is not compactification). **We do not claim a correspondence of theories from a coincidence of words.**
3. The number of principal axes $N - 1$ (15) and the parallelotope dimension d (4) are different objects (Claim 16-d). **A four-dimensional parallelotope would have to have rank 4.** The measured rank is 8–11, so **the system is not a parallelotope. The converse does not follow: rank dropping to 4 does not imply a parallelotope.** The measurement of Claim 16-b is the counterexample ($N = 16$, rank = 4 random configurations have median max/min 3.44 and do not lie on an ellipsoid). Rank dropping to d is only a **necessary** condition (Claim 16-d).

Claim 13 (What is conserved is the signed trace; the total amounts of both real and imaginary grow)

By Lagrange’s identity (Claim 5), $\sum_{i < j} d_{ij}^2 = N \operatorname{tr}(B)$, and $\operatorname{tr}(B) = \sum_a \lambda_a$ is a **signed sum** (λ_a the eigenvalues of the double-centred Gram matrix B , possibly negative). We measure it.

[Corrected in v4] Up to v3 this was written $\text{tr}(T)$. **That is wrong.** The inertia tensor $T = \sum_i v_i v_i^\top$ is built from a real configuration and is necessarily positive semi-definite, so it cannot be the object of a “signed trace” with negative eigenvalues. What is sign-indefinite is the double-centred Gram matrix B (see the notational caution in Claim 5).

Quantity	pre-transition	late	ratio
signed $\sum_a \lambda_a = \text{tr}(B)$	6.3287292×10^{-2}	6.3287292×10^{-2}	1.0000
absolute sum $\sum_a \lambda_a $	6.369864×10^{-2}	1.775412×10^{-1}	2.7872
positive part $\sum_a \lambda_a^+$	6.3493×10^{-2}	1.20414×10^{-1}	1.896
negative part $\sum_a \lambda_a^- $	2.06×10^{-4}	5.713×10^{-2}	—

Over the 968 late points, the minimum $6.328729215 \times 10^{-2}$ and maximum $6.328729267 \times 10^{-2}$ of the signed trace **agree to 8 digits** (relative width 8.3×10^{-9}), while the absolute sum grows by 2.79. The real total grows by 1.90, the imaginary total grows from 2.06×10^{-4} to 5.71×10^{-2} , and the difference is conserved.

[Classified in v4] This is **conservation within numerical precision**, not an analytically proved conservation law. We have not derived conservation of $\text{tr}(B)$ from the update rule. Up to v3 we wrote “exactly conserved”; that word is not used without an analytic proof. The relative width 8.3×10^{-9} is of the order of accumulated double-precision error and **strongly suggests a conserved quantity, but is not a proof.** Establishing it analytically requires deriving from the update rule an identity corresponding to $X^\top F(X) = 0$ for the dynamics $\dot{X} = F(X)$. This has not been done (it is precisely the reduction condition of Claim 0).

(Corresponding values at $N = 12$: signed 8.281725×10^{-2} unchanged, absolute sum $\times 2.4710$, positive part $\times 1.736$, negative part 6.094×10^{-2} . **The same qualitative structure was confirmed at the two points $N = 12$ and $N = 16$; N -independence is underived**, by the same standard as Claim 12.)

A misreading this claim forbids

“Since a conservation law holds, no expansion is occurring” — wrong.

What is conserved is the signed sum, not the total real amount. The real total grows by 1.90 and the absolute sum by 2.79. The exponential amplification is real. The scale of the three observed directions is magnified by **both** the growth of the absolute sum and the concentration into the top three directions.

The reverse misreading is equally forbidden: “ A grew by 2.8, so the whole system grew by 2.8” — also wrong. The signed trace does not move.

In the vacuum control both the signed trace and the absolute sum have ratio 1.0000, unchanged at $6.250000 \times 10^{-2} = 1/16$.

Claim 14 (Imaginary directions do not appear in the vacuum. They appear with the creation of matter)

We measure the vacuum control (same resolution, same T , the side that creates no matter).

- Over all 968 points with $\tau \geq 10000$, the number of imaginary directions is **0** and the rank of the Gram matrix is **15** (all 15 non-trivial eigenvalues non-zero). > [Corrected in v4] **Rank and degeneracy are different.** Up to v3 this read “rank 15 (i.e. no degeneracy)”, which is **backwards**. Rank 15 says only that the 15 non-trivial eigenvalues are non-zero, not that they are distinct. Indeed, as written just below, in the late period all 15 eigenvalues equal 0.0645. **This is not the absence of degeneracy but a maximally degenerate state.**
- At $\tau = 39991$ all 15 principal axes have the value 0.0645. **The spectrum is completely degenerate and isotropic.**
- The top-3 occupancy goes $0.3433 \rightarrow 0.2000 = 3/15$, exactly the fully isotropic value.
- r_{rms} is unchanged at $0.062500 = 1/16$ from $\tau = 0$ to $\tau = 39991$.

Even so the vacuum does not lie on an ellipsoid (max/min = 4.61). Isotropy and “all vertices on one ellipsoid” are different conditions: an isotropic inertia tensor makes the ellipsoid a sphere but does not require the vertices to lie on that sphere. The value at $N = 16$ exceeds that at $N = 12$ (2.57), **qualitatively consistent** with the trend observed in Claim 16-b that fixing the rank and increasing N increases the deviation. **This is not a comparison under identical conditions, so the cause is not established.**

- Matter side: 4–7 imaginary directions, rank 8–11.
- Vacuum side: 0 imaginary directions, rank 15.
- The matter side too has 2 imaginary directions only at $\tau = 0$ (immediately after seeding), disappearing by $\tau \approx 1000$. Imaginary directions appear persistently only after the transition ($\tau \approx 9000$).

Hence the negative eigenvalues of B are not built into the system from the start; they appear with the creation of matter.

[Separated in v4] **This must not be identified with the imaginary symbol of Claim 7.** As organised in Claim 5, this note has **three distinct signs**:

1. real/imaginary parts in complex zero closure (Claim 7)
2. the face-dimension sign $s_e = (-1)^{k+1}$ (Claim 4)
3. the sign of the eigenvalues of the double-centred Gram matrix B (what this claim measures)

No map connecting these has been derived (assumption (Γ), §3B). So what this claim can state is only:

“Numerically, negative eigenvalues of B were confirmed to appear together with the creation of matter. A map identifying these negative eigenvalues with the imaginary symbol of the complex representation of Claim 7 is underived, and at present this remains a qualitative candidate correspondence.”

The explanation “the vacuum has nothing moved to the unobservable side, hence nothing to attach an imaginary symbol to” **is not used, since it would silently rebuild the bridge cut in Claim 5.**

- Status: measured. Matter and vacuum sides run simultaneously as a control with the same program and the same random numbers.

Claim 15 (Quasi-oscillations of order 10^2 steps exist, but their period varies with τ)

The second axiom $U^n = I$ requires the system to return after finite order, i.e. **every orbit period divides n .**

[Corrected in v4] Up to v3 this read “if $U^n = I$ then there are no fixed points, only cycles; hence a complete steady state is impossible”. **This is wrong.** Period 1 (a fixed point) is also a divisor of n . The simplest counterexample is $U = I$, which satisfies $U^n = I$ with every point fixed. In general, if U has eigenvalue 1, points in that eigenspace do not move. **The non-existence of steady states cannot be derived from $U^n = I$.**

So $U^n = I$ only gives an **upper bound** on the period; which periods actually appear must be measured.

Measured object: the mean of the 120 relation lengths. The period is resolvable only in the dense region (every step recorded).

Run: the $N = 16$ transition is at $\tau \approx 9000$, and with the default dense region $\tau < 4000$ (designed for the $N = 12$ transition time) **the transition falls outside the window.** A dedicated run with the dense region widened to $\tau < 10000$ was therefore performed (stem `electron_T40000_d0.1_rep-dump40k16d_N16`, `DUMP_TAUC=10000`, 10968 frames, 2688 s). **The measurement below includes the transition.**

Estimation (two independent methods, reported side by side)

1. **Autocorrelation:** detrend with a 201-point moving average, take the autocorrelation in windows of 496 steps, and take the first maximum at lag ≥ 10 as the dominant period. The lag of the minimum preceding that maximum is the anti-period.
2. **Peak spacing:** count local maxima of the detrended series and take the median spacing of adjacent peaks (spacings below 8 steps excluded).

That the two estimates disagree is itself a result.

Results (verbatim output of the derivation program `make_period_figure_v1.py`)

$N = 16$, dense region $\tau < 10000$ (including the transition at $\tau \approx 9000$). **Amplitude** is the standard deviation after detrending, reported as the matter-to-vacuum ratio. Peak spacings must not be read without consulting this column (see below).

τ window	matter period	corr.	anti-period	matter peak spacing	vacuum period	vacuum peak spacing	amplitude ratio matter/vacuum
0–495	133	+0.134	67	83.0	not detected	10.5	2.2×10^4
496–991	105	+0.306	46	60.0	not detected	—	”
992–1487	91	+0.673	46	94.0	118	—	0.88
1488–1983	92	+0.542	46	92.0	193	99.5	”
1984–2479	94	+0.635	48	95.5	388	95.0	0.73
2480–2975	279	+0.335	50	93.5	94	93.5	”
2976–3471	102	+0.177	54	92.0	98	94.0	0.65
3472–3967	97	+0.276	50	90.0	202	100.0	”
3968–4463	367	+0.187	137	$\langle 55.0 \rangle$	100	98.0	1.09
4464–4959	105	+0.430	58	$\langle 61.0 \rangle$	203	99.5	”
4960–5455	271	+0.305	141	$\langle 52.5 \rangle$	307	103.5	3.52
5456–5951	116	+0.377	73	$\langle 60.0 \rangle$	299	96.5	”
5952–6447	108	+0.081	75	$\langle 47.0 \rangle$	288	99.0	9.49
6448–6943	45	+0.500	31	$\langle 50.0 \rangle$	197	98.0	”
6944–7439	294	+0.305	75	$\langle 55.0 \rangle$	212	138.0	14.1
7440–7935	333	+0.323	74	$\langle 51.5 \rangle$	209	105.0	”
7936–8431	44	+0.435	26	$\langle 45.5 \rangle$	107	104.0	96.7
8432–8927	383	+0.082	279	$\langle 71.0 \rangle$	296	99.0	”
8928–9423 (transition)	259	+0.212	63	34.0	101	97.5	2.0×10^4
9424–9919 (just after)	145	+0.258	72	28.0	217	105.0	8.5×10^4

The peak spacings in $\langle \rangle$ for $\tau \in [3968, 8927]$ **must not be read**. In this interval the matter-side amplitude has fallen to order 6×10^{-7} (no amplification before the transition), and the maxima picked up are numerical noise. The amplitude-ratio column is the basis for this judgement.

(Reference) $N = 12$ (transition inside the window)

τ window	matter period	corr.	matter peak spacing	vacuum peak spacing
0–495	180	+0.338	96.0	13.0
496–991	96	+0.421	94.0	—
992–1487	97	+0.528	70.5	—
1488–1983	188	+0.366	58.5	115.0
1984–2479	148	+0.220	52.0	162.0
2480–2975	272	+0.228	26.0	101.0
2976–3471	156	+0.165	21.0	97.0
3472–3967	113	+0.308	24.5	not detected

What the measurement supports

1. **Quasi-oscillations of order 10^2 steps are real.** Before the $N = 16$ transition the dominant period is 91–105 (four windows in $\tau \in [496, 2479]$) with median peak spacing 90–95.5. The value 124 steps is one value this quasi-oscillation takes, not a constant of the system.
2. **The period varies with τ .** The autocorrelation dominant period on the $N = 16$ matter side ranges from 91 to 279. **The two estimates also disagree** (279 by autocorrelation vs 93.5 by peak spacing at $\tau \in [2480, 2975]$).
3. **It is not a sharp period.** The autocorrelation at the dominant period is $+0.134$ – $+0.673$, not near 1; a sharp period would reach 1. So this is a quasi-oscillation, not a cycle. Some windows exceed $N = 12$ ($+0.165$ – $+0.528$), but all are far from 1.
4. **The anti-period is roughly half the period.** Before the $N = 16$ transition, period/anti-period clusters near 2.00 at 1.98–2.28 ($\tau \in [992, 2479]$), though it jumps to 5.58 at $\tau \in [2480, 2975]$.
5. **The transition shortens the period; reproduced at $N = 16$.** The matter-side median peak spacing shortens from 90–95.5 before the transition ($\tau < 4000$, where the oscillation is real) to **34.0** in the transition window (τ 8928–9423) and **28.0** just after (τ 9424–9919) — about $1/3$. The vacuum side stays at 93.5–138 throughout and does not shorten. This has the same direction as the $N = 12$ finding ($96.0 \rightarrow 21.0/24.5$). **This is reproduction at the two points $N = 12$ and $N = 16$ only; resolution-independence is underived.**
6. **The amplitude jumps by 4 orders at the transition.** The matter/vacuum ratio of the detrended standard deviation goes from 0.65–1.09 before the transition to 2.0×10^4 in the transition window and 8.5×10^4 just after. At $N = 12$ it was 98; but that is because the $N = 12$ dense region reached only just past the transition, and the magnitudes of the ratios must not be compared across N .
7. **The matter-side oscillation disappears once before the transition.** In $\tau \in [3968, 8927]$ the matter-side amplitude falls to order 6×10^{-7} , comparable to or below the vacuum side. That is, the $N = 16$ system passes through three stages: oscillation $\rightarrow$ quiescence $\rightarrow$ explosive transition. This quiescent stage is not visible at $N = 12$ (the transition is earlier and the dense region shorter). **Whether it is specific to N or a consequence of the later transition time is unresolved.**

Limits of the measurement (the period cannot be measured for $\tau > 10000$)

The recording interval for $\tau \geq 10000$ is 31 steps, so the Nyquist period is 62 steps. Moreover $124/31 = 4.000$ exactly, so a slight shift of period appears as a beat. **The period in this region cannot be measured directly from this run's data.** In particular the measured 28.0–34.0 just after the transition is **below** the Nyquist period 62, so such short periods are in principle invisible at the coarse recording. Settling whether the shortening continues or settles at some value in the late period ($\tau > 10000$) requires a separate run recording every step in the late period as well.

Open task

Whether this quasi-oscillation originates in the second axiom $U^n = I$ or is a different phenomenon is an open task.

If it originated in $U^n = I$, the period should be a constant determined by n , and its variation with τ would need explanation. Whether the finite order n varies with the state, whether several orders are superposed, or whether the quasi-oscillation has an origin unrelated to $U^n = I$ — none of these is settled. No exact period has been found.

Figures are in §5.5.

Claim 16 (When zero closure is realised as a parallelotope family, the resolution becomes $N = 2^d$. This is not a sufficient condition, and that it is necessary for all zero-closing solutions is also underived)

16-a. The constraint itself, but conditional.

When zero closure is realised as a parallelotope, the number of vertices is restricted to $N = 2^d$ (Claim 4-b). Hence numerical runs to be compared with the geometric statements must take $N \in \{2, 4, 8, 16, 32, \dots\}$. This note

adopts $N = 16 = 2^4$; the corresponding parallelotope dimension is $d = 4$.

[Corrected in v4] Up to v3 this read “a zero-closing configuration is a parallelotope, so the number of vertices is restricted to $N = 2^d$ ”. **That cannot be said from the present derivation**, since “zero closure $\Rightarrow$ parallelotope” is unproved for $d \geq 3$ (Claim 4-e).

Precisely: $N = 2^d$ is a condition for the case “zero closure realised as a parallelotope family”; that it is necessary for all zero-closing solutions is underived. There is no guarantee that non-parallelotope zero-closing solutions do not exist for $d \geq 3$. The reason for choosing $N = 16$ is “to sit on the known solution family guaranteed by Claim 4-b”, not “because other N are forbidden”.

This $d = 4$ must not be identified with the “four dimensions” of the title. The title’s four dimensions are the dimension of the ambient space with coordinates (r, t, R, Q) (Claim 2B; 3 independent degrees of freedom because of the null-cone constraint); this one is the dimension of the Euclidean space into which a zero-closing configuration is embedded. **The numerical coincidence is accidental and neither follows from the other.**

16-b. $N = 2^d$ alone is not enough. rank = d alone is not enough either.

Matching the vertex count to 2^d , dropping the configuration dimension to d , or doing both, does not put the vertices on a single ellipsoid. **In this test the only constructive family giving max/min exactly 1 was the parallelotopes; that it is the only such family is not proved** (this is a test over 200 random trials, not a general theorem). Measurements at $d = 4$ (200 random configurations; max/min of the deviation s ; 1.00 means “on the ellipsoid”):

Condition	N	rank	median max/min	worst
$N = 2^d$ and rank = d (not a parallelotope)	16	4	3.4382410192	17.534
rank = d only (vertex count not a power of 2)	13	4	3.0423369129	18.926
parallelotope	16	4	1.0000000000	1.000

Fixing the rank at $d = 4$ and varying N , more vertices make it harder to lie on the ellipsoid (1.476 at $N = 6$, 2.009 at $N = 8$, 2.932 at $N = 13$, 3.373 at $N = 16$, 3.856 at $N = 20$). Among these only parallelotopes give exactly 1.

16-c. A caveat about the measurement (not a condition on configurations).

The ellipsoid test by the deviation s has discriminating power only when rank $< N - 1$. When rank = $N - 1$ the projection matrix equals $H = VT^{-1}V^T = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$, so $H_{ii} = (N - 1)/N = c$ for every i and $s_i \equiv 1$ **for any configuration**. In that case “lying on the ellipsoid” carries no information.

Imposing rank = d and rank = $N - 1$ simultaneously collapses to $N = d + 1$ ($N = 5$ for $d = 4$, the four-dimensional simplex), but **this is the case where the test becomes vacuous, not a permitted N** . Measurement confirms that $N = 5$, rank = 4 random configurations give max/min = 1.0000000000 (trivially on). And 5 is not 2^d , so it is excluded from the outset by Claim 16-a.

For the parallelotope family, rank = d and $N - 1 = 2^d - 1$; these coincide only when $2^d - 1 = d$, i.e. $d = 1$ ($N = 2$). For $d \geq 2$, and hence for this note’s $N = 16$ (rank 4 vs $N - 1 = 15$), the test always has content.

16-d. Do not confuse $N - 1$ with d .

The number of principal axes is $N - 1$ (15 at $N = 16$); the parallelotope dimension is d ($= 4$). They are different. Rank dropping to d is only a **necessary** condition for being a parallelotope (the first row of 16-b is the counterexample).

16-e. Unresolved.

Necessity for $d \geq 3$ is unproved (Claim 4-e). So 16-a is a statement conditional on “if zero closure is realised”, and does not mean that the dynamics fails to run for systems with N not a power of two. The $N = 12$ run is not invalid; it simply **is not an object to be compared with the geometric statements**.

- Derivation program: `figures_v1/check_sufficiency_v1.py`

Claim 17 (Without the sign rule, zero closure carries no information. What selects is the sign rule)

17-a. With all reals and no signs, only the trivial solution exists.

If all x_n are real, $\sum_n x_n^2 = 0 \Leftrightarrow x_n = 0$ for all n . Lengths are non-negative, so no non-trivial configuration exists. This is a one-line consequence, not a measurement.

17-b. If both signs and configuration may be chosen freely, non-trivial solutions exist from $N \geq 3$.

A distinction must be made first. s_e is not a continuous unknown but takes **discrete values**, so the counting “one equation, hence always solvable” cannot be used as it stands.

- **With the distance set fixed, sign choice alone need not achieve zero.** Counterexample: for an equilateral triangle $d_1^2 = d_2^2 = d_3^2 = 1$, no choice of signs makes $\pm 1 \pm 1 \pm 1$ vanish.
- **If the configuration may also be chosen freely, non-trivial solutions exist from $N = 3$:** $d_1^2 + d_2^2 - d_3^2 = 0$, i.e. a right triangle. With signs fixed, this is a codimension-1 condition in configuration space and solutions exist.

The measurements show this distinction. For random configurations (distance set fixed), the median of $|\sum_e s_e d_e^2| / \sum_e d_e^2$ achievable by optimising signs is 2.2×10^{-2} at $N = 4$, 5.0×10^{-3} at $N = 6$, 1.5×10^{-3} at $N = 8$. **It decreases with N but does not reach zero.** Reaching zero requires moving the configuration.

[Corrected in v4] Up to v3 this read “if signs may be chosen freely it is always solvable for $N \geq 3$ ”, which is **wrong** for a fixed distance set. It has been restricted to “if both signs and configuration may be chosen freely”, as above. That the measured values are non-zero was itself the sign that this restriction was needed.

The essence of the claim is unchanged by this restriction. What was shown above is the following.

This suffices for Claim 17. The measure-theoretic statement “with the freedom of both signs and configuration it can **almost always** be satisfied” is **not used, since this note does not prove it.**

17-c. What selects is the sign rule.

Only once the signs are fixed by the face dimension k (Claim 4-a) does the alternating sum become uniquely determined per configuration and the freedom of choice disappear. Then it vanishes identically for parallelotopes (4-b), and does not vanish for the octahedron (+36.000) or the icosahedron (+521.000) (4-d).

Hence it is not “the form of the equation $\sum x_n^2 = 0$ ” that selects parallelotopes. What selects is the sign rule by face dimension. Carrying only the form of the equation elsewhere does not produce the same selection.

17-d. The place of the imaginary unit.

From 17-a–17-c, imaginary numbers are not needed for the existence of solutions; they are **a notation for writing signs**. Indeed, as in 4-g, $\sum_{\text{edges}} d^2 = \sum_{\text{main diagonals}} d^2$ holds exactly for parallelotopes with all lengths real, and “reading the main diagonals as imaginary” is merely a transposed expression of that real identity.

- Derivation program: `figures_v1/check_real_solutions_v1.py` (tests A, B, D, E).

Claim 18 (The orientation of the principal axes is not conserved. Orientation-independent quadratic forms are)

[Title corrected in v4] **It is not an “inner product”.** $\sum_e |x_e|^2$ is the squared norm from the Hermitian inner product, but $\sum_e x_e^2$ is **a quadratic form built from the complex symmetric bilinear form, not an inner product** (neither positive definite nor conjugate linear). To name both together, “quadratic form” is correct.

Claim 12 states the concentration into the top three directions and §0B states that the principal-axis vectors are not continuous in τ . Does the orientation stabilise over a long sweep in τ ? **What matters is neither the**

orientation nor the sign, but whether an orientation-independent conserved quantity (a quadratic form) exists. We measure both.

18-a. The orientation does not stabilise.

Individual principal axes can rotate arbitrarily under degeneracy, so what must be tested is the **subspace** spanned by the top k . We measure the projector overlap $\text{tr}(P_1 P_2)/k = \|U_1^\top U_2\|_F^2/k$ (1 identical, 0 orthogonal). **The expectation between random k -dimensional subspaces is $k/(N-1)$** , the level of “orientation completely lost”: at $N = 16$, 0.200 for $k = 3$ and 0.0667 for $k = 1$.

$N = 16$, matter side, $k = 3$ (lags in steps; $\tau < 4000$ recorded every step, thereafter every 31):

τ in- terval	2	4	8	16	32	62	124	248	496	992	1984
0– 4000 (pre- transition)	0.996	0.987	0.959	0.883	0.730	—	—	—	—	—	—
4000– 8991	—	—	—	—	—	0.485	0.164	0.218	0.228	0.231	0.235
8991– 9487 (in transi- tion)	—	—	—	—	—	0.594	0.417	0.310	—	—	—
9487– 20000	—	—	—	—	—	0.651	0.556	0.459	0.385	0.314	0.238
20000– 40000	—	—	—	—	—	0.649	0.564	0.474	0.397	0.331	0.285

(The pre-transition interval gives 0.524 at lag 64.)

What can be read off

- After the transition the correlation approaches the random level 0.200 in about 2000 steps. **The orientation diffuses.**
- The same holds for the first principal axis alone ($k = 1$, random level 0.0667): 0.176 at lag 1984 (9487–20000), 0.295 (20000–40000). The first eigenvalue is well separated with relative gap 0.70, so **this is not label swapping under degeneracy; the separated eigenvector itself loses its orientation.**
- Later windows persist slightly longer (0.238 $\rightarrow$ 0.285 at lag 1984, $k = 3$). **Whether this gentle rise continues at still longer τ is unverified.** A sweep of 36,000 steps has not reached stabilisation.
- Reference: in the same measurement at $N = 12$, the decay curves of the three windows overlapped almost perfectly (0.276/0.278/0.272 at lag 1984; random level $3/11 = 0.273$) with no sign of a rise.

On the vacuum control the orientation is lost immediately. At $k = 3$ and lag 124 it is 0.078–0.235, already fluctuating around the random level. The vacuum is isotropic in all 15 directions with degenerate eigenvalues (Claim 14), so the choice of “top three” is meaningless. **Hence the very fact of “orientation becoming defined” accompanies the creation of matter, and even that orientation is not conserved.**

18-b. Orientation-independent conserved quantities nevertheless exist. At least two kinds are confirmed.

Let x_e be the complex amplitude per relation (harmonics summed first; the same aggregation as the distance readout `lengths_from_C2`). $N = 16$, matter side, $\tau \geq 10000$, 242 points:

Quantity	Value	fluctuation / mean	Verdict
$\sum_e x_e ^2$ (Hermitian)	1.01259668	8.3×10^{-9}	conserved within numerical precision
$\sum_e x_e^2$ (bilinear, complex)	$+7.16836667 \times 10^{-3} + 2.99111734 \times 10^{-3} i$	1.5×10^{-7}	conserved within numerical precision
$\text{tr}(B) = \sum_a \lambda_a$	0.06328729	8.3×10^{-9}	conserved within numerical precision
$\text{tr}(B^2) = \sum_a \lambda_a^2$	—	1.7	not conserved
$\text{tr}(B^3)$	—	2.5	not conserved

Since $\text{tr}(B) = \sum_e |x_e|^2 / N$ (Lagrange’s identity, Claim 5), the first two are the same quantity. **Two kinds of quadratic invariant were confirmed conserved in this run:** the Hermitian $\sum |x_e|^2$ and the bilinear $\sum x_e^2$. From $\text{tr}(B^2)$ upward they are not conserved.

[Corrected in v4] Up to v3 this asserted “there are exactly two conserved quadratic invariants”. **That cannot be said.** We tested only the Hermitian, the bilinear and $\text{tr}(B^k)$; we did not show the absence of others. Not “exactly two” but **“at least these two kinds are conserved”**.

The counting also needs a definition. $\sum x_e^2$ is complex, so counted as real scalars it is **two** (real and imaginary parts). The total differs according to whether one complex quantity counts as one or as two real ones. This note counts two *kinds*.

Both are sums over relations (trace type), so their values do not change however the frame of principal axes rotates. That is, the diffusion of 18-a exacts no cost on these quantities. **Even when orientation is lost, orientation-independent quadratic quantities are not.**

18-c. $\sum x_e^2$ is conserved as a complex number through the transition (within numerical precision).

The $N = 16$ transition occurs at $\tau \in [8991, 9487]$ (Claim 10). We measure the real and imaginary parts of $\sum_e x_e^2$ across it.

τ interval	real part	imaginary part	fluctuation / mean
0–4000	$+7.16836667 \times 10^{-3}$	$+2.99111734 \times 10^{-3}$	5.9×10^{-13}
4000–8991 (pre-transition)	$+7.16836667 \times 10^{-3}$	$+2.99111735 \times 10^{-3}$	1.5×10^{-10}
8991–9487 (in transition)	$+7.16836667 \times 10^{-3}$	$+2.99111735 \times 10^{-3}$	1.7×10^{-10}
9487–20000	$+7.16836669 \times 10^{-3}$	$+2.99111738 \times 10^{-3}$	4.2×10^{-8}
20000–40000	$+7.16836706 \times 10^{-3}$	$+2.99111683 \times 10^{-3}$	9.8×10^{-8}

The value is identical over all intervals, including during the transition. Not only the modulus but also the argument is unmoved. **This too is a statement within numerical precision** (the classification of Claim 13).

On the vacuum control $\sum_e x_e^2 = 2.0 \times 10^{-15}$, i.e. **zero within numerical precision**; $\sum_e |x_e|^2 = 1.00000000$ (fluctuation 2.0×10^{-15}), $\text{tr}(B) = 0.06250000 = 1/16$.

Hence $\sum_e x_e^2$ is fixed at the complex value given by the seed at $\tau = 0$. It is 0 in the vacuum, moves to a non-zero value when a seed is inserted, and **is constant throughout the observed interval $\tau \leq 39991$.** Saying “it never moves thereafter” would require deriving an analytic conservation law (the classification of Claim 13).

18-d. Only the readout layer is conserved. The component layer is not.

The order of aggregation gives different quantities. $N = 16$, matter side:

Aggregation	Value	fluctuation / mean
harmonics summed first, then squared $\sum_e (\sum_{k,j} C_{ekj})^2$	$ \cdot = 7.767 \times 10^{-3}$	1.5×10^{-7} (conserved)
squared per component then summed $\sum_{e,k,j} C_{ekj}^2$	$ \cdot = 1.521 \times 10^{-2}$	2.7 (not conserved)

The latter coincides with the model-internal closure diagnostic closure. **The invariant appears only in the aggregation that reads out distances, not at the component layer.**

18-e. Correction to Claim 10.

Claim 10 stated, on the ground that the component-layer closure residual is non-zero, that “zero closure is violated in transient and metastable states”. **Seen at the readout layer, it is not violated.**

What exists at the readout layer is not “zero closure”. By the §0 definition, zero closure is $\sum x_n^2 = 0$, whereas the readout-layer $\sum_e x_e^2$ is a **non-zero complex constant** ($+7.168 \times 10^{-3} + 2.991 \times 10^{-3}i$ on the $N = 16$ matter side). Correctly: **“at the readout layer there is no zero closure; instead the bilinear quadratic quantity $\sum_e x_e^2$ exists as a non-zero conserved quantity.”** It is 0 within numerical precision in the vacuum, becomes a non-zero constant when a seed is inserted, and does not move through the transition.

The scope of Claim 10 is therefore restricted as follows.

- **Component layer** (the model’s closure diagnostic): not zero, and strongly varying. Claim 10 is correct as a statement about this layer.
- **Readout layer** (the aggregation giving distances): not zero, but conserved within numerical precision. The description “violated” does not apply.

Relation to Claim 11 (retracted in v4).

[Corrected in v4] Up to v3 this read “if $\sum x_e^2$ is a conserved quantity fixed by the initial conditions, the deviation will not decay however far τ is extended; the only way to reduce it is to lower δ .” **This inference does not hold.**

Even with one conserved quantity fixed, the shape of the configuration may vary freely on that conservation surface. This is exactly what Claim 0 states — “a conserved quantity does not determine the motion; it determines only the space of permitted motions” — and **the inference above contradicts Claim 0.**

To derive non-decay of the deviation from conservation of $\sum x_e^2$, at least one of the following is needed: 1. show that the deviation is a function of the conserved quantities alone; 2. show that the conserved quantity gives a positive lower bound on the deviation.

Neither is derived. So “the deviation does not decay as τ is extended” is an **empirical fact** (Claim 10, flat over the range $T = 40000$), not a consequence of the conserved quantity.

Measuring the δ dependence of $|\sum x_e^2|$ in a δ sweep remains a valid experiment for settling Claim 11. **It has not been performed.**

18-f. Implications for the readout problem.

Claim 2B left “how to read out of r ” unresolved, and Claim 12 left “can $A, B, \dots, p$ be identified with physics?” unresolved. 18-a and 18-b give these the following form.

Within this run, identification with a fixed frame is not supported. The orientation diffuses (18-a). If anything can be identified, it is orientation-independent quantities: the conserved quadratic forms and the magnitudes of the spectrum (18-b).

There are three readings. **At present no experiment discriminating among them has been designed. That is itself an open task.**

1. The top three principal axes are not the readout; a different map is required.

2. The readout uses only orientation-independent quantities; three-dimensionality and magnitude are the physical quantities, and absolute orientation never existed.
3. Diffusion of orientation is correct. A purely relational system has no external standard fixing orientation, so diffusion of an absolute coordinate frame is not a defect but the expected consequence.

From the relational premise (Claim 2B), 3 is natural. If relationality gives only lengths, the readout should be able to produce three components but not an absolute orientation, and 18-a is consistent with that.

What can be measured (not a discrimination): lowering δ to reach a steady state and testing whether the diffusion of orientation stops would show **whether the diffusion is a transient, seed-dependent phenomenon or a structure surviving in the steady limit. This has not been performed.**

This measurement does not, however, discriminate 1 from 2 and 3. If the diffusion stops, one cannot conclude that reading 1 (“the top three axes are not the readout”) is false — only that one piece of evidence supporting it disappears. Conversely, if the diffusion does not stop, the possibility that the top three axes are not the readout at all remains. **No experiment discriminating the three readings has been designed; that is itself an open task.**

- Derivation program: `__v1//_N1to20__v2/check_invariants_v1.py`
 - Measured values: `__v1//_N1to20__v2/figures_tau/invariants_electron_T40000_d0.1_rep-dump40k16_N16_v1.js`
 - Reproduce: `python3 check_invariants_v1.py electron_T40000_d0.1_rep-dump40k16_N16 --N 16 --late 10000 --windows 0,4000,8991,9487,20000,40000`
-

3. What is not claimed

- This note contains no new theorem.
- It does not claim to explain that real physical space is three-dimensional. Claim 2 is a statement about the readout of this model.
- It claims no connection with the fine-structure constant, the elementary charge, or a particle periodic table.
- It does not prove the **necessity** direction of Claim 4 for $d \geq 3$ (alternating sum zero $\Rightarrow$ parallelotope) (Claim 4-e).
- That parallelotopes satisfy zero closure is confirmed (Claim 4-b, all d). **Whether this can be identified with physical entities, cannot be, or requires an intervening map is an open task.** There is at present no ground for identification (the same map problem as Claims 2B, 12, 18-f).
- Individual semi-axes, sphericity, and the number of hidden axes are not determined by zero closure (Claims 5, 6). **Whether separate conditions determining them exist, or whether they are degrees of freedom undetermined in principle, is an open task.**
- **The dynamical law F has not been derived.** Time evolution itself is measured extensively in Claims 10–18, but F in $\dot{X} = F(X)$ has not been written down. **What F satisfies the reduction condition $X^\top F(X) = \frac{1}{2}\dot{C}$ of Claim 0 is underived, and identifying F is the first gate to passing on to dynamics.**
- It is not claimed that $A, B, C, D, E, F, h, i, j, k, l$ correspond to axes of physical spacetime. There is no ground for identification (Claim 12).
- It is not claimed that the number of principal axes (15 at $N = 16$, 11 at $N = 12$) is the same as the dimension count of superstring or other higher-dimensional theories. That 11 appeared at $N = 12$ is an accident of the choice of resolution. A coincidence of words is not a correspondence of theories (Claim 12).
- It is not claimed that lowering the seed strength or extending τ makes the deviation from the ellipsoid vanish. This is unverified (Claim 11).
- It is not claimed that the numerical model of this series reaches a steady state. It does not within $T = 40000$ (Claim 10).
- It is not claimed that the observed quasi-oscillation originates in the second axiom $U^n = I$. Since the period varies with τ , whether it comes from $U^n = I$ or is a different phenomenon is unsettled (Claim 15).
- 124 steps is not claimed as a period constant of the system. It is one value the quasi-oscillation takes (Claim 15).

- The title’s “four dimensions” (r, t, R, Q) is not identified with the parallelotope dimension $d = 4$ ($N = 16$). They are different and the numerical coincidence is accidental (Claims 2B, 16-a).
- It is not claimed that the ellipsoid interpretation of Claim 0-c has been confirmed on the numerical model. It is an analytic statement, and numerical confirmation in a state where (S), (U), (R) and fixed C hold has not been performed. In this model the candidate for such a state is the steady state (Claim 0).
- It is not claimed that the readout surface is the inertia ellipsoid itself. The readout map Λ is not given and $G^{-1} \propto T$ is unproved (Claim 0-c).
- It is not claimed that $x^2 + y^2 + z^2 - t^2 = R^2 + Q^2$ is a closed surface as it stands in four variables. Its signature is $(3, 1)$, and closure holds only for the level set of the visible three components at fixed $C = t^2 + R^2 + Q^2$. Fixing t alone does not close the surface if R, Q move (Claim 0-c).
- It is not claimed that for general complex numbers zero closure gives a single independent real constraint. There are two (equality and orthogonality). Reduction to one holds only under assumption (S: separation of the supports of real and imaginary parts) (Claim 0-b).
- It is not claimed that the numerical model satisfies (S). The model’s relation amplitudes are general complex numbers and the supports are not separated. Whether a structure corresponding to (S) appears in the model is an open task (Claim 0-b).
- It is not claimed that the reduction from M dimensions to 3 maps a surface onto a surface. A general linear surjection maps a sphere onto a solid ellipsoid. The existence of the three-dimensional subspace U is underived (Claim 0-c).
- It is not claimed that zero-closing configurations are limited to parallelotopes. Necessity for $d \geq 3$ is unproved, and $N = 2^d$ is a condition for realisation as a parallelotope family (Claims 4-e, 16-a).
- It is not claimed that $U^n = I$ implies the non-existence of a steady state. Period 1 also divides n (Claim 15).
- It is not claimed that non-decay of the ellipsoid deviation follows from a conserved quantity. The non-decay is an empirical fact (Claim 18-e).
- It is not claimed that a discrete spectrum follows from compactness. The operator must be specified separately (Claim 9).
- It is not claimed that the conserved quantity on the right-hand side is conserved for all τ . At the component layer it holds only in the steady state and deviates and oscillates in transient and metastable states (Claims 0-d, 10, 15).
- Constancy within numerical precision is not claimed to establish an analytic conservation law. Derivation from the update rule has not been performed (Claims 13, 18-b).
- It is not claimed that a correspondence has been given between t, R, Q and any measured quantities (Claims 0, 2B, 12).
- It is not claimed that the readout map from r to x, y, z has been given. It is unresolved (Claim 2B).
- It is not claimed that the orientation of the principal axes stabilises as τ is extended. A gentle rise was seen at $N = 16$, but 36,000 steps did not stabilise it (Claim 18-a).
- It is not claimed that there are only two conserved quadratic invariants. The result is only that these two kinds were conserved within the tested range (Claim 18-b).
- It is not claimed that complex zero closure gives an ellipsoid with any of (S), (U), (R) removed. Without the assumptions the constraints number two and the surface is not determined (Claim 0-c).
- It is not claimed from Claim 1 that the motion of the system decomposes into a direct sum of fixed two-dimensional planes. The decomposition is per generator, and non-commuting generators cannot be simultaneously block-diagonalised (Claim 1).
- It is not claimed that the three components of Claim 2 follow immediately from Claim 1. The aggregation rule passing each generator’s 2D block into common three components is underived (Claim 2).
- If axiom 0.5 is taken to include “different scales denote the same physical state”, it is not claimed that this follows from the homogeneity of zero closure. A separate projectivisation is required (Claim 8).
- It is not claimed that the form of the equation $\sum x_n^2 = 0$ by itself selects parallelotopes.

The selection is done by the face-dimension sign rule; without it, as long as both signs and configuration may be chosen freely, non-trivial solutions exist from $N \geq 3$ (Claim 17).

- It is not claimed that imaginary numbers are necessary for the existence of solutions. For parallelotopes, $\sum_{\text{edges}} d^2 = \sum_{\text{main diagonals}} d^2$ holds exactly with all quantities real. The imaginary unit is a notation for signs (Claims 4-g, 17-d).
- It is not claimed that satisfying $N = 2^d$ puts a configuration on an ellipsoid. It is not a sufficient condition (Claim 16-b). Nor is it derived that $N = 2^d$ is necessary for all zero-closing solutions (Claims 4-e, 16-a).
- It is not claimed that $N - 1$ (the number of principal axes) and d (the parallelotope dimension) are the same thing (Claim 16-d).

3B. Unresolved theoretical gaps

The largest gap of this note is stated explicitly.

The numerical model posits $\sum_e x_e^2 = 0$ for the complex amplitudes **as a steady-state closure condition at the component layer. It is not enforced exactly by projection at each time step.** Were it enforced as a hard constraint it could not be violated, but in measurement the closure residual is non-zero in transient and metastable states, and the number of steps with exact zero is 0 (Claim 10). The update rule **requires this closure condition as a steady solution** but does not require it to hold at each step. What the geometry of Claim 4 requires, on the other hand, is a **signed** closure condition classifying segments by face dimension k with sign $(-1)^{k+1}$.

Nothing at present connects these two. This is the hidden assumption of this note.

Layer distinction (added in v4): the phrasing “the model imposes $\sum_e x_e^2 = 0$ ” is correct only for the component layer. Measured at the **readout layer** (x_e with harmonics summed first), $\sum_e x_e^2$ is not zero but a constant fixed at the complex value given by the seed at $\tau = 0$ (Claim 18-c: $+7.168 \times 10^{-3} + 2.991 \times 10^{-3}i$ on the $N = 16$ matter side, invariant to 10^{-7} through the transition). It is exactly zero only on the vacuum side. Always state which layer is meant.

A larger gap became clear in Claim 0. We could show, under (S), (U), (R), that complex zero closure gives a closed surface of a positive-definite quadratic form G on the level set of the visible three components at fixed C ; but **that this G stands in the relation $G^{-1} \propto T$ to the inertia tensor T is unproved** (Claim 0-c). Without it one cannot say “readout surface = inertia ellipsoid”. Moreover, since the numerical model does not reach a parallelotope, showing it for parallelotopes would not reach the model.

Another gap is the readout map. Claim 2B states that the base layer is r and that x, y, z are determined by the readout, but **does not give that map.** It is very likely the same problem as Claim 12’s “can $A, B, \dots, p$ be identified with physics?” (Claims 2B, 18-f). Claim 18-a shows that at least in the form “identification with a fixed frame” there is no solution.

Related open questions:

- ~~It holds only when some of the segments are read as imaginary; the conjecture is the diagonals, but this has not been verified~~ → **resolved in v4.** See Claims 4-f and 4-g. Recovery of the face-dimension classes from lengths alone was verified for $d = 2, 3, 4$ (10/10 exact agreement), and $\sum_{\text{edges}} d^2 = \sum_{\text{main diagonals}} d^2$ was confirmed exact in all dimensions. “What becomes imaginary is the diagonals” is exact only for $d = 2$ ($N = 4$); in general the **even face-dimension classes** become imaginary. At $N = 16$ it is 64 real / 56 imaginary.
- The configurations of the numerical model do not reach a parallelotope. The rank falls from 15 to 8–11 but not down to $d = 4$ (Claims 12, 16-d). Why the system does not move toward the configurations for which Claim 4-b guarantees zero closure is not understood.
- The second axiom $U^n = I$ requires every orbit period to divide n . **Since period 1 (a fixed point) is included, the non-existence of a steady state cannot be derived from it** ($U = I$ is a counterexample; the statement up to v3 that “no complete steady state exists, only cycles” was wrong and was removed in v4). Quasi-oscillations of order 10^2 steps were measured, but the period varies with τ and no exact period has been found. Whether it originates in $U^n = I$ or is a different phenomenon is unsettled (Claim 15).

4. Prior work

Every fact used in this note is known.

Fact used	Prior work
$\sum \text{edges}^2 - \sum \text{diagonals}^2 = 4LM^2$, equality $\iff$ parallelogram	Euler’s quadrilateral theorem (18th century)
$\sum_{i<j} d_{ij}^2 = N \sum R_i^2$	Lagrange’s identity
distance matrix $\rightarrow$ configuration unique up to reflection; dimension = rank of the Gram matrix	I. J. Schoenberg, <i>Ann. of Math.</i> 36 (1935) 724
volume ² = Cayley–Menger determinant	Cayley (1841), Menger (1928)
orthogonal canonical form of real antisymmetric matrices, rank $\mathfrak{so}(N) = \lfloor N/2 \rfloor$	Gantmacher, <i>The Theory of Matrices</i> (1959)
real quadratic homogeneous map $\mathbb{C}^2 \rightarrow \mathbb{R}^3$, Hopf map, shape sphere	R. Montgomery, <i>Amer. Math. Monthly</i> 122 (2015) 299
correspondence of isotropic vectors and spinors, 2:1 covering	É. Cartan, <i>Leçons sur la théorie des spineurs</i> (1938)
three-dimensional directions emerging from relations alone	R. Penrose, “Angular momentum: an approach to combinatorial space-time” (1971)
separation of rotation and internal motion in n -body systems, shape space	R. Littlejohn, M. Reinsch, <i>Rev. Mod. Phys.</i> 69 (1997) 213
geometric quantisation (a finite-dimensional Hilbert space for a compact phase space under integrality and a polarisation). The shortcut “compact hence discrete spectrum” is not the claim of these references; see the correction in Claim 9	Kostant (1970), Souriau (1970), Woodhouse (1992)

In particular Penrose’s spin geometry theorem (1971) reconstructs three-dimensional directions from combinatorial structure without a background manifold, reaching the same conclusion as Claim 2. The contribution of Claim 2 is to reduce that same conclusion to the one line of linear algebra $\dim \text{Sym}^2(\mathbb{R}^2) = 3$, and to give a falsification condition (if the unit is d -dimensional, $d(d+1)/2$).

4B. Self-citations (papers of this series referred to in §1)

All by Noriaki Kihara, Zenodo, 2026. DOIs are Concept DOIs (always pointing to the latest version).

Tag	Title	Concept DOI
[S1]	Basic axiom system of the anonymous equal-amplitude composite wave model	10.5281/zenodo.21315735
[S2]	The geometric identity of the zero-square-sum constraint and scale invariance — projectivisation of the isotropic cone, projective quadric, intrinsic quantumness (expository note)	10.5281/zenodo.21495305
[S3]	How locality is exchanged in the exchange-interference scattering matrix	10.5281/zenodo.21333766

Tag	Title	Concept DOI
[S4]	Reading the metastable states of white, black and grey cats in a closed system	10.5281/zenodo.21353208
[S5]	Discovery of finite-order resonance in repeated exchange scattering	10.5281/zenodo.21421366
[S6]	Linear upper bound on generator rank and three-direction saturation in N -body complete two-body relation waves	10.5281/zenodo.21465898
[S7]	Plane-decomposition readout in N -body fixed-generator systems	10.5281/zenodo.21468959
[S8]	Onset and the three-way classification of outcomes of spontaneous splitting of states in N -body relation-wave closed systems	10.5281/zenodo.21486233
[S9]	The number of waves is the resolution of the system	10.5281/zenodo.21486544
[S10]	Cessation of spontaneous splitting and the emergence of a new orthogonal rotation plane (paper 6)	10.5281/zenodo.21543070
[S11]	Emergence of three-direction space (paper 7)	10.5281/zenodo.21578401
[S12]	Causal separation of the temporal structure of three-direction generation — two-stage seed removal (paper 8)	10.5281/zenodo.21614402
[S13]	The generative structure of fermions (paper 9)	10.5281/zenodo.21766706
[S14]	Geometric rapid expansion is intrinsic to unstable self-consistent closure — causal discrimination of onset modes	10.5281/zenodo.21798854
[S15]	Generation of fermionic structure occurs by induction, autocatalysis and pair correlation — assumptions and consequences of a universal inelastic map taking only waveforms as input	10.5281/zenodo.21808091
[S16]	How inflation ends is how matter creation begins — N -body embedding of a universal interaction without a timing rule, and a through-run of genesis	10.5281/zenodo.21809814
[S17]	Creation of three spatial axes and proper time — re-derivation of readout from the condensate, and the unique unreadable coordinate time	10.5281/zenodo.21816651
[S18]	A periodic table of waves — a hypothesis for particle classification by winding-number address and observation clock (v2)	10.5281/zenodo.21822358

Tag	Title	Concept DOI
[S19]	Two-layer separation of waves and fields — unification of gauge and gravitational fields by a universal field-readout function	10.5281/zenodo.21832256
[S20]	The generation conditions for space, matter and clocks do not coincide — the condition under which a seed produces particles, and the lower bound on resolution	10.5281/zenodo.21874481

This note presupposes the axiom system of [S1] and starts from the geometry of the state space identified in [S2], giving a linear-algebraic ground for the three-direction saturation found numerically in [S6][S7][S11]. Claims 8 and 9 examine the complex convention of [S1] itself, and so stand in the relation of a **proposed revision** to [S1].

5. Figures

All paths in this section and in Appendix A are **relative to the location of this file**.

5.1 Geometry figures (synthetic configurations, not the numerical model)

These four illustrate the geometry stated in Claims 4, 6 and 6B using artificially constructed configurations. **They are not results of the numerical model.**

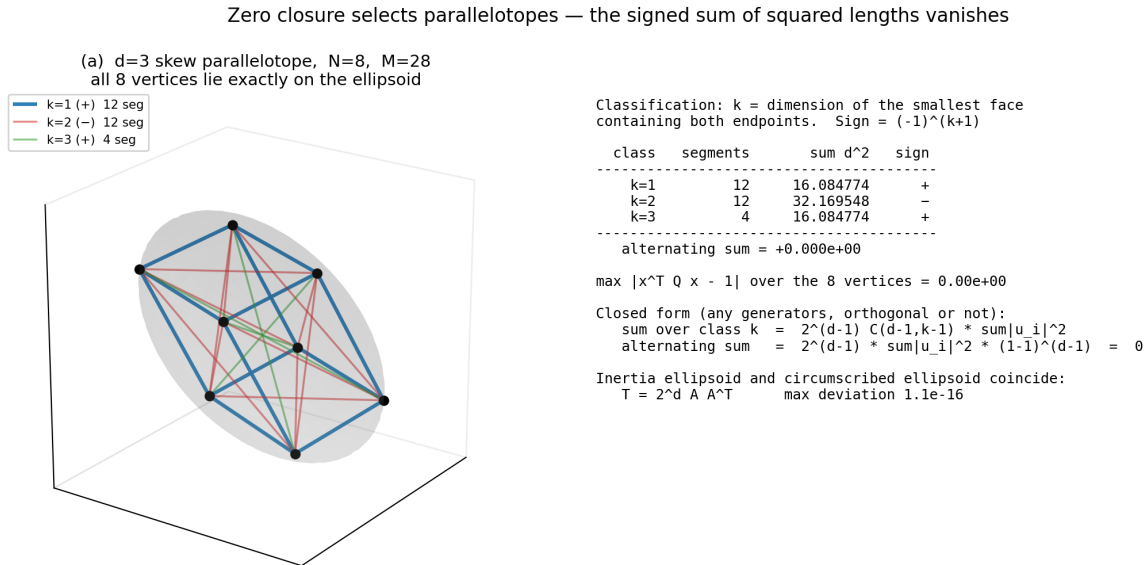


Figure 1: Figure 1

Figure 1 — A three-dimensional parallelotope and the face-dimension classification A skew parallelepiped ($d=3$, $N=2^3=8$, $M=28$). The 28 segments are coloured by the dimension k of the smallest face containing both endpoints: $k=1$ (edges) 12, $k=2$ (face diagonals) 12, $k=3$ (space diagonals) 4. The alternating sum with sign $(-1)^{k+1}$ vanishes exactly.

They must not be classified by length. A skew parallelepiped has 13 distinct values of length², whereas the face-dimension classification gives exactly 3 classes. The two coincide only for highly symmetric figures such as

the cube.

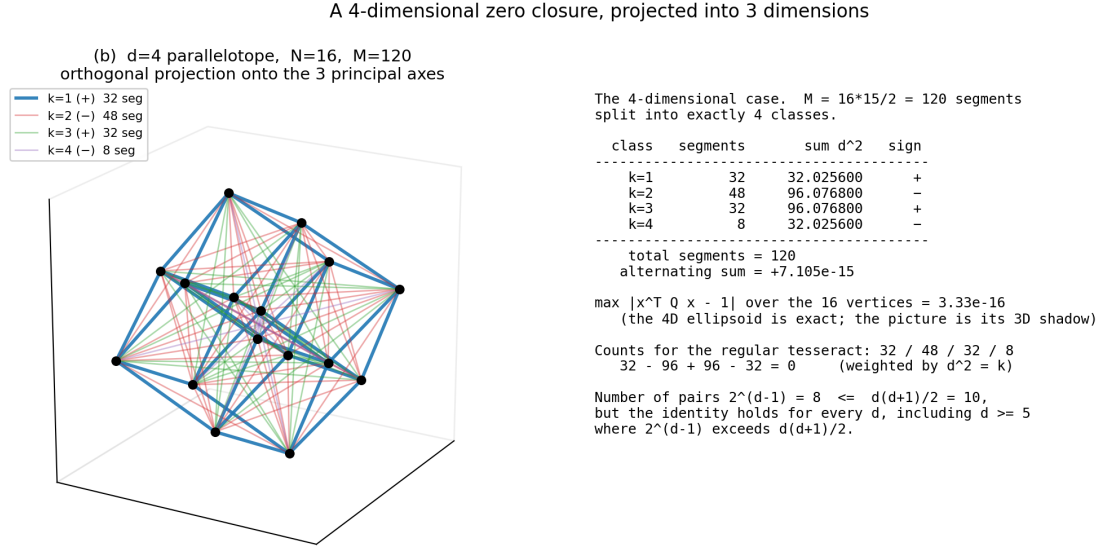


Figure 2: Figure 2

Figure 2 — A four-dimensional parallelotope (projected to three dimensions) $d = 4$, $N = 16$, $M = 120$. The 120 segments split into 4 classes (32/48/32/8). The $\sum d^2$ are 32.03/96.08/96.08/32.03 and the alternating sum is $+7.1 \times 10^{-15}$.

The picture is a shadow in three dimensions. The ellipsoid is exact in four dimensions: $|x^T Q x - 1| \leq 3.3 \times 10^{-16}$ at all 16 vertices. Any apparent departure from an ellipse in the shadow is an effect of the projection.

The number of pairs is $2^{d-1} = 8 \leq d(d+1)/2 = 10$ here, but the identity also holds for $d \geq 5$ where 2^{d-1} exceeds $d(d+1)/2$ (Claim 6B).

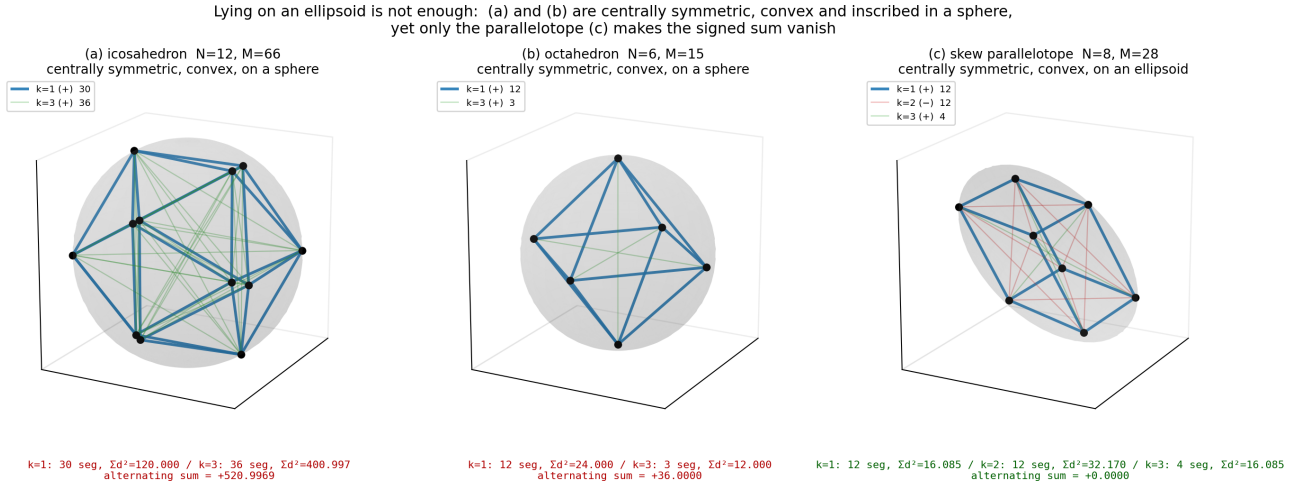


Figure 3: Figure 3

Figure 3 — Counterexamples: lying on an ellipsoid is not enough

(a) icosahedron ($N = 12$), (b) octahedron ($N = 6$), (c) skew parallelepiped ($N = 8$). (a) and (b) are **centrally symmetric, convex, and lie on a sphere**, yet their alternating sums are $+520.997$ and $+36.000$, not

zero. Only the parallelotope (c) gives zero (+0.0000).

What this figure shows is the single point that “lying on an ellipsoid” and “zero closure” are different conditions. Zero closure is strictly stronger.

The ellipsoid of the zero closure: its three semi-axes are A, B, C , and the closure fixes only $A^2+B^2+C^2 = r^2$

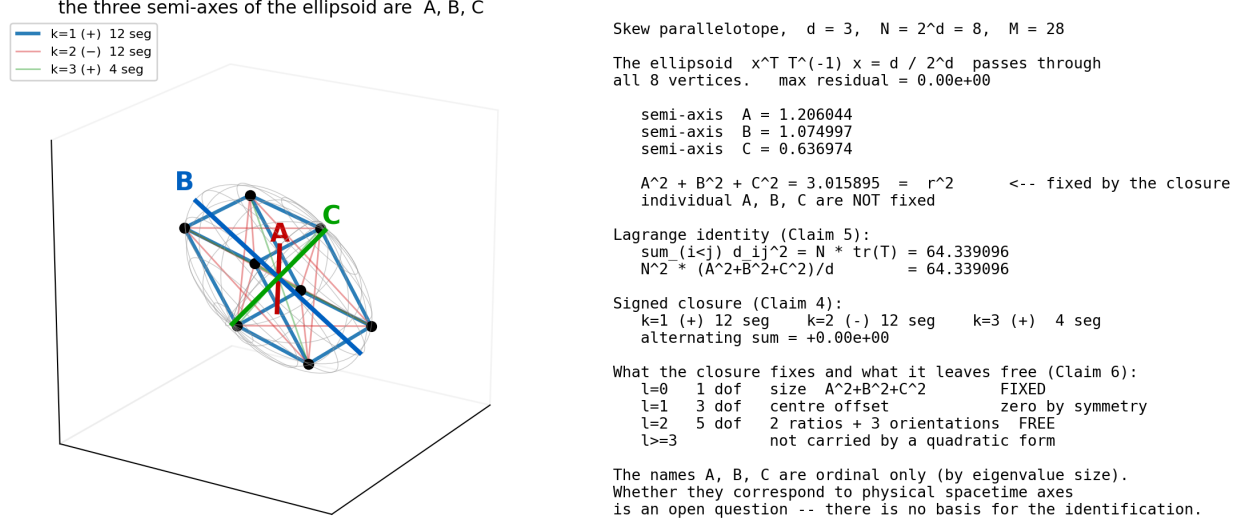


Figure 4: Figure 4

Figure 4 — The zero-closure ellipsoid and what closure fixes The circumscribed ellipsoid of a parallelotope coincides with the **normalised inertia ellipsoid** ($c = d/N$) (Claim 6B, §0 definitions). Write the semi-axes as A, B, C . **Zero closure** $S(D) = 0$ **imposes one condition on the signed distance sum but does not fix the value of $A^2 + B^2 + C^2$** . Adding the fixing of C or a normalisation fixes the unsigned total square $U(D) = 2C$, and correspondingly the trace of the inertia tensor, i.e. $l = 0$ (Claim 5). The individual A, B, C are not determined. In multipoles, one $l = 0$ degree of freedom is fixed (by fixing C or normalisation) and the five $l = 2$ degrees (2 shape + 3 orientation) remain free (Claim 6). **The decomposition of T has only the two pieces $l = 0 \oplus l = 2$; $l = 1$ (the first moment) is not a component of T and is zero by definition in centroid coordinates** (the correction in Claim 6).

The names A, B, C are ordinals by eigenvalue magnitude. Correspondence with physical spacetime is an open task (§0B, Claim 12).

5.2 Numerical model figures (run results)

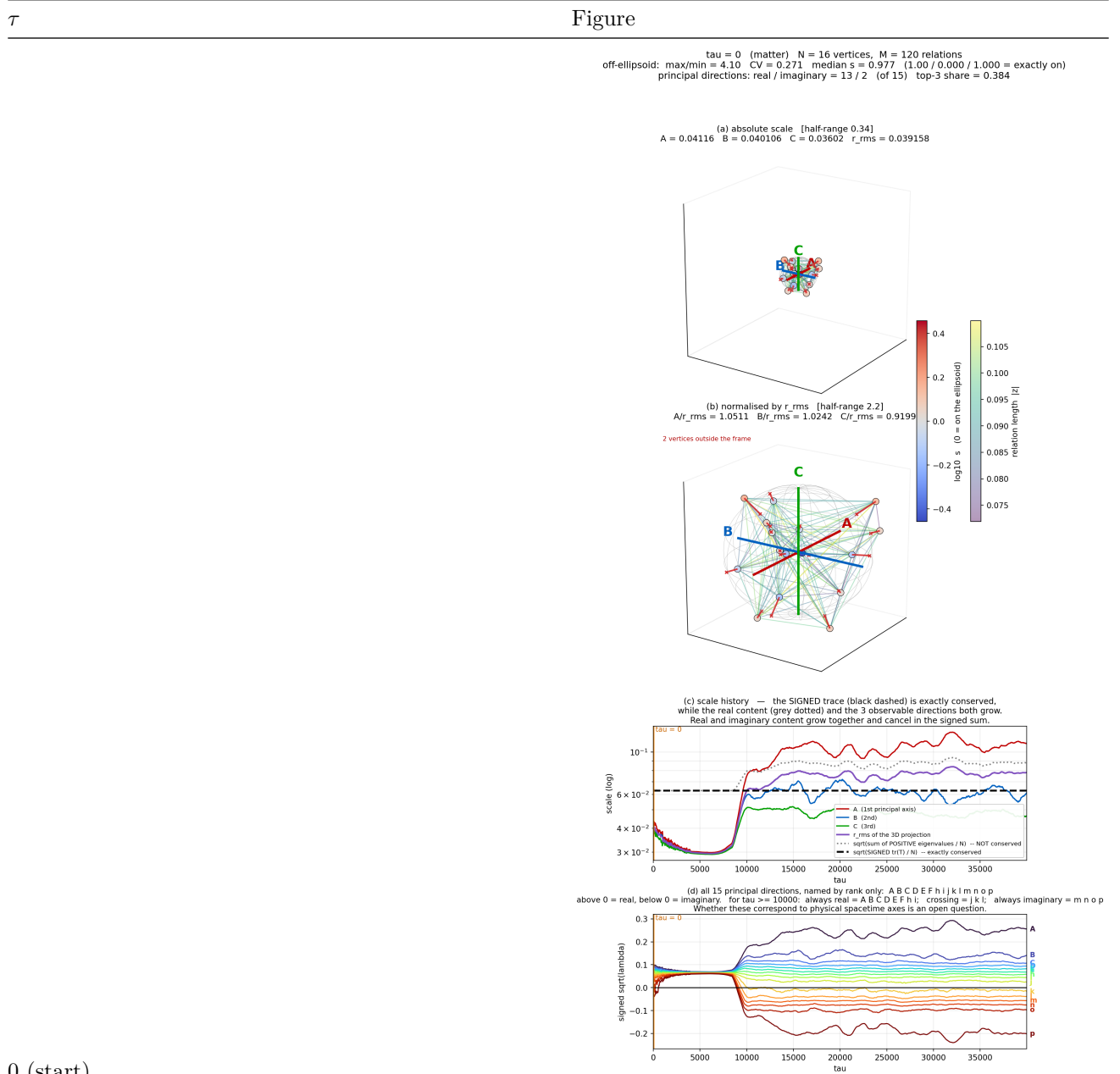
Conditions: mode = electron, $N = 16 = 2^4$, $\delta = 0.1$, $T = 40000$. Matter side ($_m_$) and vacuum control ($_v_$). Each figure has four panels.

Panel	Content
(a)	absolute scale; frame fixed at ± 0.17 . The same frame at every τ , so change of size relative to the frame is change of scale
(b)	normalised by r_{rms} ; frame fixed at ± 1.1 . Compares shape only
(c)	scale history (logarithmic); the current τ marked by a vertical line
(d)	history of all 15 principal axes; above 0 real, below 0 imaginary

On the frame values (correcting an error of v3): the half-range in the figure title is the value of the launch argument `--absmax`; the actual frame is that divided by `--zoom` (default 2.0). v3 wrote $\pm 0.28/\pm 2.2$, but the actual frames were $\pm 0.14/\pm 1.1$. The v4 $N = 16$ figures use `--absmax 0.34` `--zoom 2` (actual frame ± 0.17) so that all vertices fit inside.

Vertex colour shows the deviation s_i (blue inside the ellipsoid, red outside, white on it); segment colour shows the relation length $|z|$.

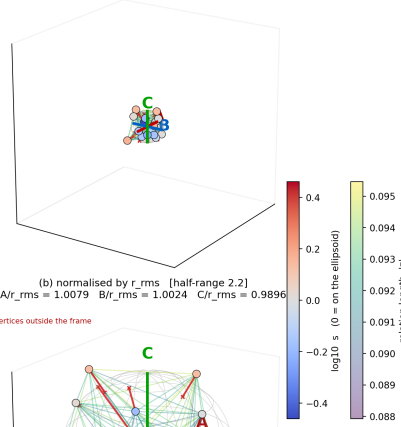
Figures 5–8 — Four time points on the matter side



0 (start)

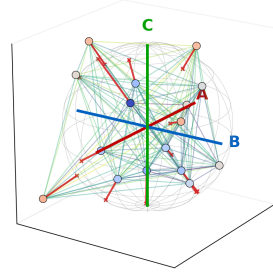
tau = 4000 (matter) N = 16 vertices, M = 120 relations
off-ellipsoid: max/min = 4.25 CV = 0.349 median s = 0.926 (1.00 / 0.000 / 1.000 = exactly on)
principal directions: real / imaginary = 15 / 0 (of 15) top-3 share = 0.227

(a) absolute scale [half-range 0.34]
A = 0.030182 B = 0.030016 C = 0.029635 r_rms = 0.029945

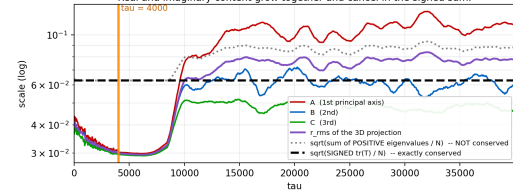


(b) normalised by r_rms [half-range 2.2]
A/r_rms = 1.0079 B/r_rms = 1.0024 C/r_rms = 0.9896

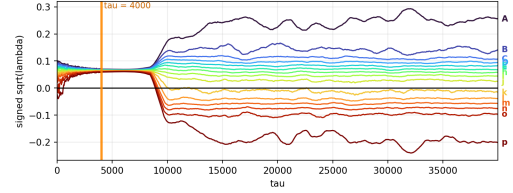
1 vertices outside the frame



(c) scale history — the SIGNED trace (black dashed) is exactly conserved, while the real content (grey dotted) and the 3 observable directions both grow. Real and imaginary content grow together and cancel in the signed sum.



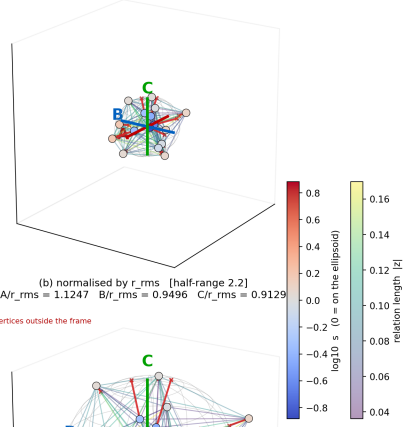
(d) all 15 principal directions, named by rank only: A B C D E F h i j k l m n o p
above 0 = real, below 0 = imaginary for tau >= 10000: always real = A B C D E F h i; crossing = j k l; always imaginary = m n o p
Whether these correspond to physical spacetime axes is an open question.



4000 (pre-transition)

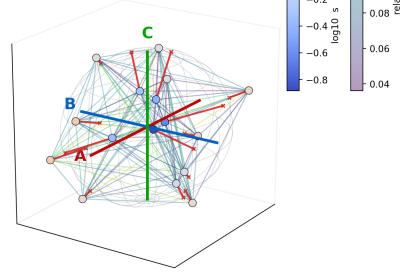
tau = 9487 (matter) N = 16 vertices, M = 120 relations
off-ellipsoid: max/min = 11.74 CV = 0.487 median s = 1.000 (1.00 / 0.000 / 1.000 = exactly on)
principal directions: real / imaginary = 10 / 5 (of 15) top-3 share = 0.579

(a) absolute scale [half-range 0.34]
A = 0.065763 B = 0.055528 C = 0.053378 r_rms = 0.058473

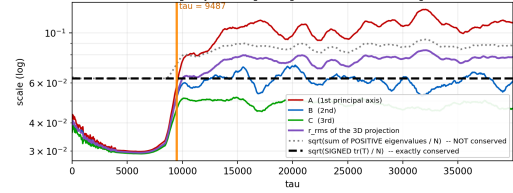


(b) normalised by r_rms [half-range 2.2]
A/r_rms = 1.1247 B/r_rms = 0.9496 C/r_rms = 0.9129

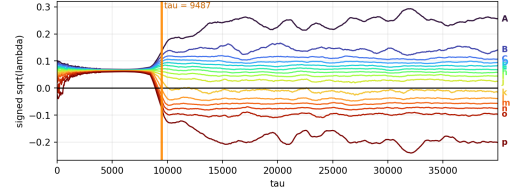
3 vertices outside the frame



(c) scale history — the SIGNED trace (black dashed) is exactly conserved, while the real content (grey dotted) and the 3 observable directions both grow. Real and imaginary content grow together and cancel in the signed sum.



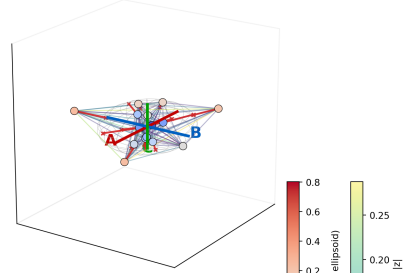
(d) all 15 principal directions, named by rank only: A B C D E F h i j k l m n o p
above 0 = real, below 0 = imaginary. for tau >= 10000: always real = A B C D E F h i; crossing = j k l; always imaginary = m n o p
Whether these correspond to physical spacetime axes is an open question.



9487 (just after transition)

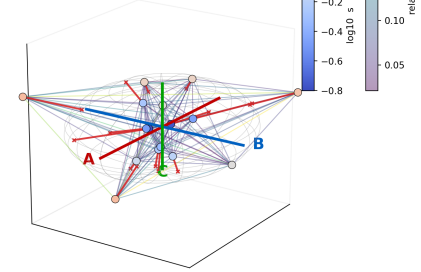
tau = 39991 (matter) N = 16 vertices, M = 120 relations
off-ellipsoid: max/min = 11.16 CV = 0.614 median s = 0.738 (1.00 / 0.000 / 1.000 = exactly on)
principal directions: real / imaginary = 9 / 6 (of 15) top-3 share = 0.812

(a) absolute scale [half-range 0.34]
A = 0.097622 B = 0.087259 C = 0.041955 r_rms = 0.079382

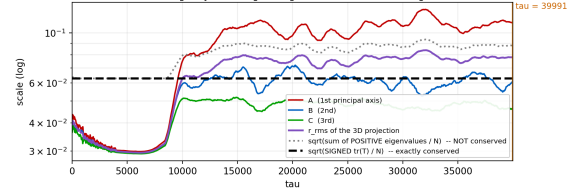


(b) normalised by r_rms [half-range 2.2]
A/r_rms = 1.2298 B/r_rms = 1.0992 C/r_rms = 0.5285

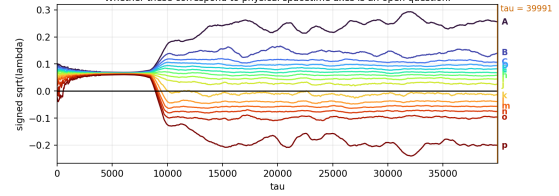
3 vertices outside the frame



(c) scale history — the SIGNED trace (black dashed) is exactly conserved, while the real content (grey dotted) and the 3 observable directions both grow. Real and imaginary content grow together and cancel in the signed sum.



(d) all 15 principal directions, named by rank only: A B C D E F h i j k l m n o p above 0 = real, below 0 = imaginary. for tau >= 10000: always real = A B C D E F h i; crossing = j k l; always imaginary = m n o p. Whether these correspond to physical spacetime axes is an open question.

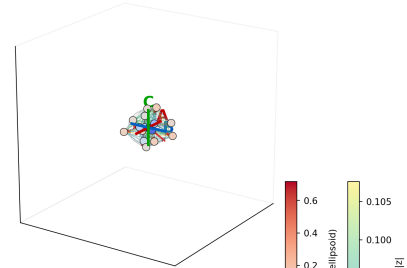


39991 (end)

Figures 9–12 — Four time points of the vacuum control

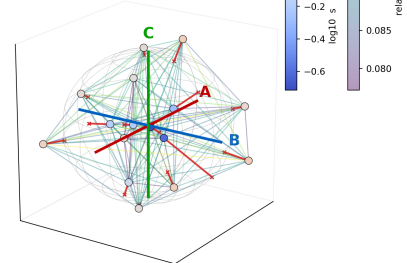
tau = 0 (vacuum control) N = 16 vertices, M = 120 relations
off-ellipsoid: max/min = 6.92 CV = 0.393 median s = 1.082 (1.00 / 0.000 / 1.000 = exactly on)
principal directions: real / imaginary = 15 / 0 (of 15) top-3 share = 0.343

(a) absolute scale [half-range 0.34]
A = 0.039117 B = 0.036972 C = 0.033562 $r_{rms} = 0.036622$

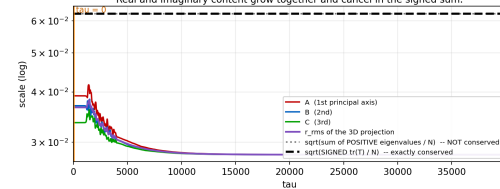


(b) normalised by r_{rms} [half-range 2.2]
A/ r_{rms} = 1.0681 B/ r_{rms} = 1.0096 C/ r_{rms} = 0.9165

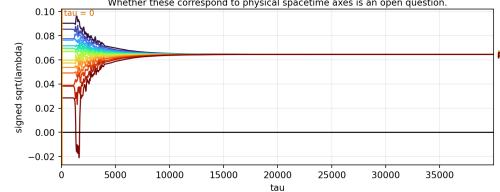
1 vertices outside the frame



(c) scale history — the SIGNED trace (black dashed) is exactly conserved, while the real content (grey dotted) and the 3 observable directions both grow. Real and imaginary content grow together and cancel in the signed sum.

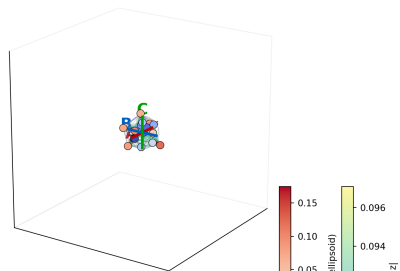


(d) all 15 principal directions, named by rank only: A B C D E F h i j k l m n o p
above 0 = real, below 0 = imaginary. for tau >= 10000: always real = A B C D E F h i j k l m n o p; crossing = -; always imaginary = -
Whether these correspond to physical spacetime axes is an open question.



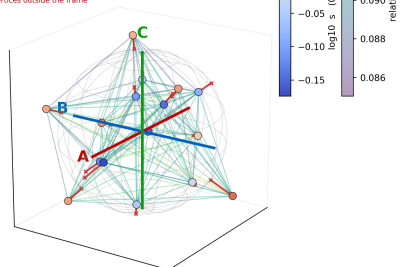
tau = 4000 (vacuum control) N = 16 vertices, M = 120 relations
off-ellipsoid: max/min = 1.98 CV = 0.225 median s = 0.922 (1.00 / 0.000 / 1.000 = exactly on)
principal directions: real / imaginary = 15 / 0 (of 15) top-3 share = 0.240

(a) absolute scale [half-range 0.34]
A = 0.031175 B = 0.030557 C = 0.030059 r_rms = 0.0306

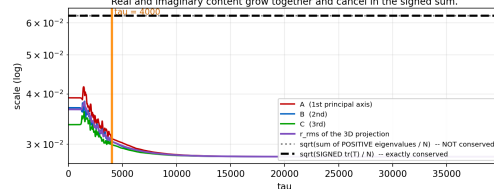


(b) normalised by r_rms [half-range 2.2]
A/r_rms = 1.0188 B/r_rms = 0.9986 C/r_rms = 0.9823

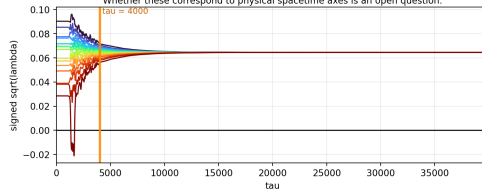
3 vertices outside the frame



(c) scale history — the SIGNED trace (black dashed) is exactly conserved, while the real content (grey dotted) and the 3 observable directions both grow. Real and imaginary content grow together and cancel in the signed sum.



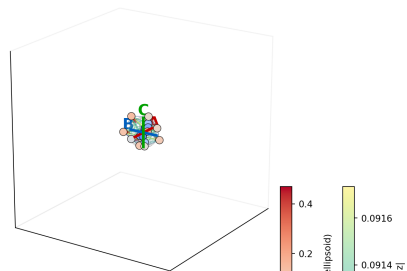
(d) all 15 principal directions, named by rank only: A B C D E F h i j k l m n o p
above 0 = real, below 0 = imaginary. for tau >= 10000: always real = A B C D E F h i j k l m n o p; crossing = -; always imaginary = -
Whether these correspond to physical spacetime axes is an open question.



4000

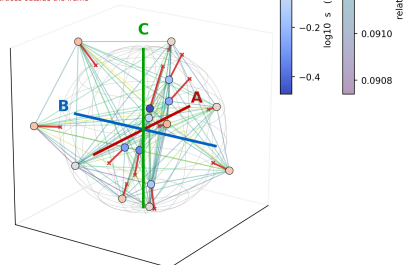
tau = 9487 (vacuum control) N = 16 vertices, M = 120 relations
off-ellipsoid: max/min = 4.31 CV = 0.370 median s = 1.009 (1.00 / 0.000 / 1.000 = exactly on)
principal directions: real / imaginary = 15 / 0 (of 15) top-3 share = 0.204

(a) absolute scale [half-range 0.34]
A = 0.028315 B = 0.028285 C = 0.028171 r_{rms} = 0.028257

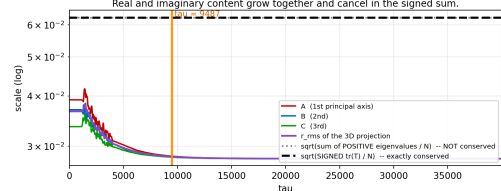


(b) normalised by r_{rms} [half-range 2.2]
A/ r_{rms} = 1.0020 B/ r_{rms} = 1.0010 C/ r_{rms} = 0.9969

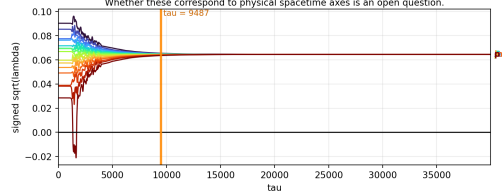
2 vertices outside the frame



(c) scale history — the SIGNED trace (black dashed) is exactly conserved, while the real content (grey dotted) and the 3 observable directions both grow. Real and imaginary content grow together and cancel in the signed sum.

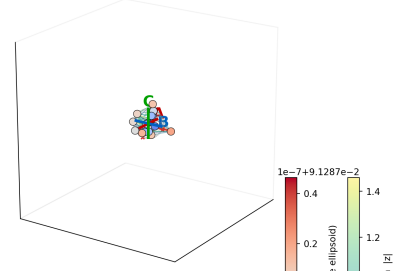


(d) all 15 principal directions, named by rank only: A B C D E F h i j k l m n o p
above 0 = real, below 0 = imaginary. for tau >= 10000: always real = A B C D E F h i j k l m n o p; crossing = -; always imaginary = -
Whether these correspond to physical spacetime axes is an open question.



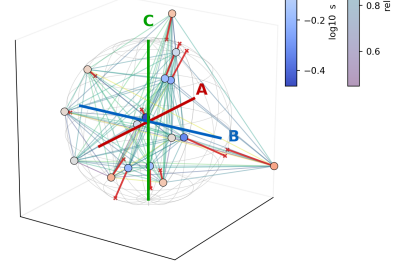
tau = 39991 (vacuum control) N = 16 vertices, M = 120 relations
off-ellipsoid: max/min = 4.61 CV = 0.378 median s = 0.945 (1.00 / 0.000 / 1.000 = exactly on)
principal directions: real / imaginary = 15 / 0 (of 15) top-3 share = 0.200

(a) absolute scale [half-range 0.34]
A = 0.027951 B = 0.027951 C = 0.027951 $r_{rms} = 0.027951$

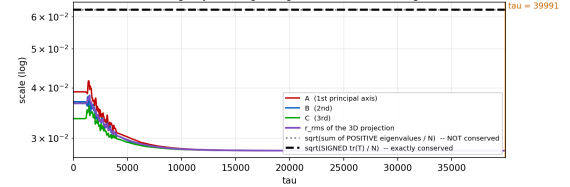


(b) normalised by r_{rms} [half-range 2.2]
A/ r_{rms} = 1.0000 B/ r_{rms} = 1.0000 C/ r_{rms} = 1.0000

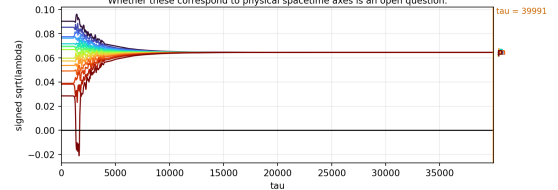
3 vertices outside the frame



(c) scale history — the SIGNED trace (black dashed) is exactly conserved, while the real content (grey dotted) and the 3 observable directions both grow. Real and imaginary content grow together and cancel in the signed sum.



(d) all 15 principal directions, named by rank only: A B C D E F h i j k l m n o p
above 0 = real, below 0 = imaginary. for tau >= 10000: always real = A B C D E F h j k l m n o p; crossing = -; always imaginary = -
Whether these correspond to physical spacetime axes is an open question.



39991

5.3 How to read the figures, part 1 — preventing misreadings of the inflation

This section is required reading. Figures 5–8 invite two opposite misreadings. Both are wrong.

Misreading 1: “A conservation law holds, so no expansion is occurring”

The **black dashed line** in panel (c) is perfectly horizontal over all τ . Reading this as “nothing is expanding” is wrong.

The black dashed line represents the **signed** trace $\text{tr}(B) = \sum_a \lambda_a$ (B the double-centred Gram matrix, not the inertia tensor T ; see the notational caution in Claim 5). By Lagrange’s identity $\sum_{i < j} d_{ij}^2 = N \text{tr}(B)$ this is the power of the relation quantities itself. In measurement, the minimum and maximum over the 968 late points agree to 8 digits (relative width 8.3×10^{-9}). **This is conservation within numerical precision, not an analytic conservation law derived from the update rule** (the classification of Claim 13).

But **conservation of the signed sum does not mean that each direction is still.**

The **grey dotted line** in the same panel (c) is the sum of the positive eigenvalues alone. It is not horizontal; it jumps by 1.90 at the transition. The absolute sum $\sum_a |\lambda_a|$ grows by 2.79. The real total grows, the imaginary total appears from near zero, and their **difference** is conserved (Claim 13).

The exponential amplification is real. It is the red, blue and green lines in panel (c) shooting up vertically at $\tau \approx 9000$. **At $N = 12$ this jump was at $\tau \approx 2500$. The transition time depends on the resolution.**

Misreading 2: “ A grew by 2.8, so the whole system grew by 2.8”

Also wrong. The signed trace does not move.

The three observed directions expand by the **product of two effects**:

1. the absolute sum grows by 2.79;
2. the top-3 occupancy rises from 0.384 to 0.812 (concentration).

Neither alone explains it.

Misreading 3: “The extra dimensions compactified”

Also wrong. m, n, o, p did not curl up small and vanish from observation.

In panel (d), p **passes through** zero at $\tau \approx 9000$ and continues down to -0.19 , **increasing in absolute value**. The ratio across the transition is 5.568. Likewise o by 2.624, n by 1.674, m by 1.123. **All four directions moving to the imaginary side increase in absolute value.** What shrinks are the alternating pair j (0.477) and k (0.380).

So what is happening is not “extra dimensions shrank” but **“three directions expanded and seven changed sign (three of them alternating)”**. The mechanism is different.

Misreading 4: “There are 15 principal axes, so this system is a 15-dimensional parallelotope”

Also wrong. The number of principal axes is $N - 1 = 15$, while the parallelotope dimension required by zero closure is $d = 4$ (from $N = 2^4$); they are different (Claim 16-d). Had the system reached a parallelotope, the rank would have to fall to 4, but the measured rank is 8–11. It has not.

Key points for reading the figures

- Panel (a) has the frame fixed at ± 0.17 for all τ . Placing the $\tau = 4000$ figure next to the $\tau = 39991$ figure shows directly that the ellipsoid itself has grown relative to the frame.
- Panel (b) is divided by r_{rms} , so size information is removed and only shape remains. At $\tau = 39991$, $A/r_{\text{rms}} = 1.230$ and $C/r_{\text{rms}} = 0.529$: it has become oblate.
- The vertical line in panel (c) is the current τ . Each figure’s time point can be located there. **Do not judge the time series from the absolute scale of panel (a) alone.**
- The letters at the right edge of panel (d) are the names of the 15 principal axes.

Comparison with the vacuum control

In Figures 9–12 (vacuum), **both** the black dashed and the grey dotted lines in panel (c) are horizontal. No transition occurs, so the absolute sum does not grow. The 15 lines in panel (d) all stay positive and converge at $\tau = 39991$ to 0.0645, fully isotropic.

The expansion and the fall into imaginary directions visible in Figures 5–8 are phenomena accompanying the creation of matter, not properties the system had from the outset. That is the conclusion of the control experiment.

5.4 How to read the figures, part 2 — what are the 15 principal axes $A, B, C, D, E, F, h, i, j, k, l, m, n, o, p$?

Why 15

A system of resolution $N = 16$ has $M = N(N - 1)/2 = 120$ relations. Reading a length d_{ij} from each and forming the Gram matrix by double centring

$$B = -\frac{1}{2} J D^{\circ 2} J, \quad J = I - \frac{1}{N} \mathbf{1}\mathbf{1}^T$$

gives $B\mathbf{1} = 0$ exactly. That trivial zero is not a principal axis, so there are $N - 1 = 15$ principal axes.

The number 15 is $N - 1$ arising from the resolution $N = 16$, not a number chosen from outside. Changing N changes it (at $N = 12$ it was 11). In this respect it differs in character from dimension counts such as the 10 or 11 of superstring theory, which are determined uniquely by consistency of the theory.

And there is a ground for the choice of N . If zero closure is realised as a parallelotope family then $N = 2^d$ (Claim 16), and this note takes $N = 16 = 2^4$. Hence the number of principal axes is $2^d - 1 = 15$. Do not confuse d with $2^d - 1$ (Claim 16-d).

What the 15 do

Group	Axes	ratio across the transition	real / imaginary	Role
A, B, C	1st–3rd	2.80 / 1.73 / 1.42	real	expand; appear in the three-dimensional projection; used for the figures nearly unchanged; real but absent from the projection cross the real/imaginary boundary; the cause of the imaginary count fluctuating between 4 and 7 fall to the imaginary side, but their absolute values grow
D, E, F, h, i	4th–8th	1.27 / 1.14 / 1.01 / 0.87 / 0.71	real	
j, k, l	9th–11th	0.48 / 0.38 / 0.73	alternating	
m, n, o, p	12th–15th	1.12 / 1.67 / 2.62 / 5.57	imaginary	

The real axes are the 8 from A to i and the imaginary ones the 4 m, n, o, p , definitively; the 3 axes j, k, l cross back and forth. “Six imaginary” is the most frequent (616 of 968 points).

What it means that the names are ordinals

The names A through p merely express the **rank order of eigenvalues at that instant**. Note the following.

1. **The names are not physical quantities.** No correspondence such as “ A is the time axis” or “ B is R ” is claimed at all.
2. **The same name does not necessarily keep pointing in the same direction.** At times of eigenvalue degeneracy the order swaps. The measured minimum gap at $N = 16$ is 3.4×10^{-6} – 5.1×10^{-5} , an order of magnitude smaller than at $N = 12$. Degeneracy really occurs.

3. **The principal-axis vectors are not continuous in time.** The absolute inner product of the first principal-axis vectors is only 0.0450 between $\tau = 4000$ and $\tau = 39991$, and 0.0233 between $\tau = 9487$ and $\tau = 39991$. The A just after the transition and the A at the end are nearly orthogonal, different directions.

Hence the 15 lines in panel (d) are not “the time evolution of 15 physical axes”. They are the series of values ranked by eigenvalue magnitude at each instant. This distinction matters.

Identification with physics is an open task

We obtained a structure in which 3 of 15 directions expand rapidly, 5 stay nearly unchanged, and 7 move to imaginary (3 of them alternating). However,

whether these can be identified with physical spacetime (1 time axis + 3 space axes, or R and Q), cannot be, or require an intervening map, is an open task. There is at present no ground for identification.

Likewise,

how the spectral structure of these 15 principal axes relates to superstring theory and other existing higher-dimensional theories is also an open task.

Up to v3 this note took $N = 12$, so there were 11 principal axes, numerically matching the 11 dimensions of superstring theory. That was an accident of the choice of resolution. At $N = 16$ it becomes 15. As stated in misreading 3 of §5.3, the mechanism also differs (this is not compactification), and so does the provenance ($N - 1$ here, consistency there). **A coincidence of words is not a correspondence of theories.**

5.5 Figure 13 — the quasi-oscillation and its period (Claim 15)

The derivation program is `__v1//_N1to20__v2/make_period_figure_v1.py`. The measured values themselves are stored in `__v1//_N1to20__v2/figures_tau/period_series_electron_T40000_d0.1_rep-dump40k16d_N16_v1.npz`.

This figure comes from the dedicated run with the dense region widened to $\tau < 10000$ (`...rep-dump40k16d_N16`) and includes the transition ($\tau \approx 9000$). It shares conditions and random seed with the ellipsoid-figure run `...rep-dump40k16_N16`, differing only in the dense-region setting.

Panel	Content
(a)	the observable itself: the mean of the 120 relation lengths, dense region only (every step recorded)
(b)	the residual after subtracting a 201-point moving average — the quasi-oscillation itself
(c)	τ dependence of the dominant period: autocorrelation (circles, solid), peak spacing (squares, dashed) and anti-period (triangles, dotted) overlaid
(d)	the autocorrelations themselves; four matter-side windows

How to read this figure

- (a): the matter side (red) is flat until $\tau \approx 8900$ and then **falls sharply**. That is the transition. The vacuum side (blue) stays flat throughout. At $N = 12$ the same drop was at $\tau \approx 2300$.
- (b): **the two sides use different axes**. The matter side (left axis) has almost zero amplitude until the transition, then rises abruptly from $\tau \approx 8900$. The vacuum side (right axis, order 10^{-5}) shows one clean oscillation at $\tau \approx 1200$ – 4000 and then decays. **Before the transition there is an interval where the matter-side oscillation is in fact smaller than the vacuum side** ($\tau \in [3968, 8927]$, item 7 of Claim 15). This panel is the basis for judging the peak spacings there to be noise.

Quasi-oscillation and its period [electron_T40000_d0.1_rep-dump40k16d_N16]
measured only where every step is recorded ($\tau < 10000$); beyond that the record stride is 31 and $124/31 = 4.000$, so the period cannot be measured there

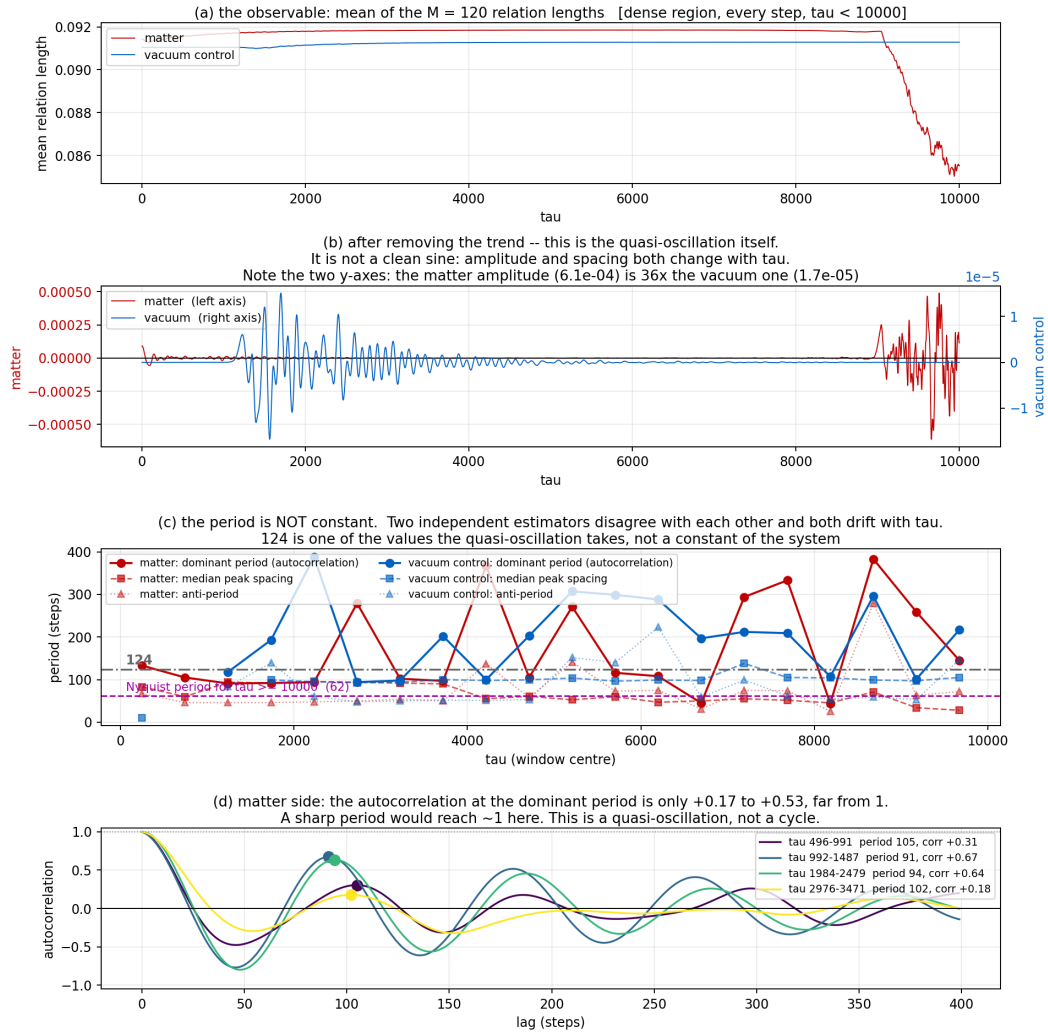


Figure 5: Figure 13

- (c): **this panel is the core of Claim 15.** The grey dash-dotted line is 124. Both estimation curves merely cross it and do not stay on it. **That the two estimators disagree with each other, and that both drift with τ , is the evidence that no single sharp period exists.** The purple dashed line is the Nyquist period 62 for $\tau \geq 10000$; below that line is unmeasurable in the late period. **The post-transition peak spacings $34.0 \rightarrow 28.0$ lie below this line**, so they could not in principle be captured by the coarsely recorded run.
- (d): **for a sharp period the autocorrelation at the dominant lag should reach 1.** The $N = 16$ measurements stay at $+0.134$ – $+0.673$ (the circles on each curve). This is the ground for judging it a “quasi-oscillation” rather than a “cycle”. The shapes of the curves also differ per window.

Misreadings this figure forbids

“There is a period of 124 steps” — not accurate. 124 is one value the quasi-oscillation takes, not a constant of the system. Note that the two estimation curves in panel (c) do not sit on the 124 line.

“There is a period, so $U^n = I$ is confirmed” — not confirmed. If it came from $U^n = I$, the period would be a constant determined by n . The measurements vary. Whether it comes from $U^n = I$ or from a different phenomenon is unsettled.

6. What this note has settled

The answers to the two questions raised at the start are as follows.

Question 1: Is representing waves by complex numbers indispensable?

It is not (Claim 9). What complex numbers add to the real indefinite form is the single equation $\sum q_n p_n = 0$ (Claim 8). **This must not be called “a restatement of scale symmetry”**, which would be to return to the generator argument retracted in Claim 8.

So the complex representation is not indispensable. But the phrasing “a convenient device saving one axiom” is not accurate: reading A of axiom 0.5 can be removed from the independent axioms because of the **homogeneity of the zero-closure equation**, not because of the complex representation (for a homogeneous equation the solution set is a cone whether complex or not). **What the complex representation adds to the real indefinite form is the orthogonality condition $\sum q_n p_n = 0$.**

Question 2: What form does central projection take in the complex representation?

The centre of the central projection is not a vertex and is unobservable from the subjective space. The imaginary symbol attaches only to the direction of the centre, and all segment lengths remain real (Claim 7).

And since assumption (S) holds in the central-projection formulation (Claim 7), the structure “the many-body problem reduces to a single scalar equality constraint” is preserved (Claim 0). In the real case, $\sum x_n^2 = R^2$ represented an arbitrary point P on the spherical shell of radius R , and the constraint reduced to the single $X^T F(X) = 0$. In the complex case it becomes

$$x^2 + y^2 + z^2 - t^2 = R^2 + Q^2$$

which in four variables has signature $(3, 1)$ — a one-sheeted hyperboloid, not closed. **Fixing the conserved quantity $C = t^2 + R^2 + Q^2$, and adding (U: the state lies in a three-dimensional subspace) and (R: the readout is linear and regular, $X = \Lambda u$), gives a closed surface (an ellipsoid) defined by $G = (\Lambda^{-1})^T \Lambda^{-1}$ (Claim 0-c).** **Fixing t alone does not close the surface, since C changes if R, Q move.**

The reduction itself holds with (S) alone and needs neither (U) nor (R). (U) and (R) are needed to relate it to observation coordinates.

Two things must not be confused, however. First, a conserved quantity does not determine the motion; it determines only **the space of permitted motions**. Second, **that this ellipsoid is the inertia ellipsoid itself is unproved**: that requires showing $G^{-1} \propto T$, which cannot be done while the readout map Λ is not given (Claim 2B). **This is the largest underived part of this note.**

What is needed for the right-hand side to be conserved is $\dot{C} = 0$, and in the component layer of this model that can be expected only for a steady solution. In transient and metastable states it deviates and may oscillate (Claims 0-d, 10, 15). The layer distinction is also needed: those deviations are at the component layer, while the readout layer has a quantity conserved through the transition (Claims 18-c, 18-d).

As configurations satisfying zero closure in this form, **parallelotopes** exist (Claim 4-b, valid for all d), with vertex number $N = 2^d$. **That zero-closing configurations are limited to parallelotopes is unproved for $d \geq 3$** (Claim 4-e). That centrally symmetric and convex is not enough was shown by counterexamples (4-d). A parallelotope is always inscribed in an ellipsoid, and its circumscribed ellipsoid coincides with the **normalised inertia ellipsoid** ($c = d/N$) (Claim 6B). The second-order information of the configuration is exhausted by the inertia ellipsoid and saturates at multipole $l \leq 2$ (Claim 6).

That is, as an **exactly constructible family satisfying central projection and zero closure simultaneously**, there are **configurations built as the Minkowski sum of d segments, inscribed in a single ellipsoid, whose ellipsoid information saturates at $l \leq 2$** . Whether it is the only such family is unproved (Claim 4-e).

Question 3, added in v3: does the numerical model actually become that configuration?

It does not.

- **At the component layer**, zero closure is an identity holding only in the steady state and is violated in transient and metastable states. The deviation from the ellipsoid does not decrease monotonically in τ and stays around $\max/\min = 12$ for 36,000 steps after the transition (Claim 10).
- This deviation depends on the seed strength; the $\delta = 0.1$ of this run is large. **Whether the deviation decreases with a weaker seed or a longer τ is unverified** (Claim 11).
- The configuration does not reach a parallelotope. The rank falls from 15 to 8–11 but not to $d = 4$ (§3B).
- The double-centring readout has $N - 1 = 15$ non-trivial principal axes, and spectral concentration into the top three A, B, C magnifies the observed scale. D - i are nearly unchanged, m, n, o, p move to imaginary, and j, k, l alternate. **That this is not a compactification of the “extra directions shrink and become invisible” type is established, but identification with physical spacetime is an open task with no supporting ground** (Claim 12).
- If zero closure is realised as a parallelotope family, the resolution becomes $N = 2^d$; this note takes $N = 16$ ($d = 4$). This is not a sufficient condition (rank = d alone is also insufficient), and it is not derived that it is necessary for all zero-closing solutions (Claim 16).
- What selects parallelotopes is not the form of the equation $\sum x_n^2 = 0$ but the face-dimension sign rule. If both signs and configuration may be chosen freely, non-trivial solutions exist from $N \geq 3$ (Claim 17).
- What is conserved is the signed trace; the real and imaginary totals both grow and cancel (Claim 13).
- Imaginary directions do not exist in the vacuum and appear with the creation of matter (Claim 14).
- **The orientation of the principal axes is not conserved.** The top-3 subspace loses correlation to the random level in about 2000 steps and does not stabilise over a 36,000-step sweep. But **orientation-independent conserved quantities (quadratic forms) do exist**: at least the two kinds $\sum_e |x_e|^2$ and $\sum_e x_e^2$, both of trace type and hence unaffected by the diffusion of orientation. In particular $\sum_e x_e^2$ is **invariant as a complex number through the transition** (Claim 18).
- Quasi-oscillations of order 10^2 steps exist, but the period varies with τ . At $N = 16$ the pre-transition peak spacing 90–95.5 shortens to 34.0 in the transition window and 28.0 just after, while the vacuum side stays at 93.5–138 and does not shorten. **The $N = 12$ finding that the transition shortens the period was reproduced at $N = 16$. Whether this is due to the second axiom $U^n = I$ or a different phenomenon is an open task** (Claim 15).

Question 4, added in v4: what are the title’s “four dimensions”?

They are the **four dimensions** with coordinates (r, t, R, Q) (Claim 2B). And since zero closure gives

$$r^2 = t^2 + R^2 + Q^2,$$

there is a **light-cone-type null cone** inside that four-dimensional space. **That is the meaning of the title.**

x, y, z belong to the subsequent three-dimensional readout and are not needed for the four-dimension claim. Relationality gives only the length r , and x, y, z are points on the surface S_r^2 that r permits (Claim 2B). The degrees of freedom on the null cone are 3; projectivised, 2.

This is a different object from the parallelotope dimension $d = 4$ ($N = 16$, Claim 16). The numerical coincidence is accidental and neither follows from the other.

Question 5, added in v4: does the orientation stabilise?

It does not (Claim 18-a). But **what matters is not the orientation but whether orientation-independent conserved quantities (quadratic forms) exist.** They do (Claim 18-b): at least the Hermitian $\sum |x_e|^2$ and the bilinear $\sum x_e^2$, the latter invariant as a complex number through the transition. Both are of trace type, so their values do not change as the orientation diffuses. **However, conservation holds only at the readout layer, not at the component layer (Claim 18-d).**

This result organises the two unresolved items — the readout map from r (Claim 2B) and the identification of $A, B, \dots, p$ with physics (Claim 12) — as **one and the same problem. Within this run**, identification with a fixed frame is not supported, and if anything can be identified it is orientation-independent quantities (Claim 18-f).

A distance remains unbridged between the geometric statements (Claims 1–9) and the measurements on the numerical model (Claims 10–18). This note measured that distance and identified where it is unbridged. The work needed to bridge it narrows to the following five items.

1. **Establish assumption (S) on the model side, or show that it does not hold** (Claim 0-b). If the supports of the real and imaginary parts separate, the orthogonality condition $\sum q_n p_n = 0$ becomes an identity and the independent constraints drop from two to **one. If this goes through, the central claim of this note — reducing the many-body problem to a single quadratic equality constraint — is completed on the model side as well. This is the top priority.**
2. **Establish assumption (Γ : geometric sign correspondence), or show that it does not hold** (Claim 5, §3B): whether the partition $\mathcal{I}_\pm$ coincides with the partition by the face-dimension sign s_e . **Until this goes through, complex zero closure and the signed closure $S(D) = 0$ of Claim 4 are not connected.**
3. **Show the existence of the three-dimensional state subspace U , give the readout map Λ , and show $G^{-1} \propto T$** (Claim 0-c). All three are underived. If all go through, “the readout surface of complex zero closure = the inertia ellipsoid” connects in one line and Claim 0 links directly to Claim 6B.
4. **Derive from the update rule an identity corresponding to $X^\top F(X) = \frac{1}{2} \dot{C}$** (Claims 13, 18-b). The present conservation is an observation within numerical precision, not an analytic conservation law.
5. **Prove “zero closure $\Rightarrow$ parallelotope” for $d \geq 3$, or produce a counterexample** (Claim 4-e). Until this is settled, $N = 2^d$ remains a conditional statement.

Appendix A. List of programs used in this note

All are stored in the repository. **Paths are relative to the location of this file, written without abbreviation.** Hashes are the first 8 hex digits of the MD5.

There are two replicas.

- `__N16_v1/` — verified replica, copied from the originals with matching md5, **not modified by a single line.**
- `__v1/` — dump-modified version. **This is what was actually run.**

Since the dependencies use `HERE.parent` and `PAPER8.parent.parent`, the replicas preserve the repository-relative nesting. The internal relative structure is identical in both, and the paths below are those of the `__v1/` side used for the runs.

A.1 Programs newly written for this note (7)

File	Hash	Role	Products
__v1//_N1to20__v2/make_scale_series_v1.py	da16b80c	extracts and caches, from the dump, the principal-axis spectrum, rank, imaginary-direction count, signed trace, and positive/negative parts at all τ	__v1//_N1to20__v2/scale_series_*.n
__v1//_N1to20__v2/make_ellipsoid_figure_v1.py	e21156478	the four-panel figure at a given τ (ellipsoid, normalised, scale history, all $N - 1$ principal-axis histories). Derivation of Claims 10, 12, 13, 14. Extended in v4 to 15 axis names $(\dots l, m, n, o, p)$ (the v3 hash was da11c229, which had names only up to 11 and drew the 12th onward by number)	Figures 5–12
__v1//_N1to20__v2/make_period_figure_v1.py	5a124d62	measures the quasi-oscillation period by both autocorrelation and peak spacing and plots it. Derivation of Claim 15. Two fixes in v4 (v3 hash a384c970): (1) the dense-region end <code>--dense-end</code> was fixed at 4000, a trap by which even a widened dense run measured only $\tau < 4000$; the default is now read from <code>dump_tauc</code> in <code>dump_meta</code> . (2) the title of panel (a) was fixed at $M = 66$, wrong for $N = 16$ ($M = 120$); it is now read from <code>m</code> in the meta	Figure 13, __v1//_N1to20__v2/figures_tau/per
__v1//_N1to20__v2/check_invariants_v1.py	a978342	measures the stability of principal-axis orientation (subspace overlap) and the conserved quadratic invariants (Hermitian, bilinear, $\text{tr}(B^k)$, component layer). Derivation of Claim 18	figures_tau/invariants_*.json
figures_v1/make_figs.py	7fc0ac70	face-dimension classification of parallelotopes ($d = 3$ and $d = 4$). Derivation of Claim 4	Figures 1, 2

File	Hash	Role	Products
figures_v1/make_fig3.py	ff721504	contrast of centrally symmetric convex counterexamples (icosahedron, octahedron) with a parallelotope. Derivation of Claim 4-d	Figure 3
figures_v1/make_fig4.py	b940a9af	the zero-closure ellipsoid, semi-axes A, B, C , and the degrees of freedom closure fixes. Derivation of Claims 5, 6, 6B	Figure 4
figures_v1/check_sufficiency_v1.py	eb59b4f	that $N = 2^d$ and rank $= d$ are insufficient singly and jointly, the trivialisation of the test at rank $= N - 1$, and the degeneracy at $N = d + 1$ (simplex). Derivation of Claims 16-b, 16-c	check_sufficiency_v1.json
figures_v1/check_real_solutions_v1.py	4549259	that sign choice alone cannot zero a fixed distance set and that it is solvable when both signs and configuration are free (test B), $\sum_{\text{edges}} d^2 = \sum_{\text{main diagonals}} d^2$ (test D), the real/imaginary breakdown by face-dimension class (test E), and class recovery from lengths alone (test F). Derivation of Claims 4-f, 4-g, 17	check_real_solutions_v1.json

A.2 Run programs — modified (2)

The originals (on the `__N16_v1/` side) are not rewritten by a single line. Modifications were made only on the `__v1/` side and are marked in place. The hashes are those before modification (the originals).

File (<code>__v1//_N1to20__v2/</code>)	hash of R	Modification
run_nsweep_three_series_v2.py	3e8e06c8	added a two-stage sampling dump: every step up to <code>DUMP_TAUC</code> (default 4000), thereafter every <code>DUMP_STRIDE</code> (default 31), writing C_2 to a memmap. The correspondence table <code>dump_taus</code> is saved in the meta. The physics, recorded items, decision logic and figure generation are untouched
run_tb_nsweep_1to20_v1.py	bfa5d854	added a path saving the final-frame diagnostic dictionary as per-key arrays

A.3 Run programs — unmodified (22)

Byte-identical between `__N16_v1/` and `__v1/`. These are what determine the physics.

Core (unified universal functions)

File (<code>__v1//_v1/</code>)	Hash
<code>unified_interaction_v1.py</code>	<code>bf45d4c7</code>
<code>unified_interaction_v2.py</code>	<code>728e79a7</code>
<code>unified_readout_v3.py</code>	<code>bea700fe</code>
<code>unified_dimension_v1.py</code>	<code>1416d9ad</code>
<code>selection_v1.py</code>	<code>5c23ced4</code>

Universal inelastic map, many-body connection, parent white wave

File	Hash
<code>__v1//_managed_v1/universal_inelastic_map_v1.py</code>	<code>7c07a52e</code>
<code>__v1//_managed_v1/universal_inelastic_map_v3.py</code>	<code>bfb15801</code>
<code>__v1//_managed_v1/run_ignition_fate_exact_v3.py</code>	<code>a95f3327</code>
<code>__v1//_v1/run_stage2_vertex_engine_v1.py</code>	<code>55f88e26</code>
<code>__v1//_v1/run_stage3_shared0_v2_and_hair_v1.py</code>	<code>75bc31db</code>
<code>__v1//make_parent_white_managed_v1/make_parent_white_monics_n_only_v3.py</code>	<code>323e0457</code>

Run environments of earlier papers (loaded as dependencies)

File	Hash
<code>__v1//8_seed/code/run_preliminary_seed_ablation_v1.py</code>	<code>45c4b42a</code>
<code>__v1//9_/___v1/20260713/run_exchange_scattering_simulation_fermionic_localization_transfer_preliminary_v1.py</code>	<code>3ab01fd1</code>
<code>__v1//9_/___v1/20260715/run_system_A_localization_exchange_R_sweep_preliminary_v1.py</code>	<code>941e7b21</code>
<code>__v1//9_/___v1/ab_invariant_theta_toy_v1/run_ab_invariant_theta_toy_v1.py</code>	<code>682e41ca</code>
<code>__v1/Q/_/5__v1/run_n_scaling_lowrank_v1.py</code>	<code>1b63c1ec</code>
<code>__v1/Q/_/5__v1/run_plane_flow_exact_v1.py</code>	<code>444ea69f</code>
<code>__v1/Q/_/5__v1/run_plane_flow_approx_v1.py</code>	<code>5d6a5e8a</code>
<code>__v1/Q/_/5__v1/exact_lowN_eigenspectrum_v2/code/run_exact_lowN_dimension_saturation_v2.py</code>	<code>11f1b300</code>
<code>__v1/Q/_/5__v1/exact_lowN_eigenspectrum_v2/paper7_5color_time/code/run_paper7_5color_timeseries.py</code>	<code>75b44d1e</code>
<code>__v1/Q/_/5__v1/exact_lowN_eigenspectrum_v2/paper7_125color_time/code/run_paper7_transverse.py</code>	<code>7c01b255</code>

Condition-fixing runner

File	Hash	Role
<code>__v1//_N1to20__v2/run_sample4figs_delta_n_N12_d0.1_T4000_v1.py</code>	<code>184e5356</code>	a wrapper launching the unmodified original with the conditions fixed. The ground for the choice of δ level is recorded in the docstring

A.4 Reproduction procedure

The starting directory is the one containing this file, `//`. The `cd` below are relative to it.

```
# 1. Run ( $N=16=2^4$ , dumped with two-stage sampling)
cd __v1//_N1to20__v2
```

```

DUMP_TAUC=4000 DUMP_STRIDE=31 \
python3 run_nsweep_three_series_v2.py electron 16 16 40000 0.1 dump40k16

# 2. Extract the principal-axis spectrum (matter and vacuum)
python3 make_scale_series_v1.py electron_T40000_d0.1_rep-dump40k16_N16 m v

# 3. Figures 5-12 (four time points x matter/vacuum)
#   tau is chosen to match the N=16 transition (tau approx 9000-9500)
for side in m v; do
python3 make_ellipsoid_figure_v1.py \
--stem electron_T40000_d0.1_rep-dump40k16_N16 --side $side \
--absmax 0.34 --tau 0 4000 9487 39991
done

# 4. Figure 13 (quasi-oscillation period) - needs a dedicated run
#   with the transition inside the dense region.
#   With the default DUMP_TAUC=4000 the N=16 transition (tau approx 9000)
#   falls outside the window.
DUMP_TAUC=10000 DUMP_STRIDE=31 \
python3 run_nsweep_three_series_v2.py electron 16 16 40000 0.1 dump40k16d
python3 make_period_figure_v1.py electron_T40000_d0.1_rep-dump40k16d_N16
#   --dense-end defaults to dump_tauc (=10000) read from dump_meta

# 5. Claim 18 (orientation stability and conserved quadratic forms)
python3 check_invariants_v1.py electron_T40000_d0.1_rep-dump40k16_N16 \
--N 16 --late 10000 --windows 0,4000,8991,9487,20000,40000

# 6. Figures 1-4 (synthetic geometric configurations)
#   We are now in __v1//_N1to20__v2,
#   so three levels up returns to the starting directory.
cd ../../../../figures_v1
python3 make_figs.py && python3 make_fig3.py && python3 make_fig4.py
python3 check_sufficiency_v1.py && python3 check_real_solutions_v1.py

```

A.5 Generated data

There are **two** $N = 16$ runs, under identical conditions and identical random numbers, differing only in the dense region of the two-stage sampling. The directory is `__v1//_N1to20__v2/`.

stem	DUMP_TAUC	frames	run time	use
electron_T40000_d0.1_rep-dump40k16_N16	4000	165162	2300 s	ellipsoid figures (5–12), principal-axis spectrum, Claims 10, 12, 13, 14, 18
electron_T40000_d0.1_rep-dump40k16d_N16	10000	160968	2688 s	period figure (13), Claim 15. Includes the transition $\tau \approx 9000$ in the dense region

File	Size	Note
<code>dump_C2_{stem}_{m,v}_v1.npy</code>	1268.6 MB each (2695.5 MB for 16d)	frames $\times$ 120 relations $\times$ 16 $\times$ 8 complex. gitignored (regenerable by steps 1 and 4)
<code>dump_meta_{stem}_{m,v}_v1.npz</code>	small	contains the τ table <code>dump_taus</code> , <code>dump_tauc</code> , <code>dump_stride</code> , and the relation count <code>m</code>
<code>scale_series_{stem}_{m,v}_v1.npz</code>	small	principal-axis spectrum (15), rank, imaginary count, signed trace at all τ
<code>figures_tau/period_series_electron_T40000_d0.1_rep-dump40k16d_N16_v1.npz</code>	small	the measured values of Claim 15
<code>result_nsweep_electron_T40000_d0.1_rep-dump40k16_v2.json</code>	small	decision values such as the closure residual (<code>closure_med</code> etc.)
<code>figures_v1/check_sufficiency_v1.json</code>	small	the numbers of Claims 16-b, 16-c
<code>figures_v1/check_real_solutions_v1.json</code>	small	the numbers of Claims 4-f, 4-g, 17

Run time: 2300 s (`dump40k16`) / 2688 s (`dump40k16d`) for the two series at $N = 16$, $T = 40000$. About 1.5 times longer than $N = 12$ under the same conditions.

Determinism check: the run script outputs `determinism_max_abs`. Step 1 gives the same output for the same input.